

# Rack Conveyor Dishwasher



## Installation/Operation Manual with Service Replacement Parts



Single Tank Model  
CH-44, MD-44, I-44

### Rack Conveyor Dishwasher

CH-44, MD-44  
44" Single Tank with built-in electric booster

CH-66, MD-66  
44" Single Tank with 22" Prewash  
with built-in electric booster

### International Models

I-44  
44" Single Tank with built-in electric booster

I-66  
44" Single Tank with 22" Prewash  
with built-in electric booster



Single Tank w/Prewash  
Model CH-66, MD-66, I-66



Dishwasher serial no.



[www.championindustries.com](http://www.championindustries.com)

Issue Date: 8.7.14

**Manual P/N 115279 rev. B**

For machines beginning with S/N RM13020073 and above

3765 Champion Boulevard  
Winston-Salem, NC 27105  
336/661-1556 Fax: 336/661-1660  
Toll-free: 800.858.4477

2674 N. Service Road, Jordan Station  
Ontario, Canada L0R 1S0  
905/562-4195 Fax: 905/562-4618  
Toll-free: 800.263.5798

Printed in USA

For future reference, record your dishwasher information in the box below.

Model Number \_\_\_\_\_ Serial Number \_\_\_\_\_  
Voltage \_\_\_\_\_ Hertz \_\_\_\_\_ Phase \_\_\_\_\_  
Champion Service Agent \_\_\_\_\_ Tel: \_\_\_\_\_  
Champion Parts Distributor \_\_\_\_\_ Tel: \_\_\_\_\_

### National Service Department

**In Canada:**

Toll-free: 800/ 263-5798  
Tel: 905/ 562-4195  
Fax: 905/ 562-4618  
email: [service@moyerdiebellimited.com](mailto:service@moyerdiebellimited.com)

**In the USA:**

Toll-free: 800/ 858-4477  
Tel: 336/ 661-1992  
Fax: 336/ 661-1660  
email: [service@championindustries.com](mailto:service@championindustries.com)

**ATTENTION:**

The dishwasher model number, serial number, voltage, hertz and phase are needed to identify your machine and to answer questions.

Please have this information on-hand if you call for service assistance.

For all models:

The data plate mounts to one side of the top-mounted control cabinet.

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# **REGISTER YOUR PRODUCT ONLINE**

*Make sure you are connected to the internet, then enter the address below:*

## **IN THE USA**

### **CHAMPION**

<http://www.championindustries.com/register>

### **MOYER DIEBEL**

<http://www.moyerdiebel.com/register>

## **IN CANADA**

### **CHAMPION AND MOYER DIEBEL**

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# PRODUCT REGISTRATION BY FAX

**COMPLETE THIS FORM AND FAX TO:**

**(336) 661-1660 in the USA**

**1-(800) 204-0109 in Canada**

## PRODUCT REGISTRATION CARD

Model

Serial #

Date of Installation: \_\_\_\_/\_\_\_\_/\_\_\_\_

Company Name: \_\_\_\_\_

Address: \_\_\_\_\_  
(Street) Province Postal Code

Telephone #: (    ) \_\_\_\_\_ --- \_\_\_\_\_

Contact: \_\_\_\_\_

Installation Company: \_\_\_\_\_

Address: \_\_\_\_\_

Telephone #: \_\_\_\_\_

Contact: \_\_\_\_\_

**FAILURE TO REGISTER YOUR PRODUCT MAY VOID YOUR WARRANTY**

**IMPORTANT**

**IMPORTANT**

## Revision History

A revision might be a part number change, new instructions, or information that was not available at print time. We reserve the right to make changes to this manual without notice and without incurring any liability by making the changes. Dishwasher owners may request a revised manual, at no charge, by calling (800) 858-4477 in the USA or (800) 263-5798 in Canada.

| <b>Revision Date</b> | <b>Revised Pages</b> | <b>Serial Number Effectivity</b> | <b>Description</b>                          |
|----------------------|----------------------|----------------------------------|---------------------------------------------|
| 4.1.13               | All                  | RM13020073                       | Released First Edition                      |
| 6.23.14              | 40-41                | All                              | Added International tank heater elements    |
|                      | 58-59                | All                              | Reversed items 11 and 13                    |
|                      | 62-63                | All                              | Added International motors                  |
|                      | 68-69                | All                              | Added international booster heater elements |
|                      | 70-71                | All                              | Added 27kW 200-220/50-60/1 booster          |
| 8.12.14              | 51, 53               | All                              | Changed upper manifold part numbers         |

# LIMITED WARRANTY

Champion Industries, Bi-Line Systems (herein referred to as ("The Companies")), 3765 Champion Blvd., Winston-Salem, North Carolina, 27105) warrants machines and parts, as set out below:

**WARRANTY OF MACHINES:** The Companies warrant all new machines of its manufacture bearing the name Champion or Bi-Line and installed within the United States to be free from defects in material and workmanship for a period of one (1) year from the date of installation or fifteen (15) months from the date of shipment by The Companies, whichever occurs first. This Limited Warranty does not cover products shipped outside of the United States. The warranty registration card must be returned to The Companies within ten (10) days after installation or registered online at [www.championindustries.com/warranty-registration](http://www.championindustries.com/warranty-registration) for the United States; or by fax using the form provided at the front of this manual. If the warranty card, fax, or email are not sent to The Companies within fifteen (15) days, then the warranty will expire after fifteen (15) months from the date of shipment. The Companies will not assume any responsibility for additional installation costs in any area with jurisdictional problems with local trades or unions. The Companies reserves the right to repair or replace a defective part or the entire machine, if a defect in workmanship or material is identified within the warranty period. Alternatively, The Companies may elect to accept the return of the machine for a full credit. In the event if The Companies elect to repair then the labor and work performed in connection with the warranty shall be done by The Companies' authorized service agent during regular working hours and at regular labor rates. Overtime charges are the responsibility of the equipment purchaser. Warranty travel is be covered up to fifty (50) miles from the authorized service technician's servicing office. If travel exceeds fifty (50) miles, the end user will be responsible for any additional travel expense. Service calls initiated under warranty and found not to contain any defects in materials or workmanship, will not be covered by The Companies warranty. Defective parts become the property of The Companies. Use of non-OEM replacement parts, not authorized by The Companies, will relieve The Companies of all further liability in connection with its warranty. In no event, will The Companies' warranty obligation exceed the charge for the machine. Machines that come with a factory-paid start-up will be limited to one (1) authorized service call for start-up. Installation problems or delays, of any kind, will not be covered by The Companies' warranty and will be the sole responsibility of the equipment purchaser.

## THE WARRANTY DOES NOT COVER:

- a. Chemical tubing, chemical squeeze tubes, O-rings, or curtains.
- b. Vacuum breakers.
- c. Adjustments to structural or mechanical components covered by recommended maintenance procedures.
- d. Replacement of fuses, resetting of overload breakers, or high-limit thermostats.
- e. Adjustments of thermostats or other temperature controlling devices.
- f. Adjustments of clutches.
- g. Adjustments of water pressure(s).
- h. Adjustments of factory chemical pumps and settings.
- i. Opening or closing of utility supply valves or switching of electrical supply current.
- j. Cleaning of valves, strainers, screens, nozzles, or spray pipes.
- k. Regular maintenance and cleaning as outlined in the operator's guide.
- l. Damages resulting from water conditions, accidents, alterations, improper use, abuse, tampering, improper installation, under or over voltage conditions, power surges, inadequate wiring, outdoor use, or failure to follow maintenance and operation procedures.
- m. Pulper cutter blocks, pulse vanes, and auger brush due to wear and tear.
- n. Damages due to improper storage.
- o. Special installations or applications, including remote locations, are limited in coverage by this warranty.
- p. Any installation that requires additional work and/or travel to gain access to a machine for service is the sole responsibility of the equipment purchaser.

## THE FOLLOWING DEFECTS ARE NOT COVERED BY THE WARRANTY:

1. Damage to the exterior or interior finish.
2. Damage caused by improper connection to utility service other than that designated on the rating plate.
3. Inadequate or excessive water pressure.
4. Corrosion due to foreign materials, improper water supplies, improper chemicals, or chemicals dispensed in excess of recommended concentrations.
5. Failure of components due to the connection of third-party chemical dispensing equipment installed by others.
6. Leaks and damage due to the use of non-specified water quality.
7. Leaks and damage caused by the installer, including machine table connections.
8. Leaks or damage caused by chemical dispensing equipment connections installed by others.
9. Failure to comply with all local building codes.
10. Damage caused by labor dispute.

**WARRANTY OF PARTS:** The Companies warrant all new machine parts produced or authorized by The Companies to be free from defects in material and workmanship for a period of ninety (90) days from the date of invoice. If any parts defect in material and workmanship is found to exist within the warranty period, then The Companies will refund the cost of the defective part.

## DISCLAIMER OF WARRANTIES AND LIMITATIONS OF LIABILITY.

THE COMPANIES' WARRANTY IS ONLY TO THE EXTENT REFLECTED ABOVE. THE COMPANIES MAKE NO OTHER WARRANTIES, EXPRESS OR IMPLIED, INCLUDING, BUT NOT LIMITED TO ANY WARRANTY OF MERCHANTABILITY OR FITNESS OF PURPOSE. THE COMPANIES SHALL NOT BE LIABLE FOR INCIDENTAL OR CONSEQUENTIAL DAMAGES. THE REMEDIES SET OUT ABOVE ARE THE EXCLUSIVE REMEDIES FOR ANY DEFECTS FOUND TO EXIST IN MACHINES AND PARTS OF THE COMPANIES. ALL OTHER REMEDIES ARE EXCLUDED, INCLUDING ANY LIABILITY FOR INCIDENTALS OR CONSEQUENTIAL DAMAGES.

*Champion Industries or Bi-Line Systems does not authorize any other person, including persons who deal in Champion Industries or Bi-Line Systems machines, to change this warranty or create any other obligation in connection with Champion Industries or Bi-Line Systems machines.*

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## *Model Descriptions*

### **Models CH-44 and MD-44**

Electric high temperature single tank rack conveyor dishwasher with a built-in electric booster in 40°F/22°C rise or optional 70°F/39°C rise. This model's direction of operation is convertible in the field.

### **Models CH-66 and MD-66**

Electric high temperature single tank with prewash tank rack conveyor dishwasher with a built-in electric booster in 40°F/22°C rise or optional 70°F/39°C rise.

### **Models I-44 and I-66**

Electric high temperature single tank and single tank with prewash tank rack conveyor dishwasher with a built-in electric booster in 40°F/22°C rise or optional 70°F/39°C rise. Voltages in 50 Hz.

The installation of your dishwasher must be performed by qualified electricians, and plumbers and serviced by authorized service technicians trained in commercial dishwashers.

Defects and repairs caused by unauthorized installers will not be covered by the dishwasher warranty.



## Receiving

1. Inspect the outside of the dishwasher carton for signs of damage.
2. Remove the carton and inspect the dishwasher for damage.
3. Check for any options or accessories that may have shipped with your dishwasher.
4. **Turn to the front of this manual and follow the instructions to register your product online or by fax.**

**NOTE:**

The installation of your dishwasher must be performed by qualified service personnel. Problems due to improper installation are not covered by the Warranty.

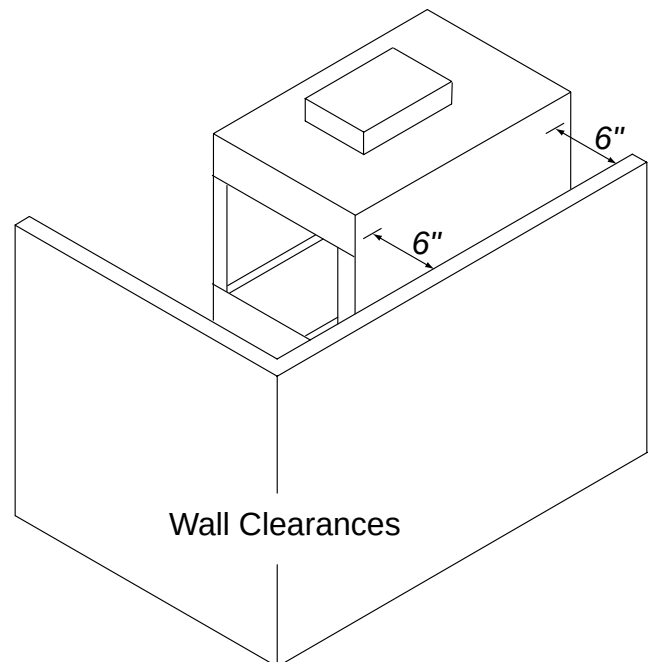
**NOTE:**

The installation of the dishwasher must comply with all local electrical, plumbing, health and safety codes or in the absence of local codes, installed in accordance with the applicable requirements in the National Electrical Code, NFPA 70, Canadian Electrical Code (CEC), Part 1, CSA C22.1; and the Standard for Ventilation Control and Fire Protection of Commercial Cooking Operations, NFPA 96.

**!!ATTENTION!!**

**Use caution** when moving or lifting the dishwasher to prevent damaging the dishwasher or the installation site. Check doorway and passageway clearance before moving the dishwasher. Remove dishwasher front panels and check under the machine base for obstructions before moving.

1. Inspect the dishwasher for shipping damage.
2. Check the dishwasher interior for curtains, panels and other supplies.
3. Lift the dishwasher off the shipping pallet and move the machine near its permanent location.
4. Leave a minimum of 6" between walls and the rear of the dishwasher.
5. Level the dishwasher side-to-side and front-to-back using a bubble level. The dishwasher legs are adjusted by screwing them in or out.
6. Do not remove tags attached to the utility connections.
7. Remove the protective film from the dishwasher exterior.
8. Remove any foreign material from the dishwasher interior.



## *Utilities*

### *Hot Water Connections*

**NOTE:**

Only qualified personnel should make dishwasher plumbing connections.

Connections must meet local plumbing and sanitary codes.

Improper installation is not covered by the dishwasher warranty.

**Hot Water Requirements:**

1. Connect a 3/4" NPT hot water supply line to the line strainer located at the top rear of the dishwasher. A water hardness of 3 grains/gal (US) [51.3 mg/L] or less is recommended.
2. For a dishwasher without a booster heater, the hot water connection must supply a minimum of 180°F/82°C measured at the dishwasher.
3. For a 40°F/22°C rise booster heater, the hot water connection must supply a minimum of 140°F/60°C measured at the dishwasher.
4. For a 70°F/39°C rise booster heater, the hot water connection must supply a minimum of 110°F/43°C measured at the dishwasher.
5. Install a pressure regulating valve (PRV) before the dishwasher supply connection to maintain a flowing pressure of 20-25 PSI/137.8-172.4 kPa.
6. Install a service shut-off valve in the supply line, as close to the dishwasher as possible. The size of the valve must be the same size or larger as the supply line.

### *Cold Water Connections*

**Cold Water Requirements:**

1. Connect a 1/2" NPT cold water supply line for a dishwasher equipped with a prewash cold water tempering option. Connection is located at the top of rear of the dishwasher load end. A water hardness of 3 grains/gal (US) [51.3 mg/L] or less is recommended.
2. Connect a 1/2" NPT cold water supply line for dishwashers required to have a drain water temperature tempering option. A water hardness of 3 grains/gal (US) [51.3 mg/L] or less is recommended. Request a P/N 452891 for a drain tempering water kit.

## Drain Connections

1. The 1-1/2" drain line was removed and packed inside the dishwasher prior to shipping. Install the drain line once the dishwasher has been placed in its final location.
2. Connect the 1-1/2" NPT drain line to above a drain sink or to a 1-1/2" or larger drain line connection.
3. Observe all local plumbing and sanitary codes when installing.

## Ventilation Connections

1. DO NOT VENT THE DISHWASHER INTO WALLS, CEILINGS OR ENCLOSED PLACES.
2. Vent stacks with adjustable dampers are supplied with the dishwasher to connect house vent.
3. Connect stainless steel water-tight duct inside the 4" x 16" / 106mm x 407mm vent stacks supplied with the dishwasher.
4. A minimum of 6 air changes per hour of kitchen is recommended

### Ventilation Guidelines:

Dishwasher without a prewash tank option:

Load end- 200 CFM @ 1/4" SP/ 95 Liters/second

Unload end- 400 CFM @ 1/4" SP/ 189 Liters/second

Dishwasher with a Prewash tank option:

Load end- 150 CFM @ 1/4" SP/ 95 Liters/second

Unload end- 400 CFM @ 1/4" SP/ 189 Liters/second

## Electrical Connections



### **WARNING:**

Electrocution or serious injury may result when working on an energized circuit.

Disconnect power at the main breaker or service disconnect switch before working on the circuit.

Lock-out and tag the breaker to indicate that work is being performed on the circuit.

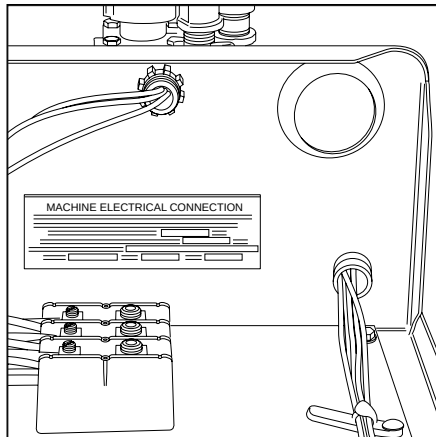
### **ATTENTION**

A qualified electrician must connect the main incoming power to the dishwasher in accordance with all local codes and regulations or in the absence of local codes in accordance with the National Electrical Code.

### **!!ATTENTION!!**

Electrical and grounding connections must comply with the National Electrical Code or in the absence of a National Code then all Local Electrical Codes.

A qualified electrician **MUST** compare the electrical power supply with the machine electrical specifications stamped on the MACHINE ELECTRICAL CONNECTION PLATE located inside the control cabinet before connecting the main power to the dishwasher.

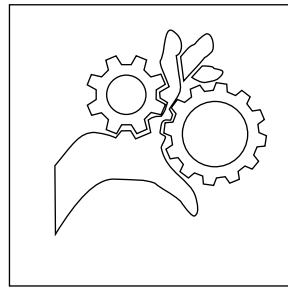


1. The main incoming power to the dishwasher is made at the top of machine in the control cabinet.
2. The electrician must connect the incoming power based on the information that is stamped on the Machine Electrical Connection Plate.
3. Any change to the Machine Electrical Connection Plate must be approved by the factory in advance.
4. The dishwasher main power terminal block is located inside the top-mounted control cabinet.
5. A knock-out plug is provided at the rear of the control cabinet for electrical service connections.
6. Electric blower-dryers have a separate main power connection.

## Electrical Connections (continued)

### Motor Rotation

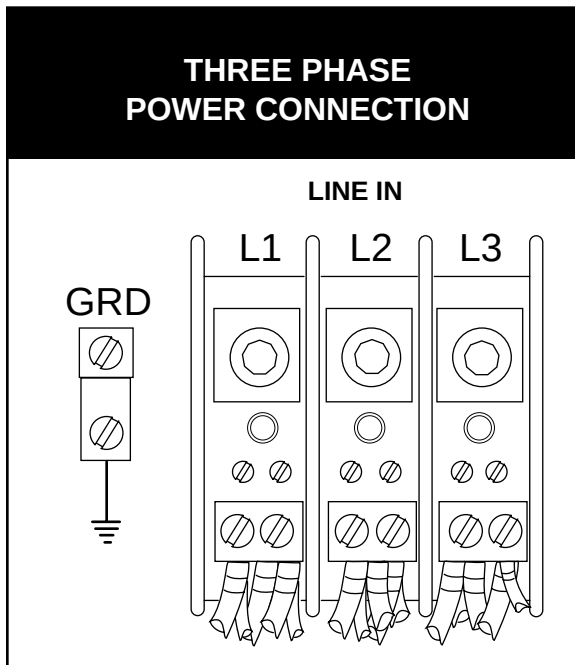
1. Motor rotation was set at the factory.
2. The conveyor drive motor rotation is indicated by a red arrow located on the side of the motor.
3. Check if all motors are running in the wrong direction.
4. Reverse the L1 and L2 wires on the output side of the dishwasher Main Terminal Block (MTB) located inside the top-mounted control cabinet.
5. The photograph below shows the conveyor drive motor with its direction arrow.
6. A wash pump/motor assembly can be seen in the background of the photo.



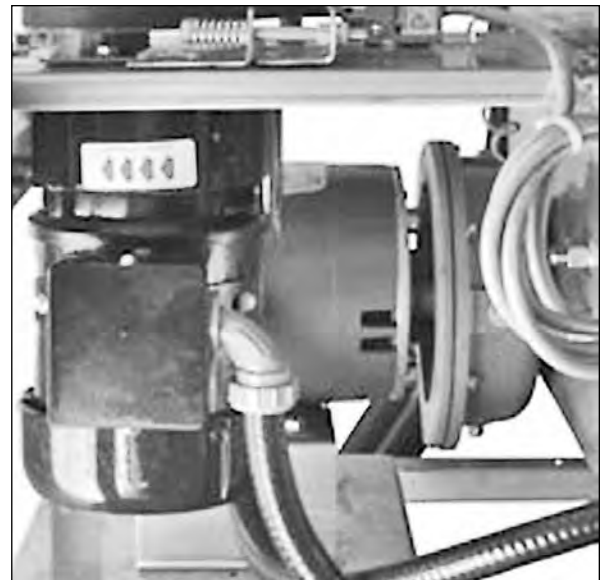
#### **WARNING:**

Moving Conveyor Parts may cause INJURY OR DEATH.  
Keep hands and clothing clear of the conveyor when the conveyor is moving.

USE EXTREME CAUTION WHEN THE CONVEYOR IS MOVING.



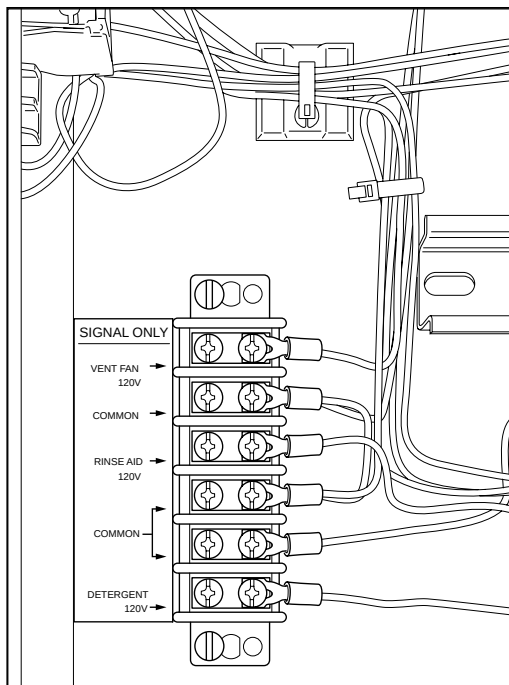
Main Terminal Block Inside the Top-mounted Control Cabinet.



Motor Direction Rotation Label on Motor Frame.

## Chemical Signal Connections

1. Use a qualified detergent/chemical supplier for detergent/chemical and dispensing equipment needs.
2. Labeled detergent control circuit connection terminals are provided in the control cabinet for detergent and rinse agent/sanitizer dispensing equipment (supplied by others).
3. The illustration at right, shows the terminal board for the machine.
4. The signal connection points include:
  - Detergent signal 120VAC, 1A max load.
  - Rinse aid/Sanitizer signal 120VAC, 1A load.
5. A removable black plug is provided in the load end side of the wash tank for installation of the detergent conductivity cell.



## Vent Fan Signal Connection

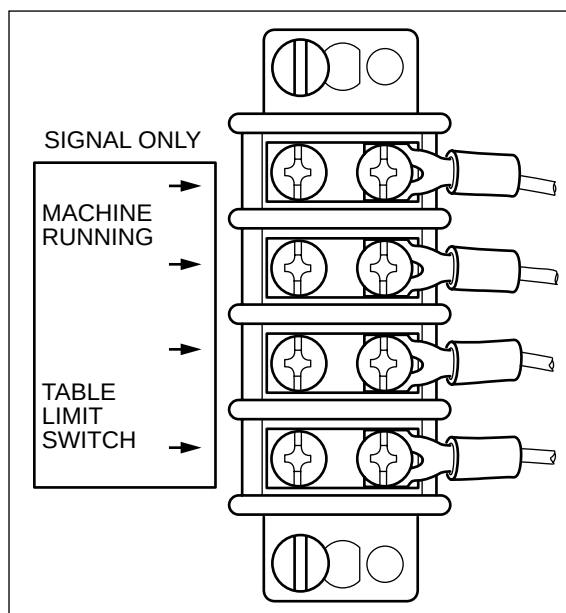
### NOTE:

The Vent Fan Signal Connection supplies 120VAC to a control relay (supplied by others) when the dishwasher is ON and 0 VAC when the dishwasher is OFF. Power to operate the vent fan (supplied by others) must be supplied separately.

1. The Vent Fan signal connection point is located on the same terminal board as the chemical connections (see illustration above). The connection is:
  - Vent Fan 120VAC, 1Amp max amp load

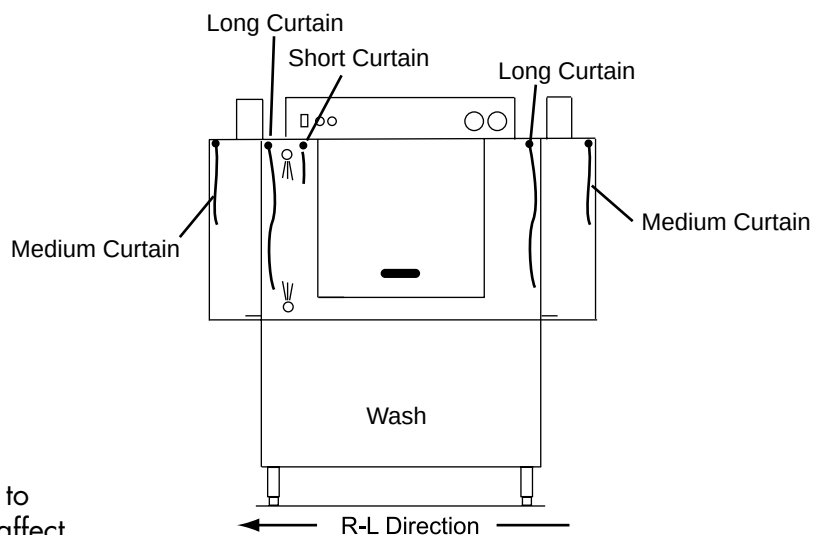
## Running Signal and Table Limit Switch Connections

1. Connections are provided for systems that require a signal to indicate the dishwasher is running.
2. A signal connection is provided to indicate that the dishwasher has stopped due to a conveyor jam or when the clean dish table is full of racks and additional racks cannot exit the machine.
3. The table limit switch option installation is recommended for all dishwashers and can be ordered from the factory by P/N 407400.



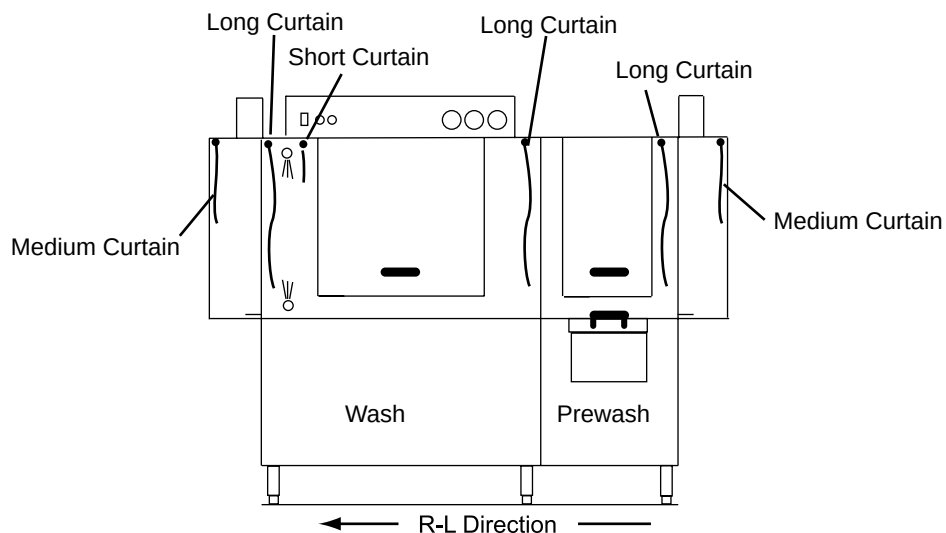
## Curtain Locations

1. Refer to the illustrations below and hang the curtains as shown.  
J-hooks are located in the corners of each section to accept the curtain rods.
  - Standard long curtains 24" x 20-1/4"
  - Standard medium curtains 24" x 13-1/4"
  - Standard short curtain 24" x 6-1/4"
2. Make sure the that the short flaps of the curtains face the load end of the dishwasher.



### NOTE:

Misplacing a curtain or failing to install a curtain will adversely affect the proper operation of the machine.



## Door Safety Switches

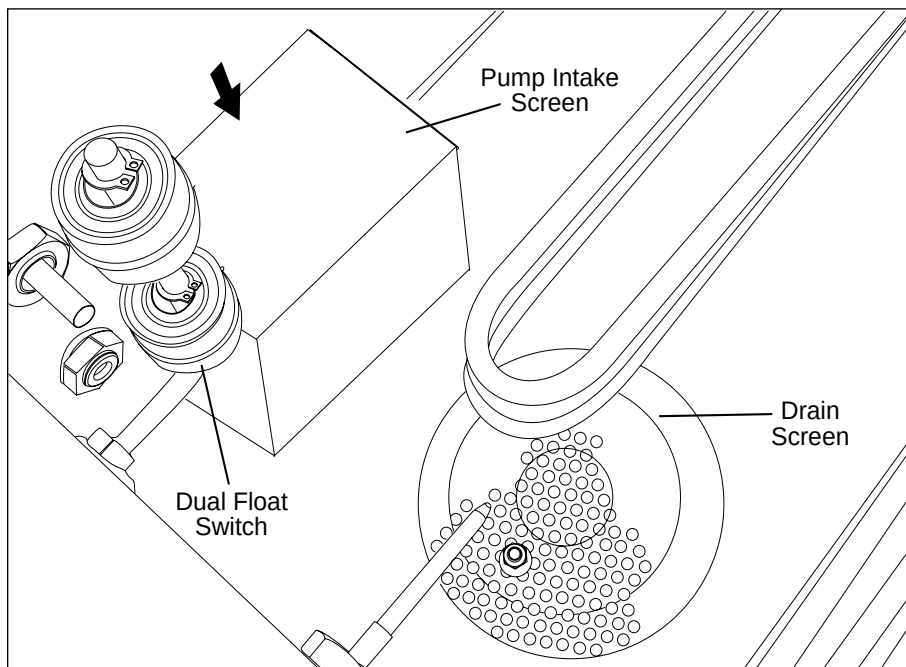
Dishwasher access doors are equipped with a door safety switch that automatically stops the dishwasher pumps and conveyor drive if a door is raised while the dishwasher is running. In addition, the dishwasher will not start if a door is left open.

1. If the dishwasher is running and a door is raised, then lighted GREEN START pushbutton goes out and the pumps and conveyor drive stop.
2. Check the interior of the dishwasher for any dish racks still in the machine. These dish racks must be washed again to ensure they are washed and sanitized completely.
3. To restart the dishwasher, make sure all doors are closed, then push the GREEN START pushbutton.

## Pump Intake Screen and Dual Float Switch

Refer to the illustration below and note the location of the pump intake screen and dual float switch.

1. Make sure the pump intake screen is installed by sliding it on the bracket located in front of the wash pump intake.
2. Make sure the float balls on the dual float switch move freely on the float stem.
3. Check the interior of the tank for any foreign objects and make sure the drain screen is clean.



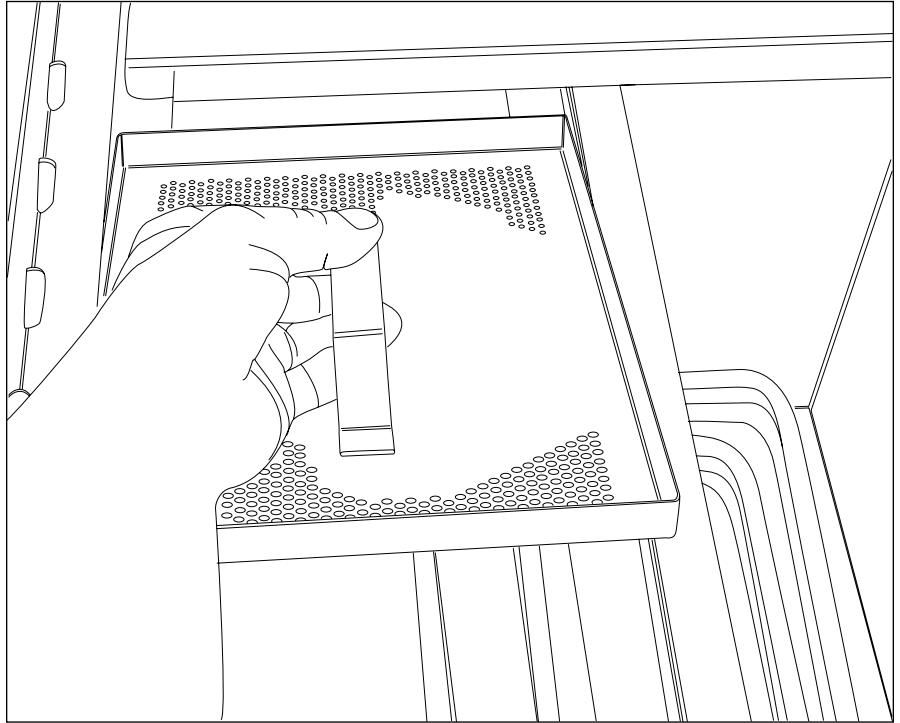
*Make sure tank is clean, the pump intake screen is installed and the dual float switch moves freely.*



## Scrap Screens

1. All models have scrap screens in the top of the wash tank.

Install four scrap screens in the wash tank making sure they fit securely without large gaps between them.



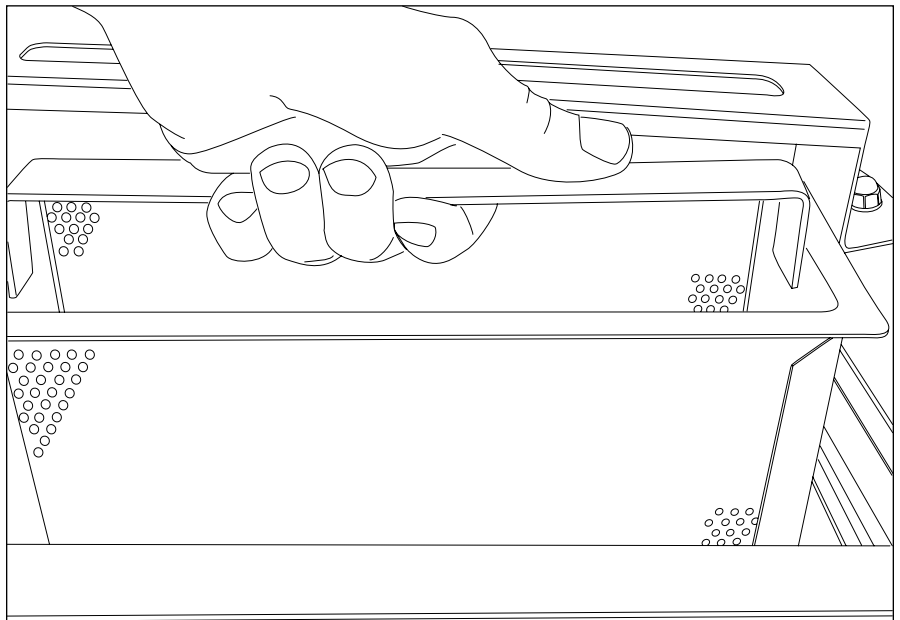
*All models have four scrap screens in the wash tank.*

2. The model CH-66 and the MD-66 has a prewash tank containing a single large scrap screen which extends from front to back in the prewash tank.

The prewash screen also holds a refuse basket to facilitate cleaning when the machine is off.

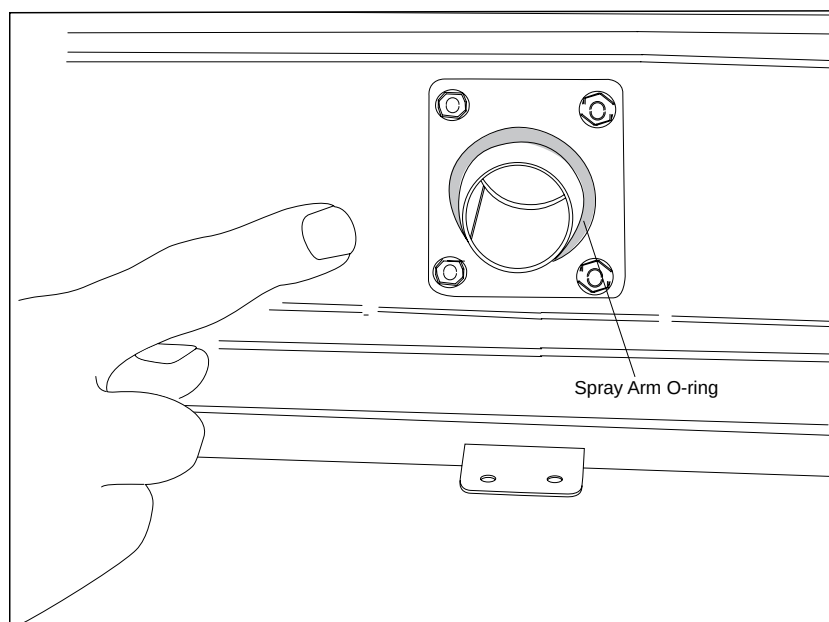
Install the large scrap screen making sure the screen fits securely in the front and back of the tank then install the refuse basket as shown in the illustration to the right.

Never remove the prewash scrap screen while the dishwasher is running.

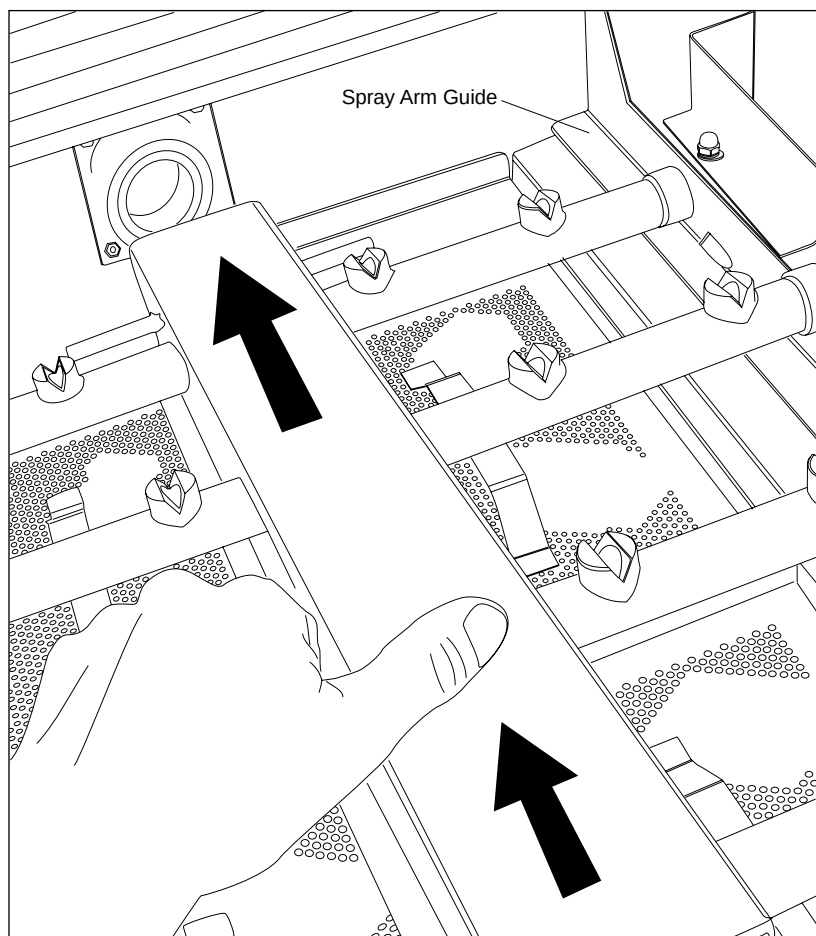


*Model CH-66 and MD-66 has a scrap screen and refuse basket located in the prewash tank.*

## Installing the Lower Spray arm Assembly



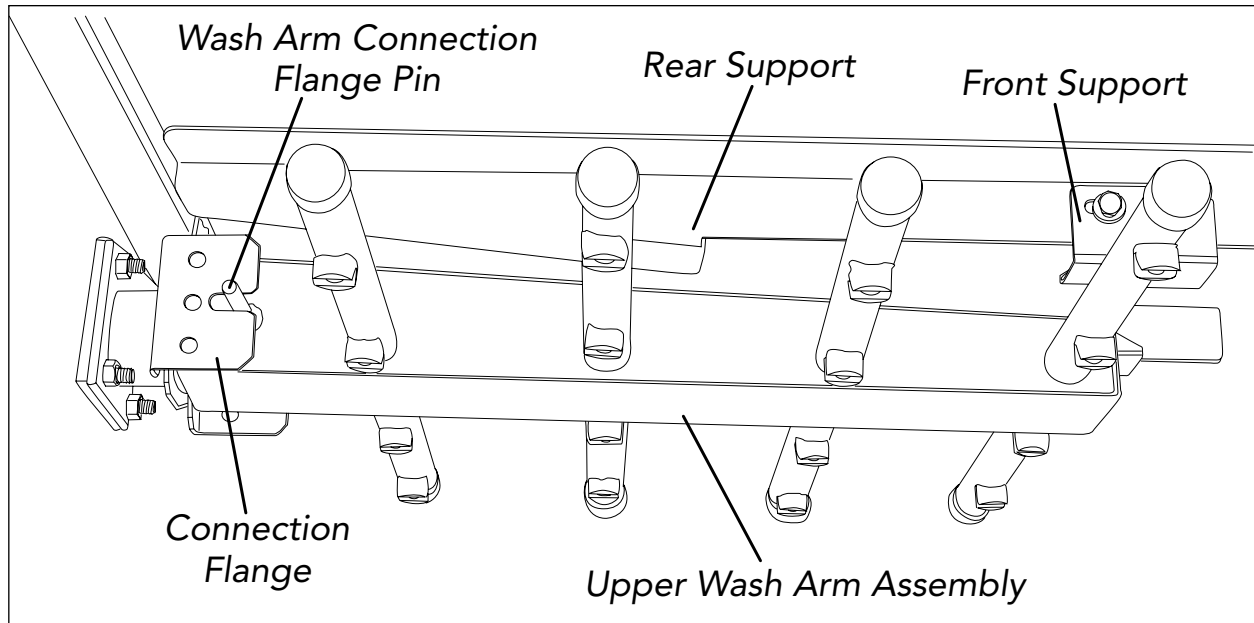
1. The lower spray arm assembly is connected to the rear wall of the wash tank.
2. The spray arm o-ring makes a water-tight seal when the spray arm is properly seated.



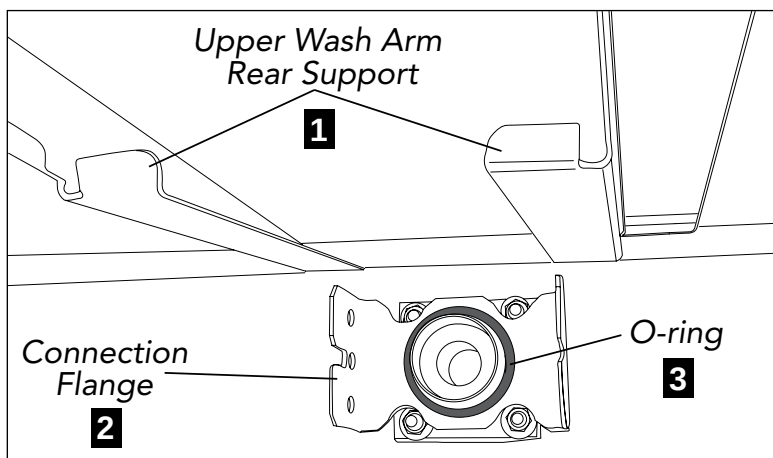
3. Slide the lower arm into place making sure the spray arm is fully back and contacting the spray arm o-ring.
4. The spray pipes are supported on each side by the spray arm guides and will lock the spray arm securely in place.

## Installing the Upper Spray Arm Assembly

The upper wash arm assembly is suspended from the top of the dishwasher hood interior by front and rear supports. Follow the instructions and illustrations below to install the upper wash arm assembly.



Upper wash arm assembly shown installed.  
(Note that the Wash Arm Connection Flange Pin is seated in the Connection Flange.)



Make sure the upper spray arm o-ring is in good condition and fits securely in the connection flange before installing the upper wash arm assembly.

### Components

1. Upper Wash Arm Rear Support  
The rear support holds the wash arm and serves as a guide when the upper wash arm is pushed toward the connection flange at the rear of the dishwasher.
2. Connection Flange  
The connection flange holds the wash arm securely in place when the upper wash arm is placed in position. In addition, the connection flange holds the wash arm o-ring in position.
3. O-ring  
The wash arm o-ring rests in the center of the connection flange and forms a water-tight seal when the upper wash arm is in place.

## Installing the Upper Spray Arm Assembly (continued)

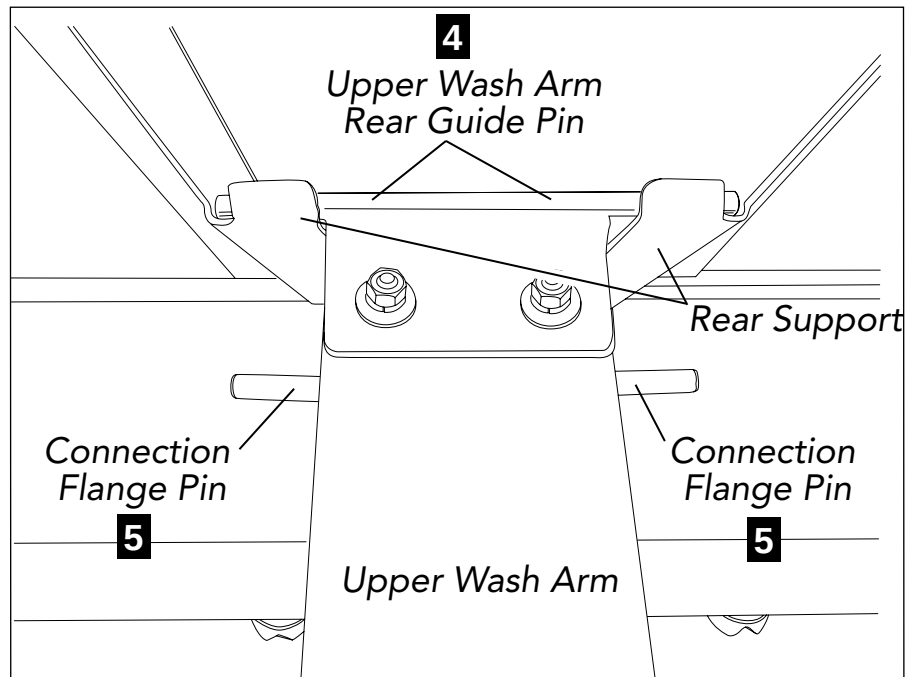
1. Upper Wash Arm

Rear Guide Pin

The rear guide pin is attached to the rear of the upper wash arm. It mates with the upper wash arm rear support.

2. Connection Flange Pins

The connection flange pins are welded to the sides of the upper wash arm. These pins mate with two slots in the connection flange to hold the wash arm in place when installed.



*The upper wash arm is shown with the rear guide pin placed on the rear support before sliding the upper wash arm into position.*

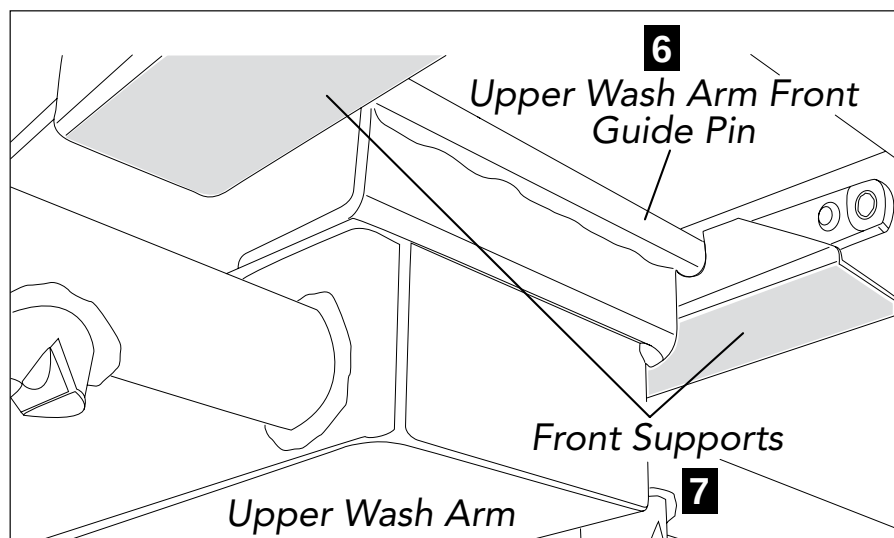
3. Upper Wash Arm

Front Guide Pin

The front guide pin is attached to the front of the upper wash arm. It mates with the front supports.

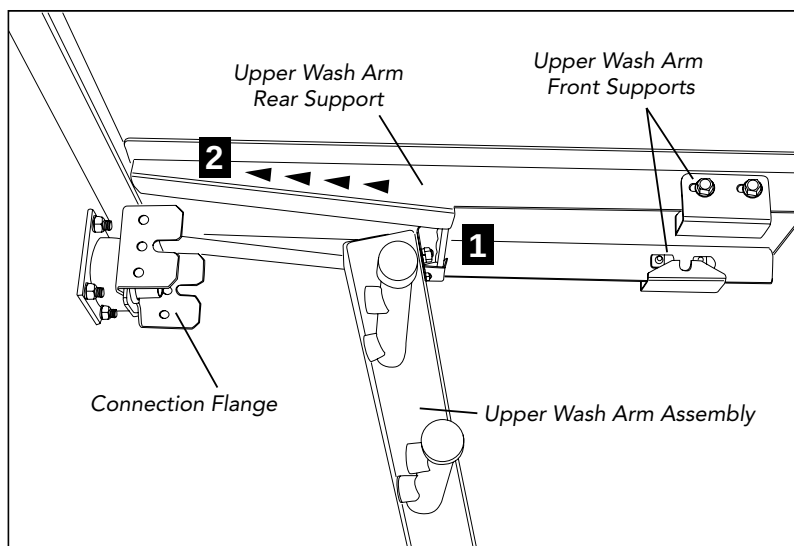
4. Front Supports

There are two front supports which hold the front of the upper wash arm in place when the wash arm is installed.

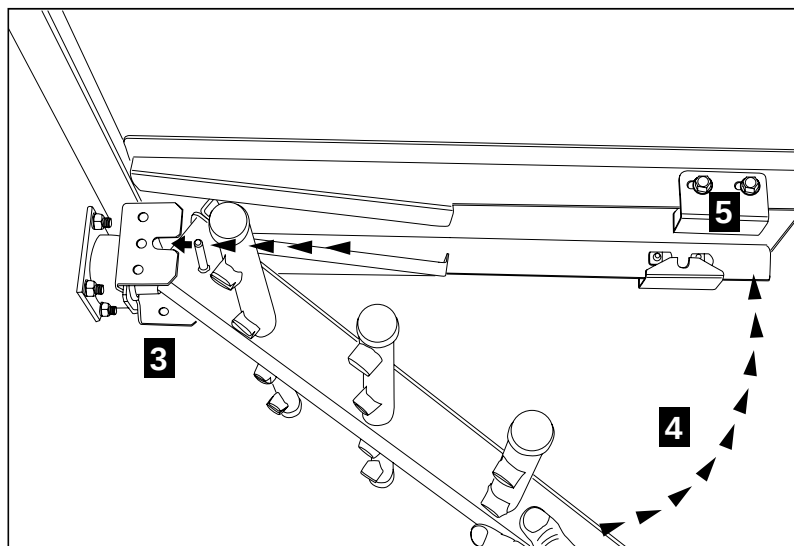


*The upper wash arm is shown with the front guide pin resting in the front supports.*

## Installing the Upper Spray Arm Assembly



The upper wash arm assembly is shown resting on the upper wash arm rear support as it slides toward the connection flange.



The upper wash arm assembly is shown resting on the upper wash arm rear support as it slides toward the connection flange.

### Installation

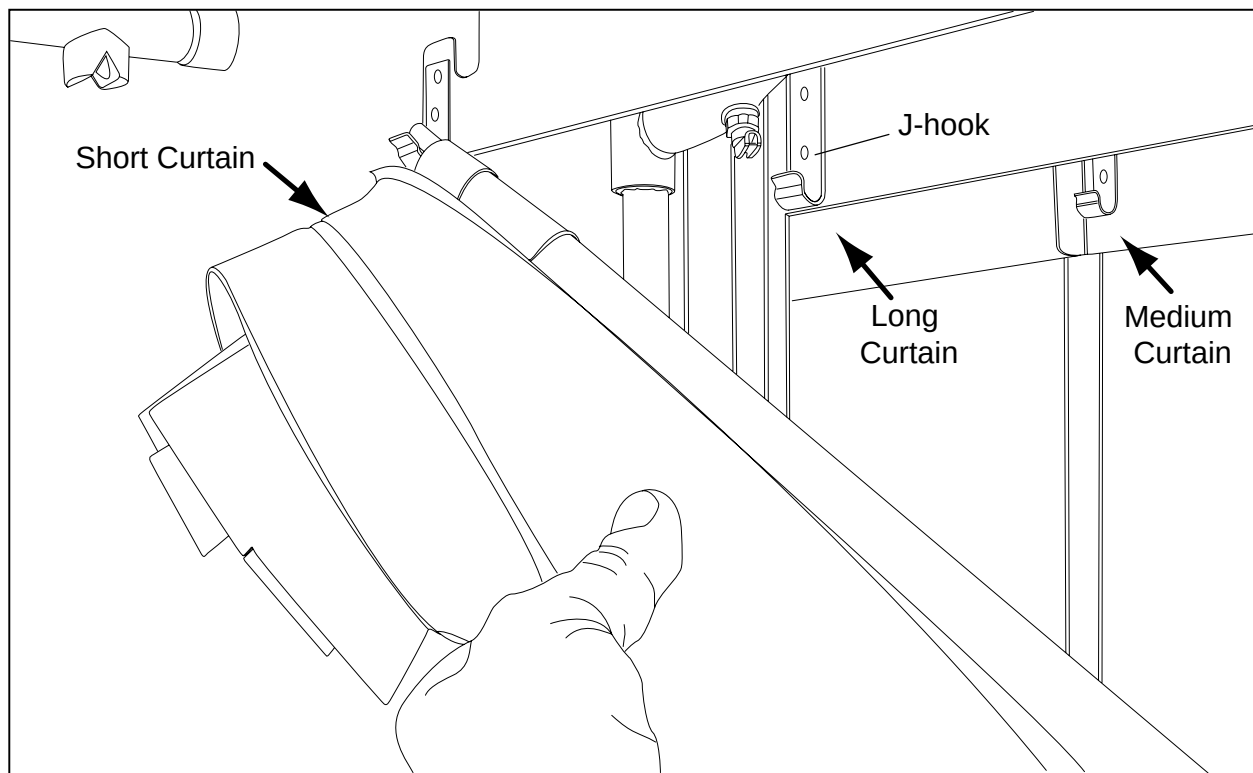
1. Open the dishwasher access doors and check the location and condition of the connection flange o-ring. Replace the o-ring if it is worn or damaged.

Place the upper wash arm rear guide pin in the rear support so the wash arm hangs in the support.

2. Slide the upper wash arm toward the rear of the dishwasher making sure the rear guide pin is centered in the rear supports.
3. Raise the front of the wash arm so the connection flange pin mates with the connection flange pin slots.
4. Swing the front of the wash arm up toward the front of the dishwasher.
5. Slide the upper wash arm forward as far as it will go making sure that the front guide pin is above the front support notched cut-out.

Lower the upper wash arm so the front guide pin seats securely in the front support notch. Installation is complete.

## Installing the Curtains



*Curtain Installation Shown at the Final Rinse End of the Dishwasher*

1. Curtains are equipped with curtain rods. The rods are hung on J-hooks located in the top of the dishwasher.
2. Make sure that the short flaps on the curtains face the load end of the dishwasher.
3. Refer to page 7, *Curtain Locations*, for additional information on curtain placement.

## Check List

1. Remove white protective film from the dishwasher exterior.
2. Install lower panels to the dishwasher.
3. Remove any foreign material from inside of the machine.
4. Check dishwasher drain/overflows are closed and in securely seated.
5. Install scrap screens, spray arms.
6. Turn main utilities to the dishwasher ON. (Power, water).
7. Make sure doors are closed.
8. Turn dishwasher power switch ON. Machine fills with water and tank heat comes on.
9. Check tank water temperatures reach proper levels.
10. Check for leaks.
11. Push the Start button. Green light comes on.
12. Insert an empty dish rack into the load end of the dishwasher.  
The pumps and conveyor will run.
13. Allow the dish rack to travel to the unload end of the dishwasher. The final rinse will run.
14. Allow the rack to exit the dishwasher. The dishwasher should have run for 90 seconds.
15. Place another dish rack in the machine to start another cycle.
16. Open each dishwasher door to make sure the safety switch stops the conveyor.
17. Immediately close the door, the dishwasher should restart.
18. Open the door and leave it open for 5 minutes. The machine should stop.  
Push the Start button and the machine should restart when a rack is inserted into the machine.
19. Push the Stop button when the machine is running and the dishwasher should stop.
20. Push the dishwasher Power Switch OFF. The dishwasher should shut down.
21. Drain the dishwasher and check that floor drains handle the water volume leaving the dishwasher.

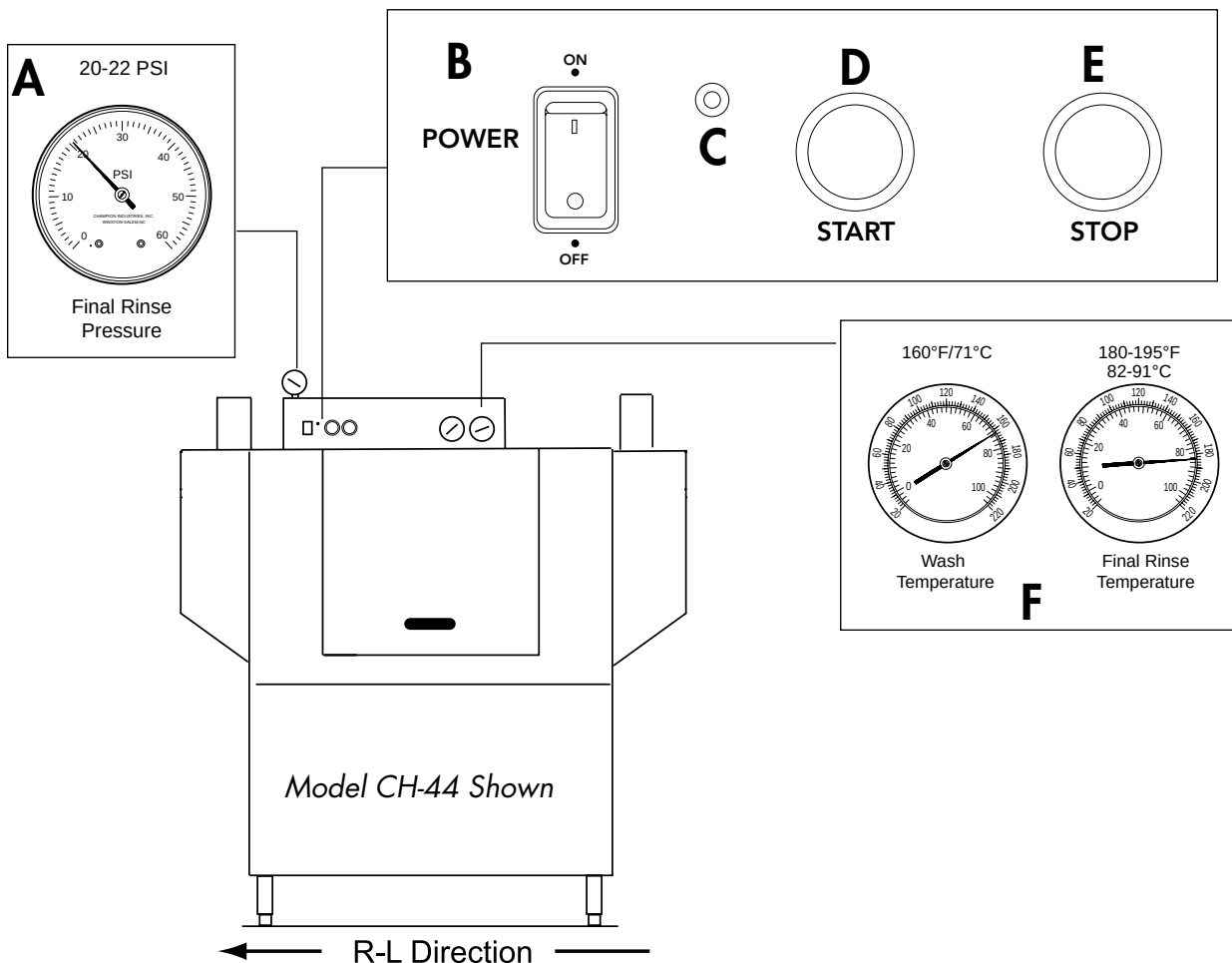
### NOTE:

**If the drain valve is open when the dishwasher is filling, the machine will fill for 10 minutes, then the machine will shut-down and the green light will blink on and off. Flip the dishwasher power switch off and close the drain valve. Flip the power switch on and fill the machine.**

## Control Panel Operation - CH-44, CH-66

The final rinse pressure gauge is located behind the control panel. The top-mounted control panel contains the Power ON/OFF switch, start push button, stop push button, and the water temperature gauges for the wash tank and the final rinse. The function of these controls are:

- A Final Rinse Pressure Gauge -** Indicates the flowing water pressure during the final rinse. Proper reading is 20-22 PSI.
- B Power Switch -** Turns power on and off to the dishwasher. The machine fills automatically when the power is turned on.
- C Cycle Light -** Indicates when the dishwasher is ready for automatic operation.
- D Start Push button -** Pushing the Start push button sets the machine for automatic operation. The conveyor and pumps start when a dish rack is inserted into the machine.
- E Stop Push button -** Pushing the Stop push button stops the conveyor and pumps.
- F Temperature Gauges -** Temperature gauges indicate the water temperature in the wash tank. The final rinse water temperature is indicated during the final rinse.

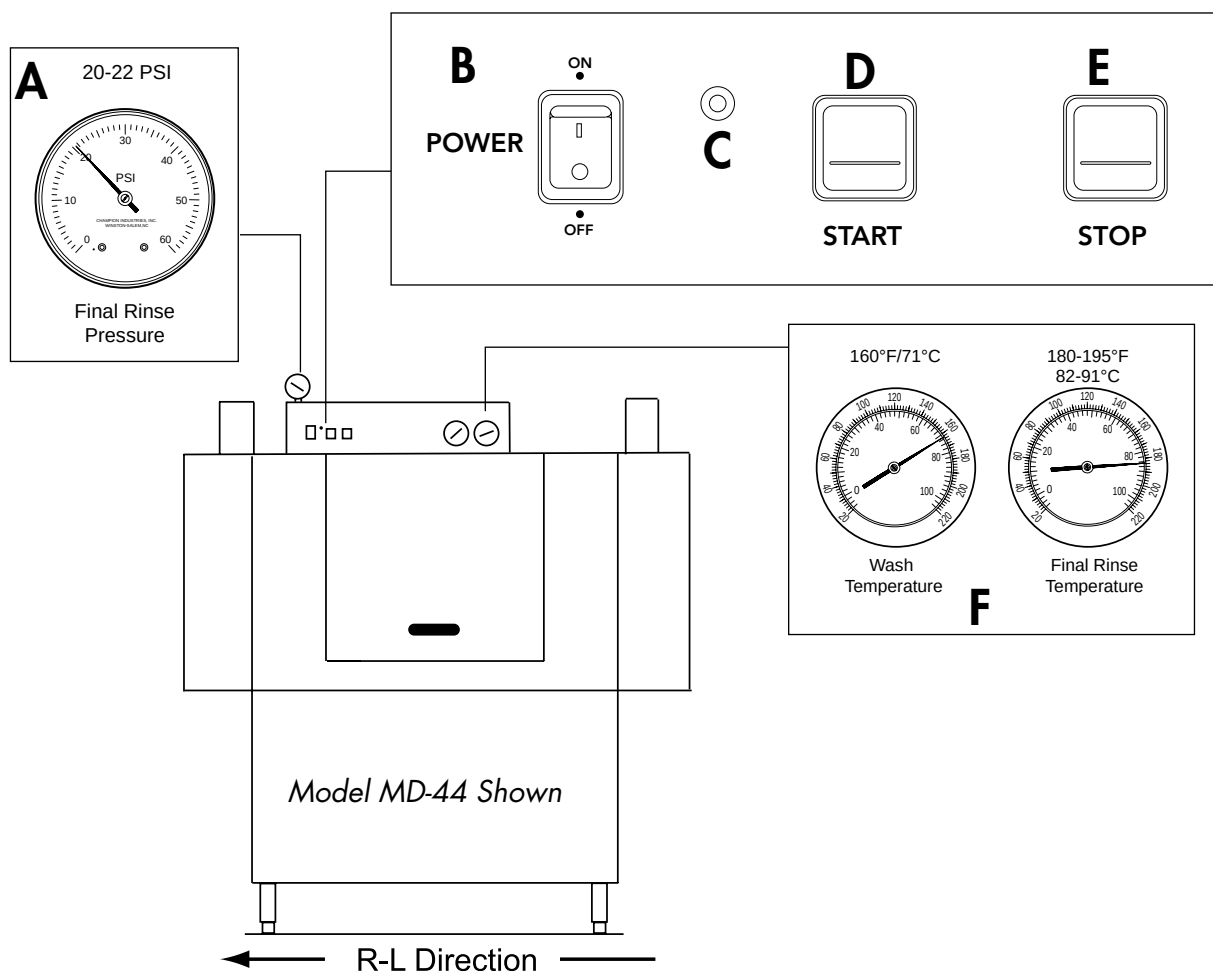




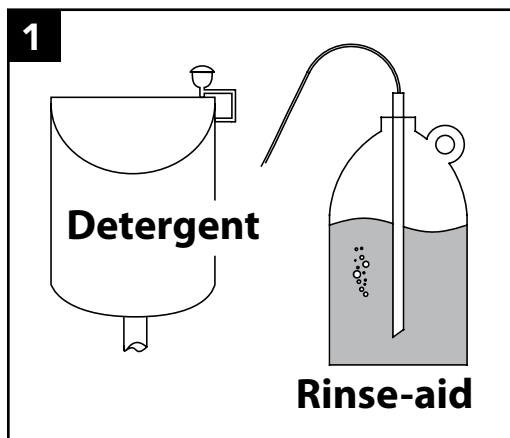
## MD-44, MD-66 - Control Panel Operation

The final rinse pressure gauge is located behind the control panel. The top-mounted control panel contains the Power ON/OFF switch, start push button, stop push button, and the water temperature gauges for the wash tank and the final rinse. The function of these controls are:

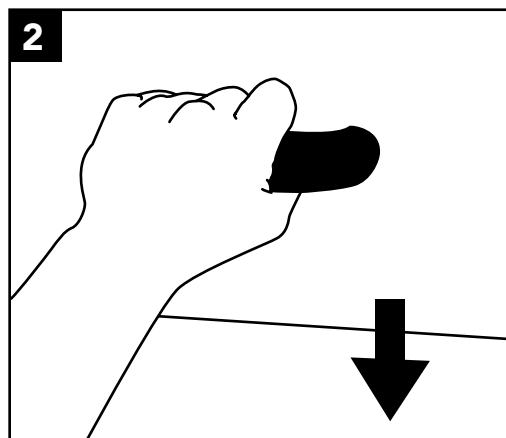
- A Final Rinse Pressure Gauge -** Indicates the flowing water pressure during the final rinse. Proper reading is 20-22 PSI.
- B Power Switch -** Turns power on and off to the dishwasher. The machine fills automatically when the power is turned on.
- C Cycle Light -** Indicates when the dishwasher is ready for automatic operation.
- D Start Push button -** Pushing the Start push button sets the machine for automatic operation. The conveyor and pumps start when a dish rack is inserted into the machine.
- E Stop Push button -** Pushing the Stop push button stops the conveyor and pumps.
- F Temperature Gauges -** Temperature gauges indicate the water temperature in the wash tank. The final rinse water temperature is indicated during the final rinse.



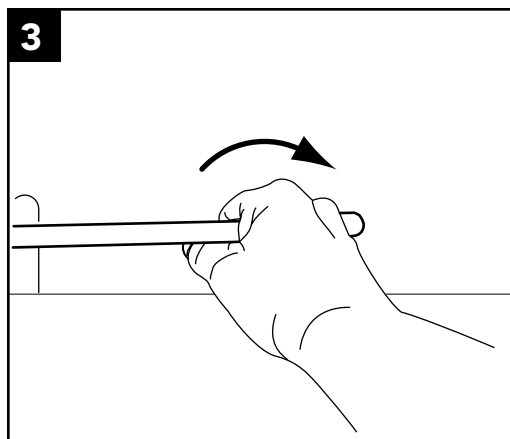
## Operation



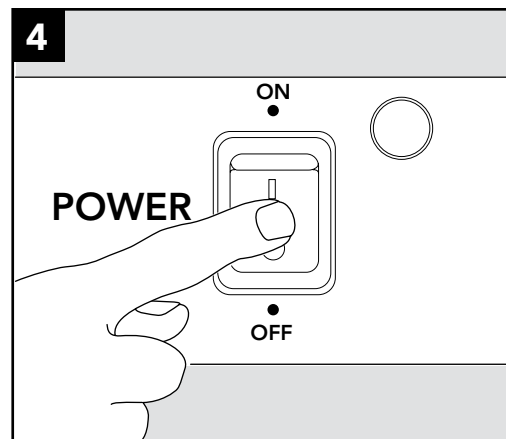
Check the chemical containers and refill as necessary.



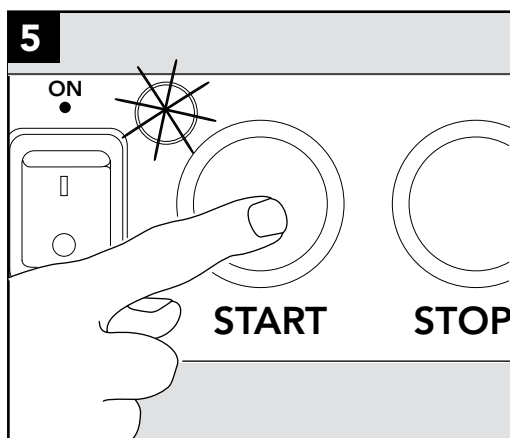
Close the doors.



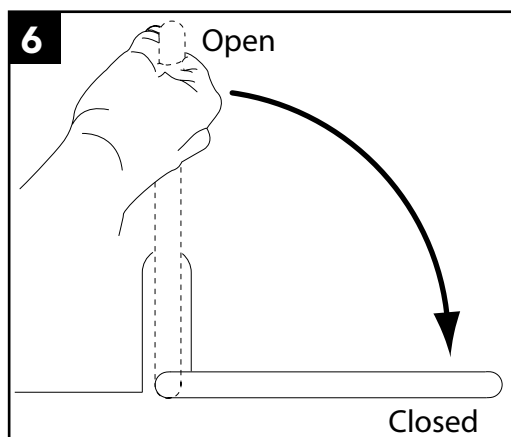
Turn drain handle clockwise to close the drains.



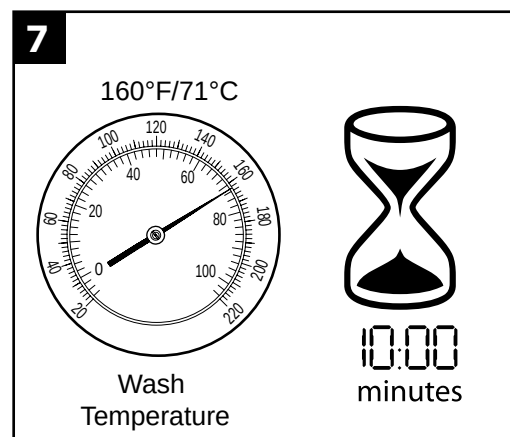
Flip power switch UP to the ON position.



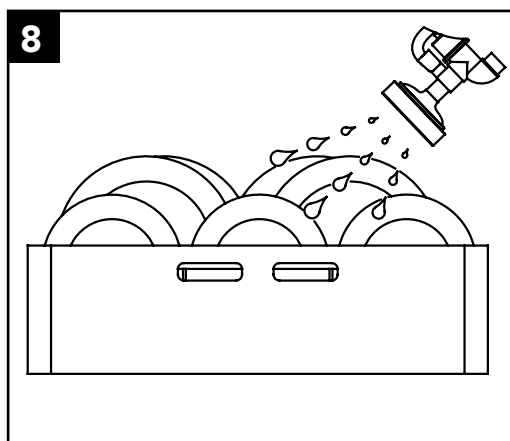
Push START switch. The green light illuminates. The machine fills with water and the tank heat comes on.



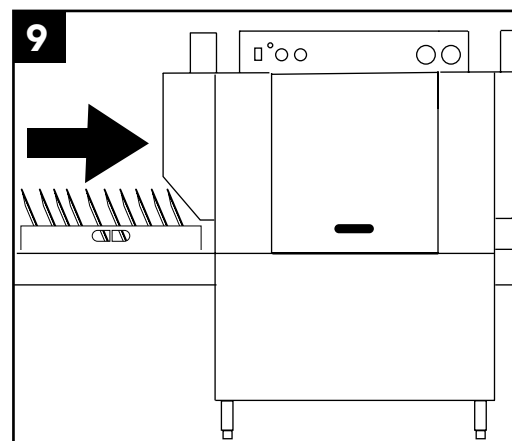
If the drain is open or the fill runs for more 10 minutes then the dishwasher shuts down & the green light blinks on and off. Close the drain or check the fill, then push the power switch off and back on to reset. Push the Start switch to restart operation.



Wait 10 minutes for the wash gauge to indicate 160°F/71°C.



Prescrap & load wares into dish racks. Plates and glasses go in peg rack, cups and bowls go in a flat rack. Place silverware in a single layer in flat rack.



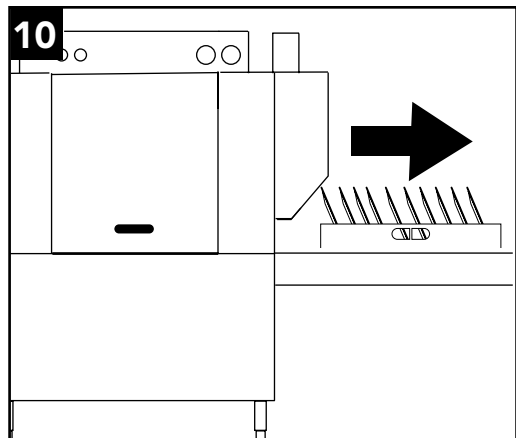
Insert a dish rack of wares into the dishwasher. The pumps and conveyor will run.

**NOTE:**

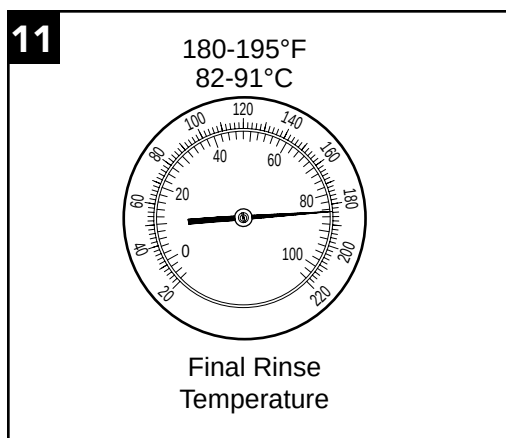
Opening a door will stop the pumps and drive. If the door is closed within 5 minutes the machine will resume operation where it left off. After 5 minutes the START switch must be pressed and a dish rack inserted into the machine for normal operation.

*continued on next page*

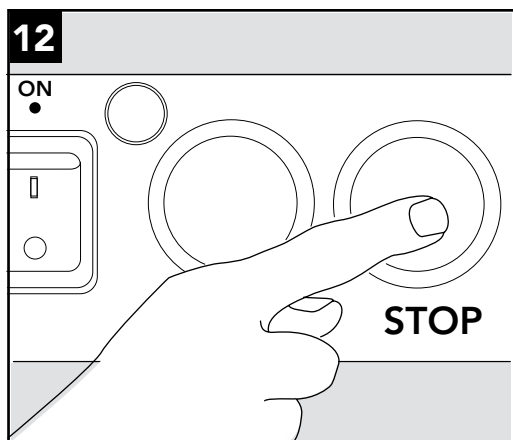
## Operation (continued)



The dishwasher pumps and drive runs 90 seconds after the rack enters the machine. Loading additional racks before the first rack exits the machine resets the timer.



Check the final rinse temperature during the final rinse cycle to ensure the gauge indicates 180-195°F/82-91°C.



Push the STOP switch to stop the pumps and the drive. The green light goes out. Push the START switch and insert a dish rack to restart operation.

### NOTE:

In the event of a conveyor jam or a tripped table limit switch, the dishwasher pumps and drive will stop. The operator should check the conveyor for articles stuck in the drive or in the case of the table limit switch, remove dish racks from the clean-end table. The START switch must be pressed and a dish rack inserted into the machine to resume normal operation.

### NOTE:

To shut the dishwasher down, press the stop button. Flip the POWER switch down to the OFF position. Open the doors and remove any dish racks left in the machine. Open the drain to drain the water from the tanks.

## Cleaning

Cleaning your dishwasher is the best maintenance you can do. The cleaning intervals below are the minimum requirements for most dishwashers. You may need to clean your dishwasher more often when washing heavily soiled wares or during long periods of continuous operation.

### Daily or every 2 hours of operation

1. Turn power switch to OFF.
2. Pull drain lever(s) to drain water. Remove scrap screens and scrap baskets. Clean inside of the tanks and flush with clean water. Back flush the scrap screens until clean.

### **DO NOT STRIKE SCREENS OR BASKETS AGAINST SOLID OBJECTS**

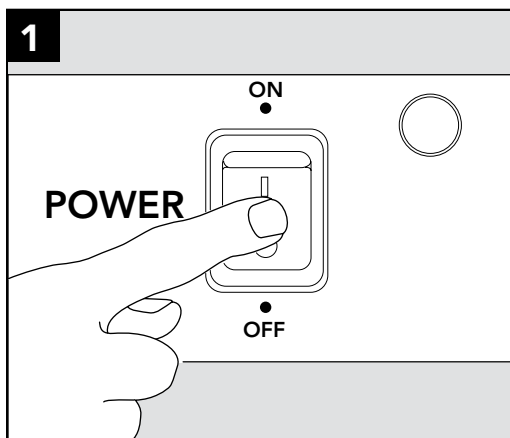
3. Remove the spray arm assemblies. Remove the end cap from each spray arm.
4. Flush the spray arms and nozzles to remove any debris.
5. Replace the end caps. Check the condition of the manifold O-ring.
6. Reinstall the spray arms.
7. Remove and clean the curtains. Allow them to dry at the end of the day.
8. Leave the doors open between operations, allowing the machine to dry.
9. Make sure that the final rinse nozzles are clear of mineral deposits.
10. Straighten a metal paper clip to clean the nozzles.
11. Check the temperature and pressure gauge readings during operation.
12. Inspect the machine for signs of water leaks.
13. Check the chemical supplies and refill as necessary.

### At the End of the Day

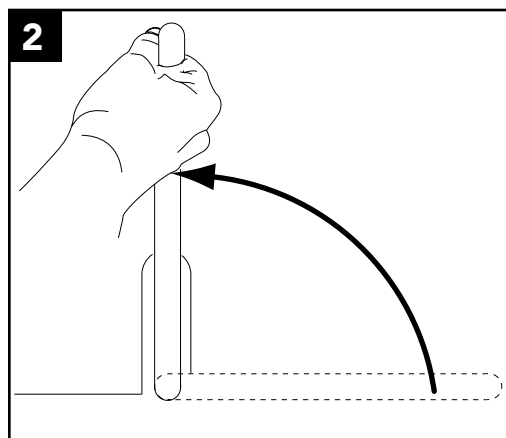
1. Perform Steps 1-10 on the previous page.
2. Remove the upper and lower rinse and wash spray arms and end plugs and flush with fresh water.
3. Clean the final rinse arm nozzles using a small paper clip.
4. Remove the curtains and clean with fresh water.  
**DO NOT USE STEEL WOOL TO CLEAN THE INTERIOR OF THE MACHINE.**
5. Wipe the interior and exterior of the machine with a soft cloth and a mild detergent.  
**DO NOT HOSE THE EXTERIOR OF THE MACHINE WITH WATER.**
6. Reassemble the dishwasher and leave the door open to allow overnight drying.
7. Contact the chemical supplier for de-liming if required.

*continued on next page*

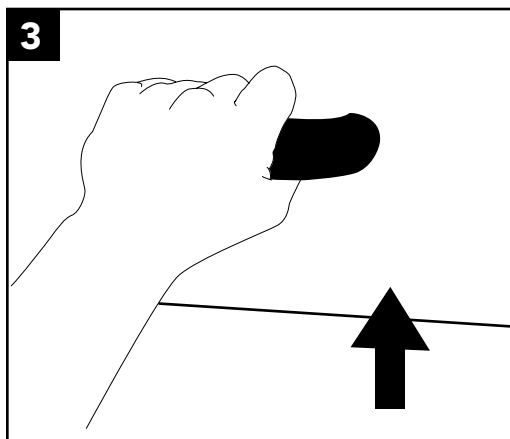
*Cleaning (continued)*



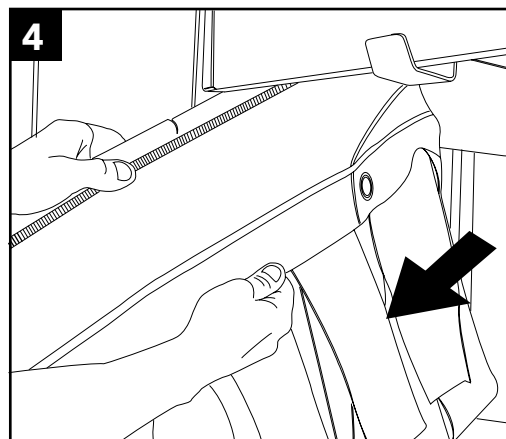
Flip power switch down to the OFF position.



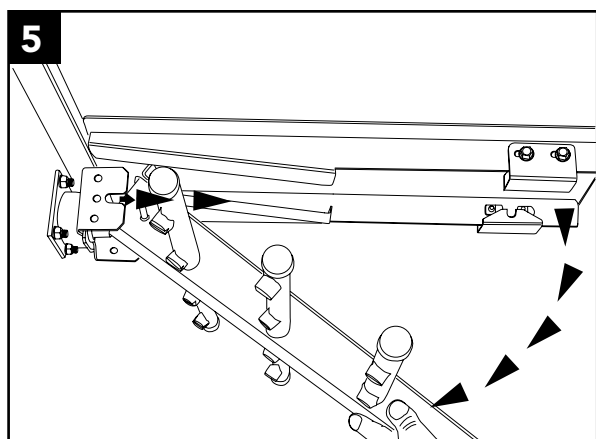
Turn drain handle counter-clockwise to the vertical position to drain the tanks.



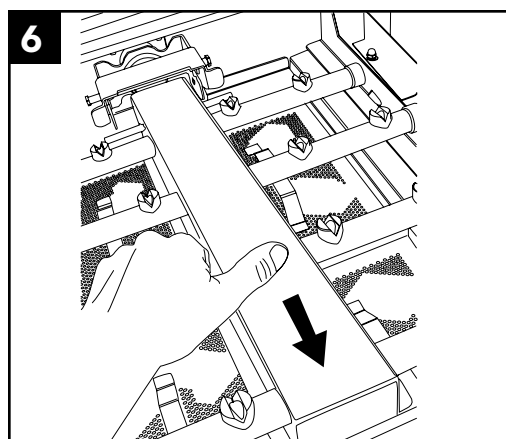
Open doors.



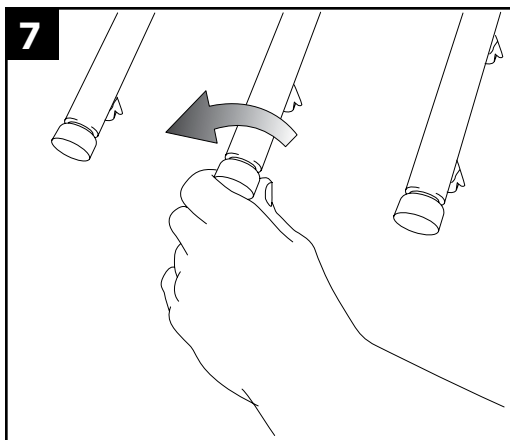
Remove curtain assemblies and rinse clean in a sink.



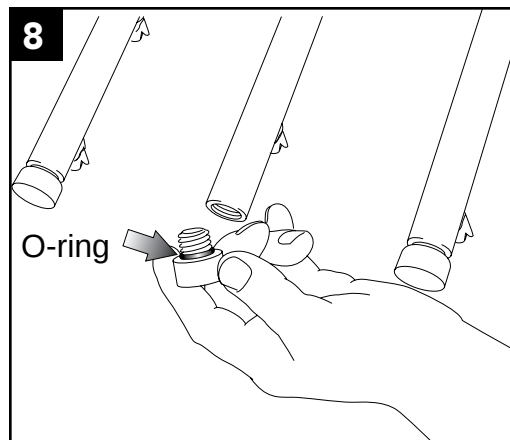
Remove the upper wash arm assemblies and flush clean in a sink.



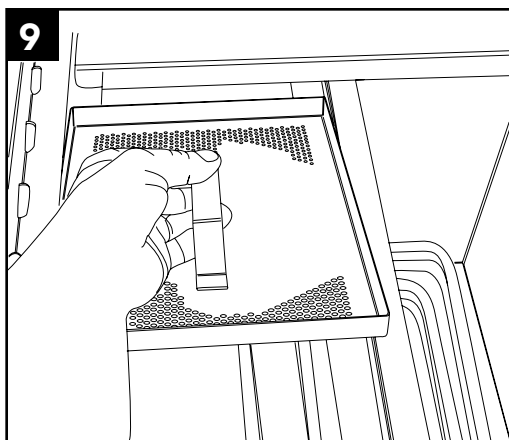
Remove lower wash arm assemblies and flush clean in a sink.



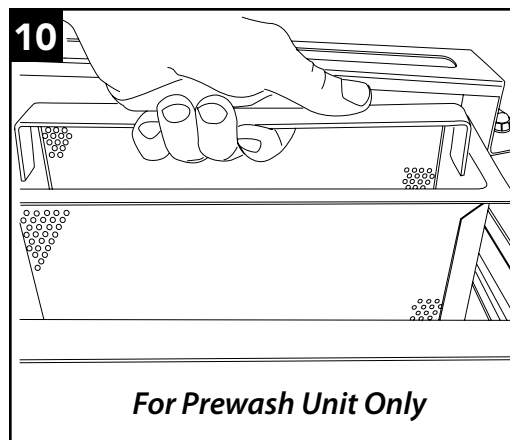
Remove the end plugs on each wash arm and flush the pipes clean.



Check the o-ring on each plug and replace if necessary.

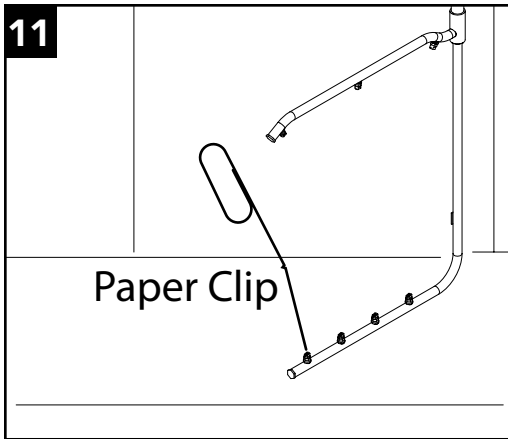


Remove the scrap screens and back-flush in a sink. Do not strike the screens on hard surfaces.

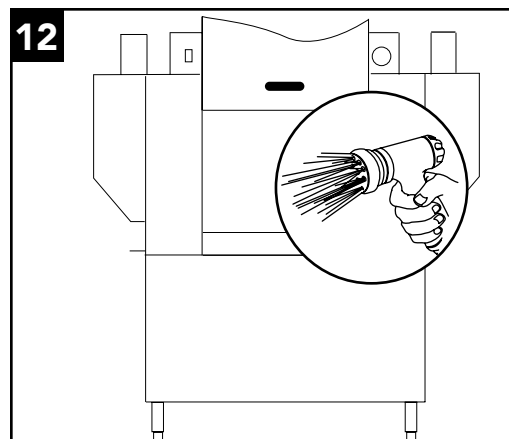


Remove the refuse basket and back-flush in a sink. Do not strike the basket on hard surfaces.

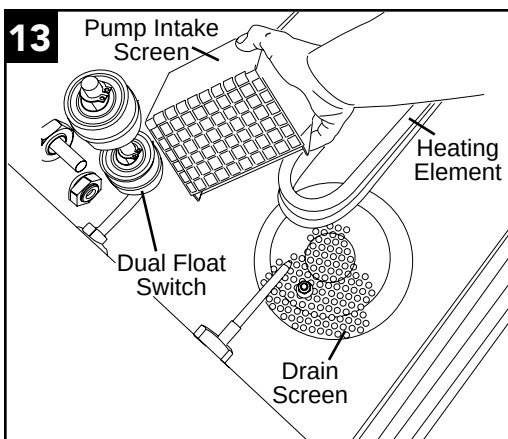
Cleaning (continued)



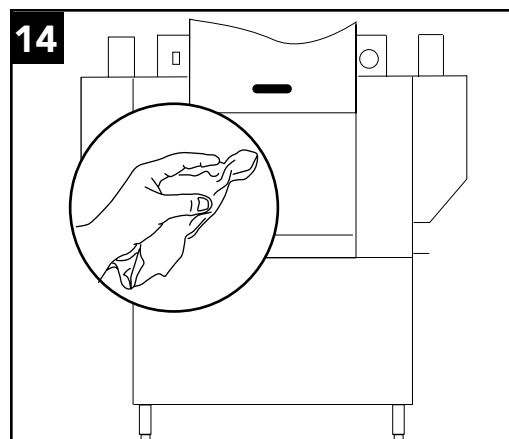
Check the final rinse nozzles and clean with a small paper clip as required.



Thoroughly spray the interior of the machine with fresh clean water.



Remove the pump intake screen and flush clean in a sink. Check and clean the dual float switch, heating element and drain screen.



Wipe the exterior of the machine with mild detergent and soft cloth. DO NOT HOSE DOWN WITH WATER.

**NOTE:**

Leave the doors open to aid in drying overnight.



Lime (scale) deposits are the result of minerals contained in the water feeding the dishwasher and appear as a white haze on the surface of the dishwasher. Severe scaling can appear as a granular deposit. These deposits are the result of the mineral content in the geographic area in which the machine is located.

**WARNING:**

**Death or injury can result from toxic fume when de-liming agents come in contact with Chlorine Bleach, or other chemicals that contain iodine, bromine, or fluorine.**

**USE EXTREME CAUTION WHEN HANDLING ANY DE-LIMING AGENT.**

**CAUTION:**

**De-liming agents can cause chemical burns.**

**Wear rubber gloves, eye protection and any other protective clothing as instructed by a qualified chemical supplier and follow the instructions provided by the chemical supplier.**

## Maintenance

### Weekly

1. Inspect all water lines for leaks and tighten at joints if required.
2. Clean any detergent residue from the exterior of the machine.
3. Check that the drain/overflow pipes seat tightly in their drains.
4. Clean any accumulated scale from the heating element.
5. Inspect the spray arms for any damage or missing parts.
6. Inspect the final rinse arms for missing parts.
7. Inspect the pawl bar and drive assembly for damage or missing parts.
8. Check that float switches move freely.
9. Check the idle pump actuator and the final rinse actuator for freedom of travel.

### Monthly

1. Inspect interior of machine for lime deposits and clean.
2. Check o-rings on spray arm piping connections.
3. Check that the drain/overflow o-rings
4. Clean any accumulated scale from the heating element.
5. Inspect the spray arms o-rings and ensure all plugs are installed.
6. Inspect the final rinse arms for missing parts and are clean.
7. Inspect the pawl bar and drive assembly for damage or missing parts.
8. Check that float switches move freely.
9. Check the idle pump actuator and the final rinse actuator for freedom of travel.
10. Check the drive clutch and adjust as necessary.
11. Check the pump rotation and direction of rotation.
12. Check the operation of temperature gauges or displays.
13. Perform complete operation check.

### Yearly

1. Contact authorized service agent to perform complete maintenance review of machine.
2. Correct any abnormal situations as recommended.

Before calling for service check the following conditions.

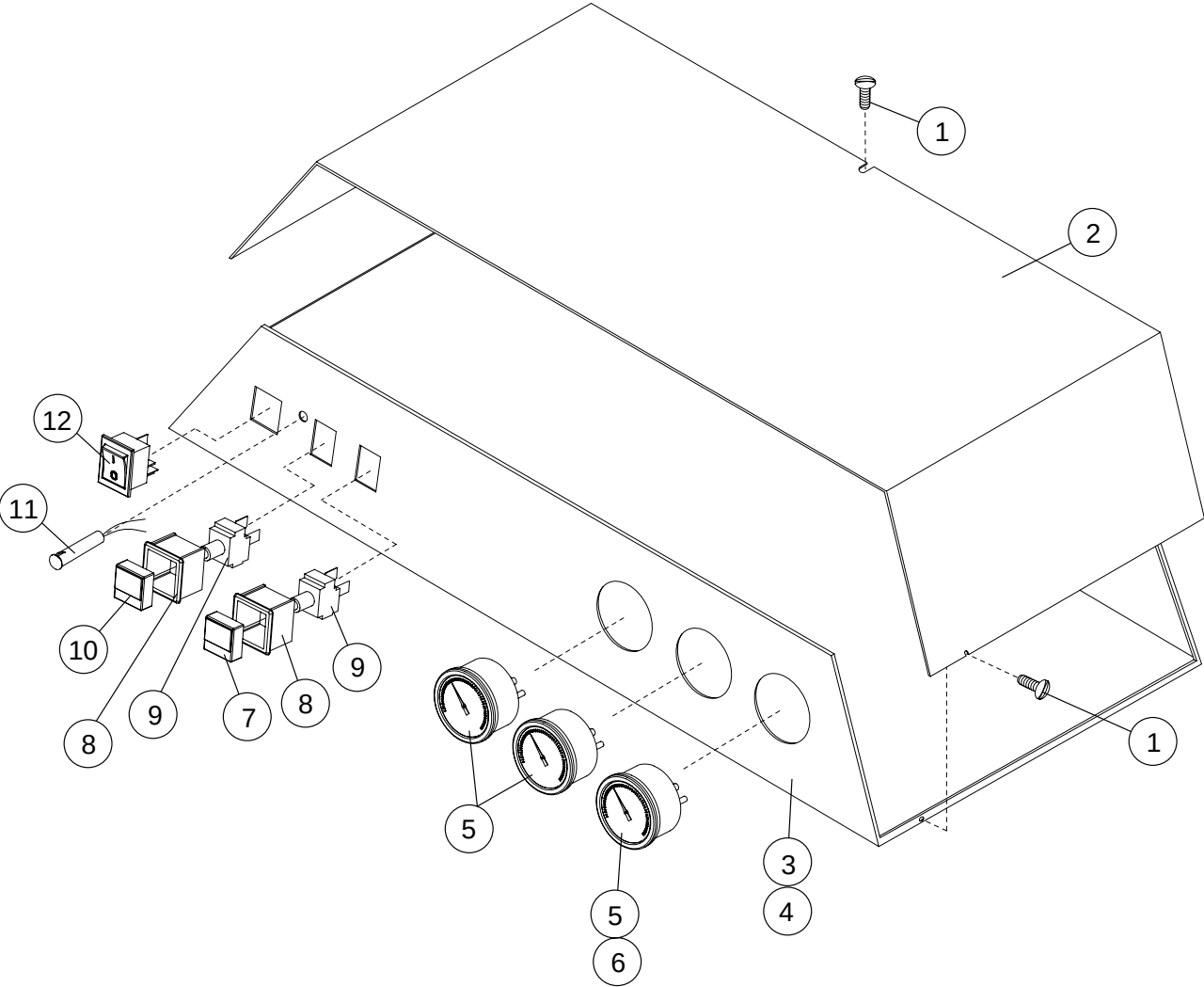
1. Dishwasher main power and water supply is on.
2. Machine has been assembled correctly.
3. Conveyor is clear of any obstructions.
4. Drains are closed.
5. Screens and pump intake screens are clear.
6. Doors are closed and secure.

| Condition                            | Cause                                                                                                                                                                                                             | Solution                                                                                                                                                                                                                     |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dishwasher will not run.             | Door not closed.<br>Main power OFF.<br>Dishwasher OFF.<br>Dish rack inserted wrong                                                                                                                                | Close door completely.<br>Check breaker on panel.<br>Turn dishwasher ON.<br>Check dish rack loading.                                                                                                                         |
| Low or no water.                     | Main water supply off.<br>PRV setting incorrect<br>Solenoid strainer clogged.<br>Solenoid valve defective.                                                                                                        | Open supply valve.<br>Adjust the PRV setting<br>Clean strainer.<br>Contact Service Agent.                                                                                                                                    |
| Poor wash results.                   | Detergent not added to tank<br><br>Wares incorrectly loaded.<br>in dishrack.<br><br>Clogged screens .<br>Clogged spray arms.<br><br>Thermostat defective.<br><br>End plugs missing.<br><br>Water temperature low. | Check detergent supply.<br><br>Reposition wares or<br>reduce amount of wares.<br><br>Clean screens..<br>Clean spray arms.<br><br>Contact Service Agent.<br><br>Check spray arms.<br><br>Check incoming water<br>temperature. |
| Dishwasher stays in<br>wash cycle.   | 90-second timer defective                                                                                                                                                                                         | Contact Service Agent                                                                                                                                                                                                        |
| Dishwasher conveyor<br>will not run. | Conveyor jammed or table<br>limit switch has stopped<br>conveyor.                                                                                                                                                 | Check conveyor for jams<br>Remove racks from load<br>end of table.                                                                                                                                                           |

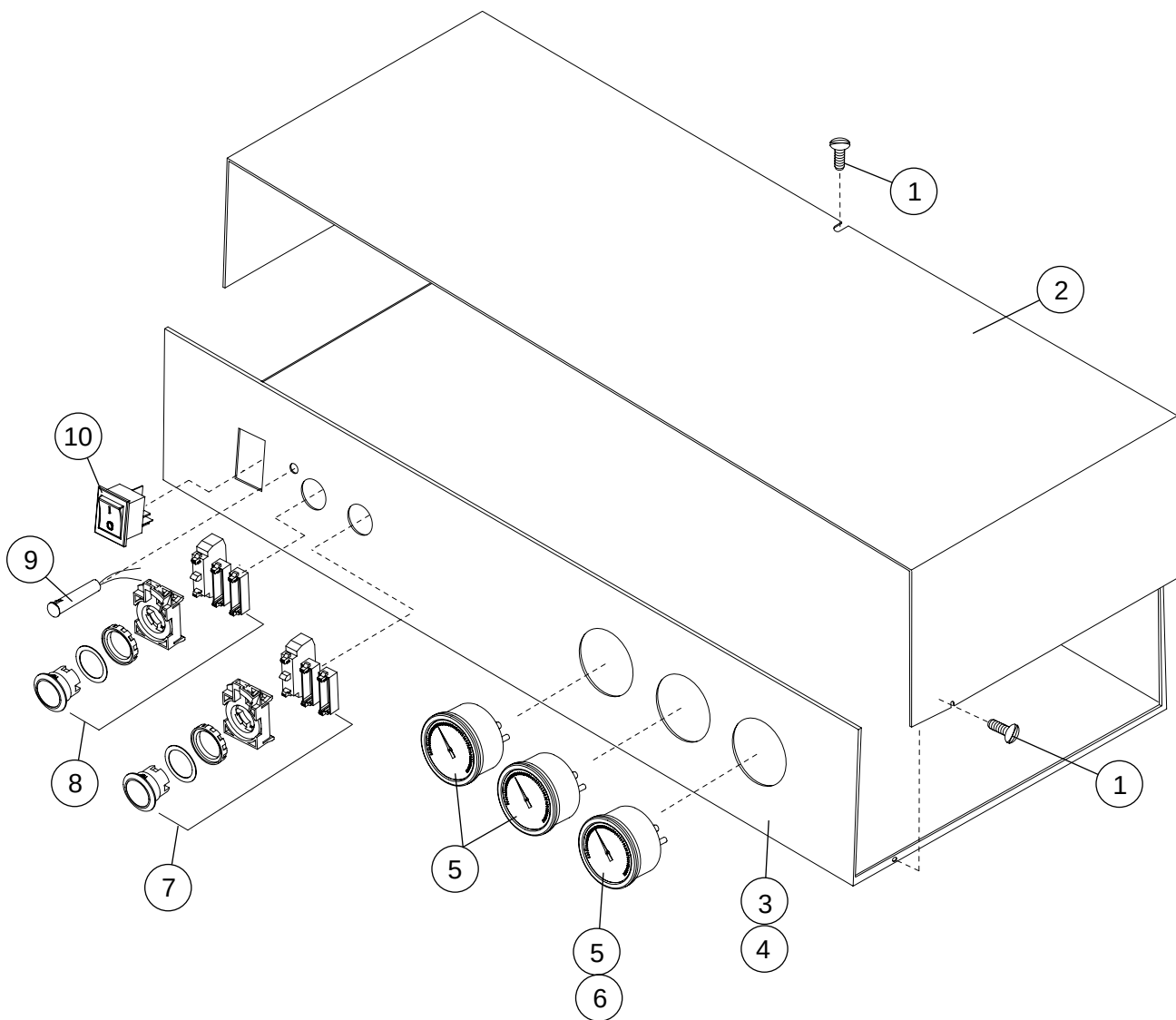
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# Service Replacement Parts

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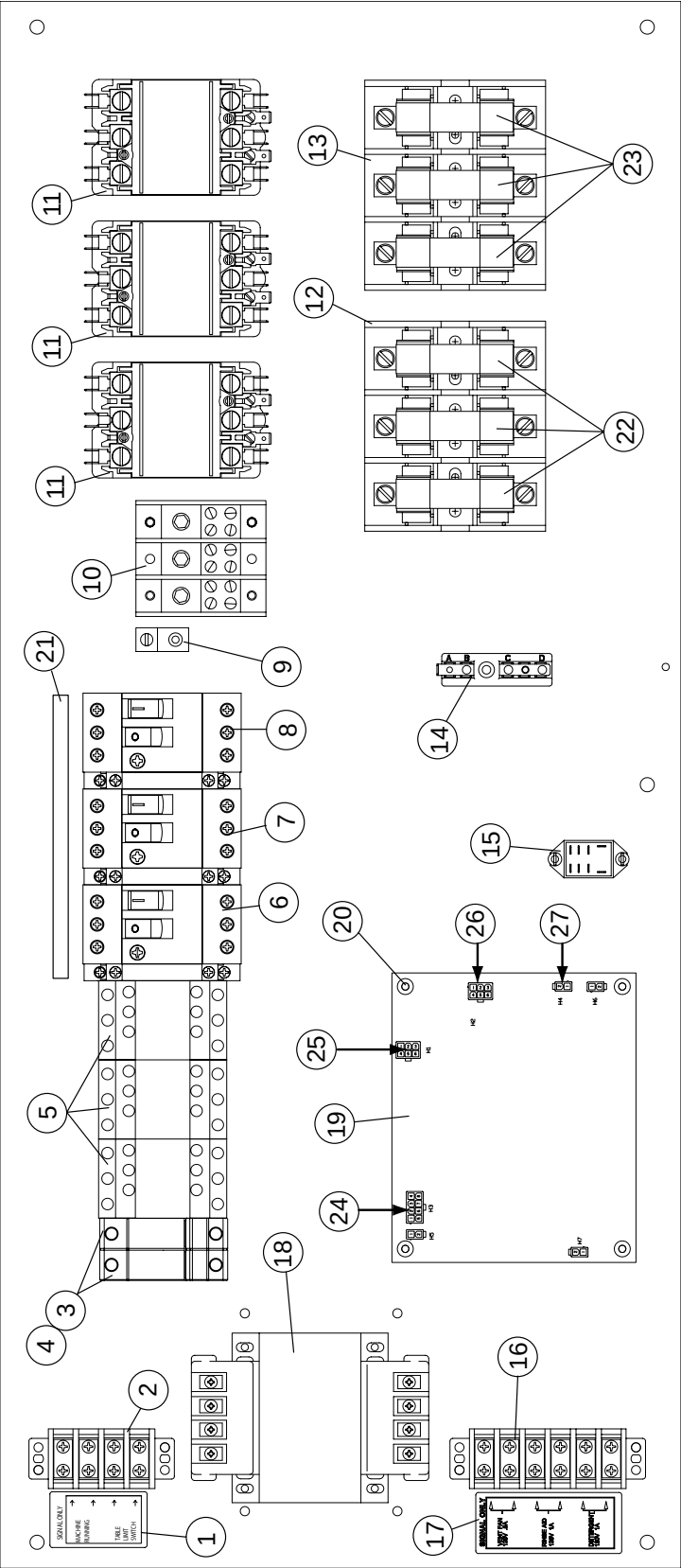


| Item No. | Part No. | Description                                              | Qty. |
|----------|----------|----------------------------------------------------------|------|
| 1        | 100097   | SCREW, TRUSS HD., 10-32 X 1/2" SST                       | 3    |
| 2        | 333533   | COVER, CONTROL CABINET                                   | 1    |
| 3        | 114768   | DECAL 36" CONTROL CABINET 3-HOLE, PW-W-R (MD66)          | 1    |
| 4        | 114767   | DECAL 36" CONTROL CABINET 2-HOLE, W-R (MD44) (not shown) | 1    |
| 5        | 107440   | THERMOMETER, 8FT. FLANGE (Qty. 1 MD44) (Qty. 3 MD66)     | A/R  |
| 6        | 113622   | THERMOMETER, 4FT. GAS FILLED (Qty. 1 MD44 Only)          | 1    |
| 7        | 114758   | PUSHBUTTON, RED SQUARE                                   | 1    |
| 8        | 0512217  | HOUSING, SWITCH                                          | 2    |
| 9        | 0512216  | CONTACT BLOCK, N.O.                                      | 2    |
| 10       | 0512218  | PUSHBUTTON, GREEN SQUARE                                 | 1    |
| 11       | 114506   | LIGHT, GREEN LED 24V                                     | 1    |
| 12       | 0512922  | SWITCH, ON/OFF                                           | 1    |

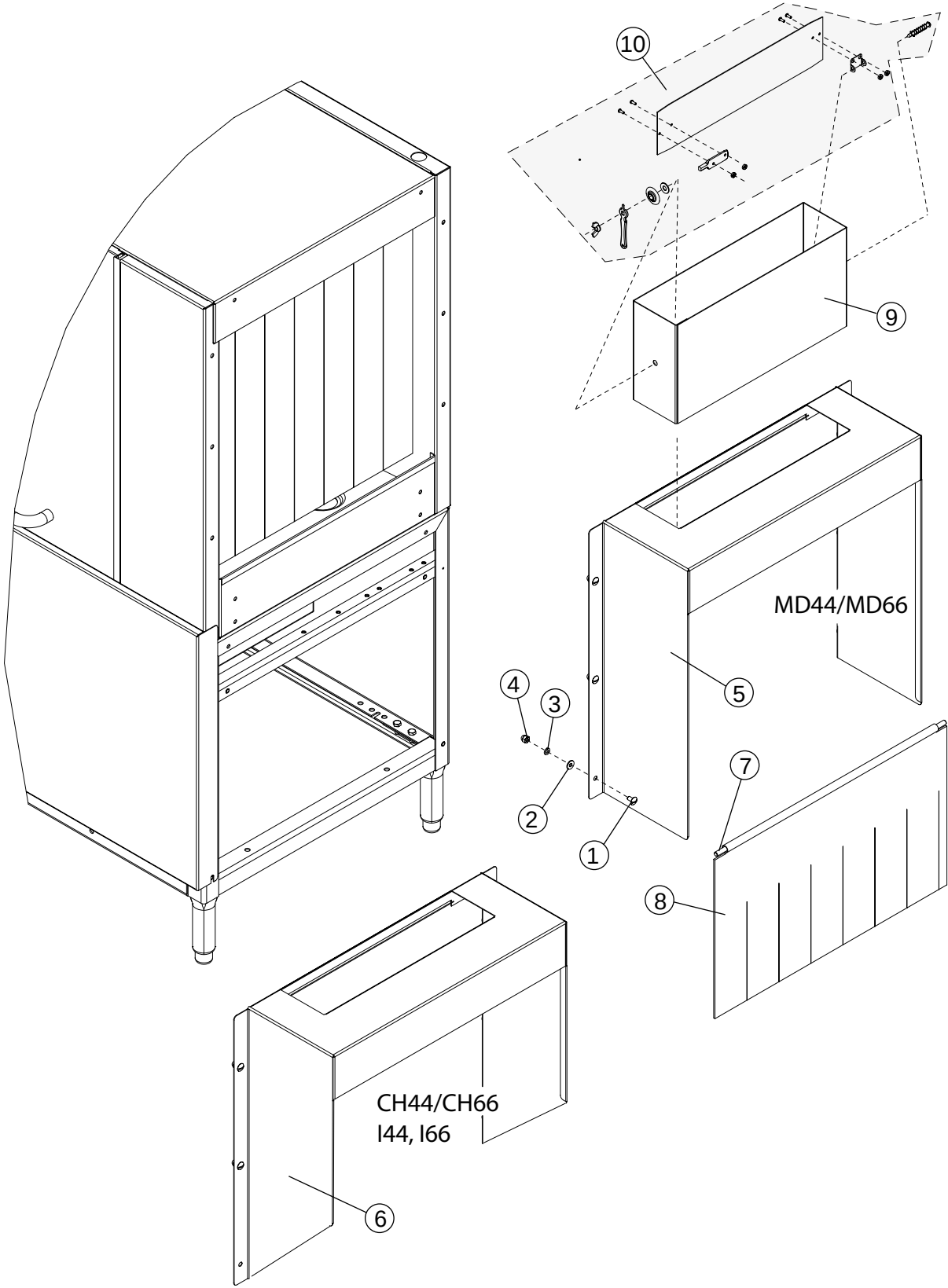




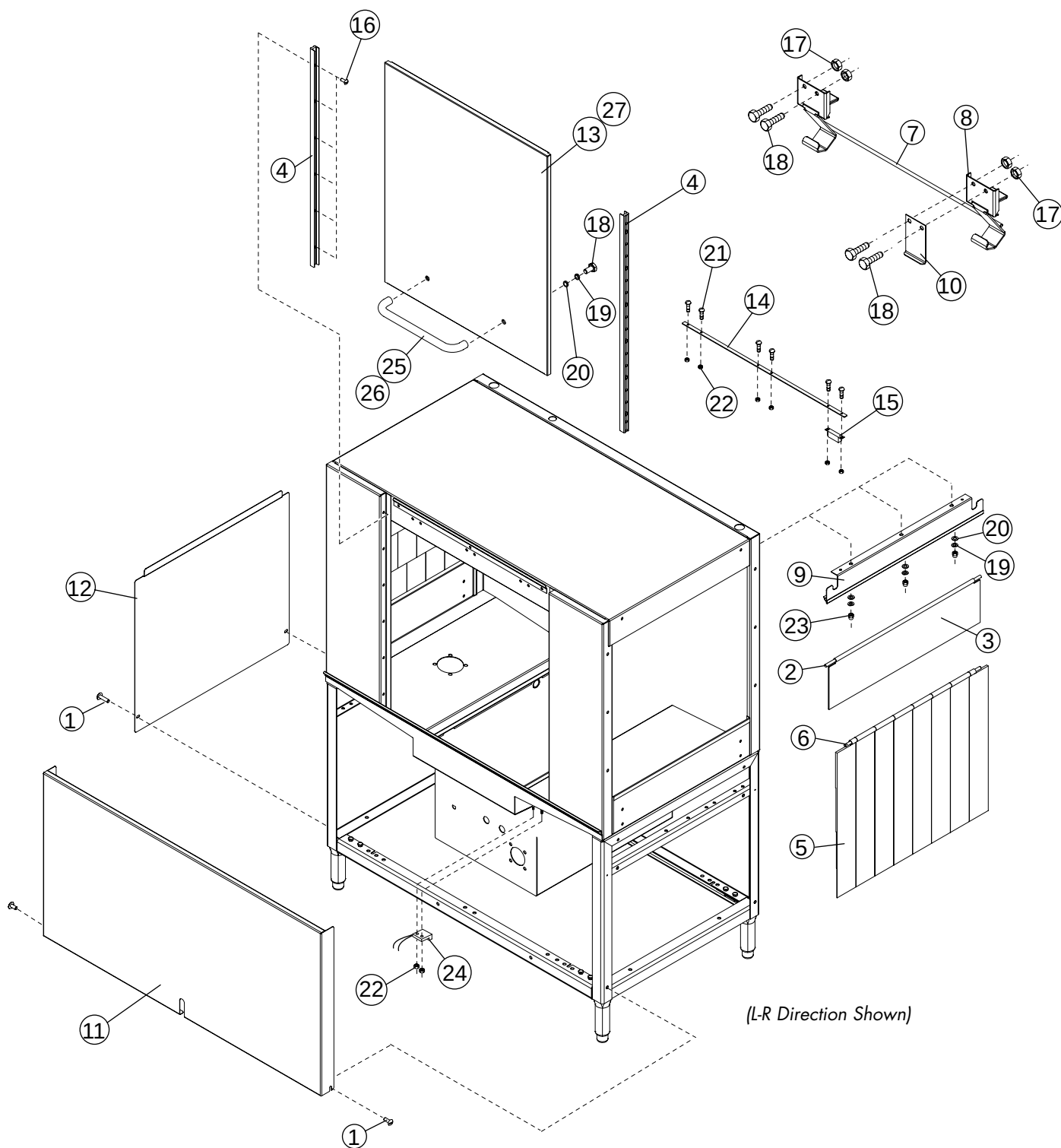
| Item No. | Part No. | Description                                                    | Qty. |
|----------|----------|----------------------------------------------------------------|------|
| 1        | 100097   | SCREW, TRUSS HD., 10-32 X 1/2" SST                             | 3    |
| 2        | 334880   | COVER, CONTROL CABINET                                         | 1    |
| 3        | 115239   | DECAL 36" CONTROL CABINET 3-HOLE, PW-W-R, CH66, I66            | 1    |
| 4        | 115238   | DECAL 36" CONTROL CABINET 2-HOLE, W-R, CH44 (not shown)        | 1    |
| 5        | 107440   | THERMOMETER, 8FT. FLANGE (Qty. 1 CH44, I44) (Qty. 3 CH66, I66) | A/R  |
| 6        | 113622   | THERMOMETER, 4FT. GAS FILLED (Qty. 1 CH44, I44 Only)           | 1    |
| 7        | 115018   | PUSHBUTTON, RED                                                | 1    |
| 8        | 115019   | PUSHBUTTON, GREEN                                              | 1    |
| 9        | 114506   | LIGHT, GREEN LED 24V                                           | 1    |
| 10       | 114505   | BREAKER, 14A, ON/OFF 24V                                       | 1    |



| Item No. | Part No. | Description                                               | Qty. |
|----------|----------|-----------------------------------------------------------|------|
| 1        | 113936   | LABEL, TABLE LIMIT SWITCH/ MACHINE RUN                    | 1    |
| 2        | 107171   | TERMINAL BLOCK, 4-POLE                                    | 1    |
| 3        | 111153   | FUSE BLOCK, 600V 30A                                      | 2    |
| 4        | 112482   | FUSE, 3.5A, LPCC 200-220-230-240/60/3                     | 2    |
| —        | 114797   | FUSE, 2.0A, LPCC 460-480/60/3                             | 2    |
| 5        | 108122   | CONTACTOR, MTR. 12A (Qty. 2 for MD44, Qty. 3 for MD66)    | A/R  |
| 6        | 111624   | OVERLOAD, MTR. 200-220-230-240/60/3 (0.63-1.0A) Drive     | 1    |
| —        | 111623   | OVERLOAD, MTR. 460-480/60/3 (0.4-.63A) Drive              | 1    |
| 7        | 111629   | OVERLOAD, MTR. 200-220-230-240/60/3 (6-10A) Prewash MD66  | 1    |
| —        | 111627   | OVERLOAD, MTR. 460-480/60/3 (2.5-4.0A) Prewash MD66       | 1    |
| 8        | 111629   | OVERLOAD, MTR. 200-220-230-240/60/3 (6-10A) Wash          | 1    |
| —        | 111627   | OVERLOAD, MTR. 460-480/60/3 (2.5-4.0A) Wash               | 1    |
| 9        | 103310   | LUG, GROUND                                               | 1    |
| 10       | 111833   | BLOCK, TERMINAL 175A, 3-POLE                              | 1    |
| 11       | 111827   | CONTACTOR, HEAT 60A (Qty. 2 for 40° Rise, Qty. 3 for 70°) | A/R  |
| 12       | 180171   | FUSE BLOCK, 600V/60A 200-220-230-240-460-480V WASH        | 1    |
| 13       | 180171   | FUSE BLOCK 600V/60A 3P J-TYPE FOR 40° RISE BOOSTER        | 1    |
| —        | 108424   | FUSE BLOCK 600V/100A 3P T-TYPE FOR 70° RISE BOOSTER       | 1    |
| 14       | 0503749  | TERMINAL BLOCK, 11-POLE                                   | 1    |
| 15       | 111068   | RELAY, 2 POLE 120V                                        | 1    |
| 16       | 100294   | TERMINAL BLOCK, 6-POLE                                    | 1    |
| 17       | 114760   | LABEL, DETERGENT/RINSE AID/VENT FAN                       | 1    |
| 18       | 109064   | TRANSFORMER, 250VA 200-220-230-240-460-480:120V           | 1    |
| 19       | 114756   | RACK CONTROL MODULE                                       | 1    |
| 20       | 114759   | SUPPPORT, PCB                                             | 4    |
| 21       | 111633   | BUS SYSTEM (2-UNIT) FOR CH44,MD44                         | 1    |
| —        | 111671   | BUS SYSTEM (3-UNIT) FOR CH66, MD66                        | 1    |
| 22       | 180175   | FUSE, 600V/50A J-TYPE 200-220V/50-60/3                    | 3    |
| —        | 180174   | FUSE, 600V/45A J-TYPE 230-240/50-60/3                     | 3    |
| —        | 180243   | FUSE, 600V/25A J-TYPE 460-480/60/3                        | 3    |
| 23       | 180176   | FUSE, 600V/60A J-TYPE 200-220V/50-60/3 40° Rise Booster   | 3    |
| —        | 180175   | FUSE, 600V/50A J-TYPE 230-240V/50-60/3 40° Rise Booster   | 3    |
| —        | 180243   | FUSE, 600V/25A J-TYPE 460-480V/60/3 40° Rise Booster      | 3    |
| —        | 108448   | FUSE, 600V/90A T-TYPE 200-220V/50-60/3 70° Rise Booster   | 3    |
| —        | 180059   | FUSE, 600V/80A T-TYPE 230-240V/50-60/3 70° Rise Booster   | 3    |
| —        | 180173   | FUSE, 600V/40A J-TYPE 460-480V/60/3 70° Rise Booster      | 3    |
| 24       | 114771   | WIRE HARNESS, MOLEX 8-PIN INPUT                           | 1    |
| 25       | 114769   | WIRE HARNESS, MOLEX 6-PIN INPUT                           | 1    |
| 26       | 114770   | WIRE HARNESS, MOLEX 6-PIN OUTPUT                          | 1    |
| 27       | 114772   | WIRE HARNESS, MOLEX 6-PIN OUTPUT HEAT                     | 1    |

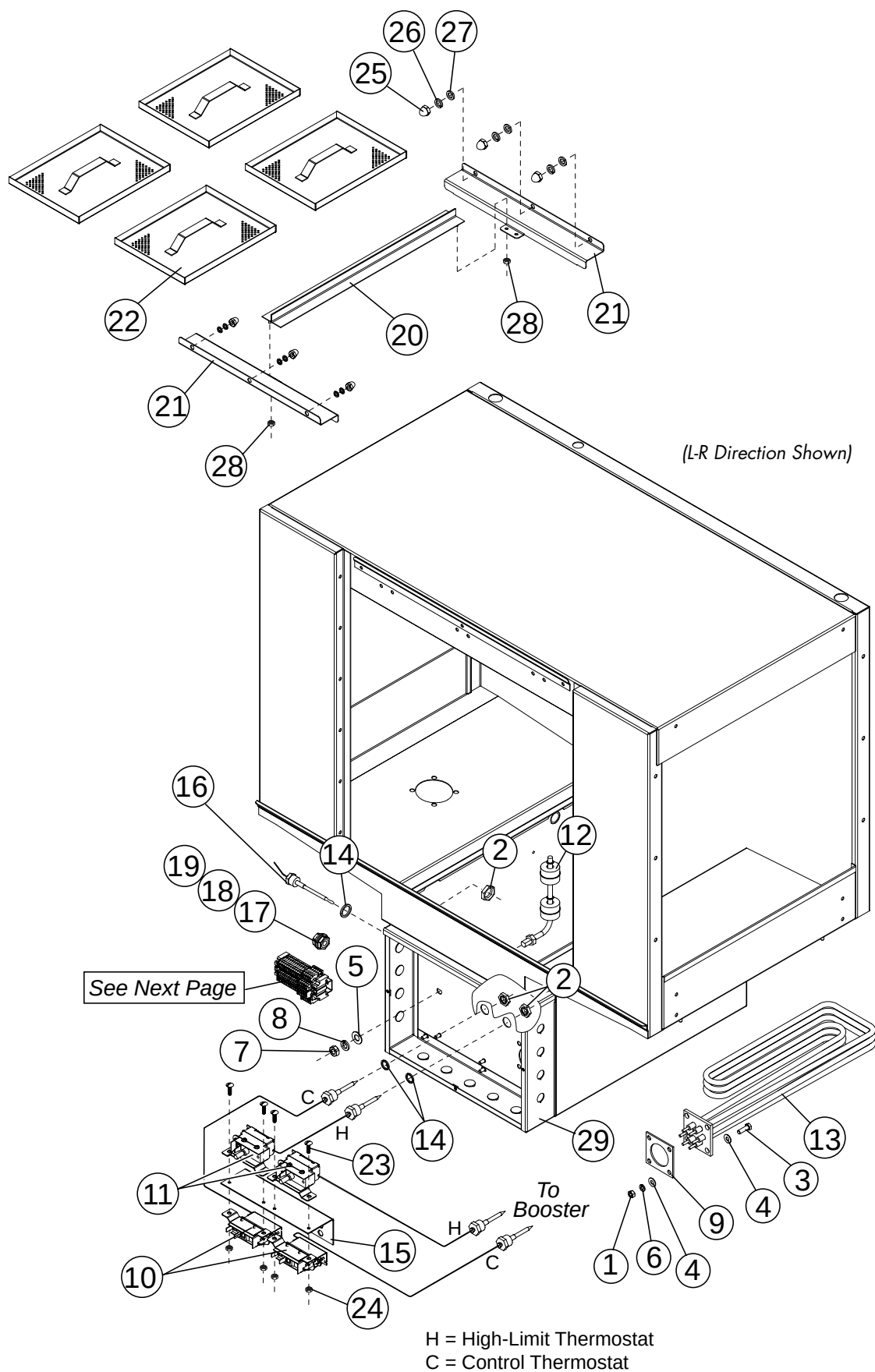


| Item No. | Part No. | Description                                            | Qty. |
|----------|----------|--------------------------------------------------------|------|
| 1        | 100073   | SCREW, 1/4-20 X 1/2" TRUSS HD., SST                    | 6    |
| 2        | 106026   | WASHER, 1/4" FLAT                                      | 6    |
| 3        | 106482   | WASHER, LOCK 1/4" SPLIT                                | 6    |
| 4        | 106014   | NUT, ACORN 1/4-20                                      | 6    |
| 5        | 333092   | SPLASH SHIELD, VENT 8" EXTENDED (MD44, MD66)           | 1    |
| 6        | 334917   | SPLASH SHIELD, VENT 8" EXTENDED (CH44, CH66, I44, I66) | 1    |
| 7        | 113828   | ROD, CURTAIN 5/16" X 21-1/2" SST                       | 1    |
| 8        | 114622   | CURTAIN, 21-3/4" X 13-1/4"                             | 1    |
| 9        | 401487   | VENT STACK 4" X 16"                                    | 1    |
| 10       | 201589   | DAMPER REGULATOR ASSY.                                 | 1    |



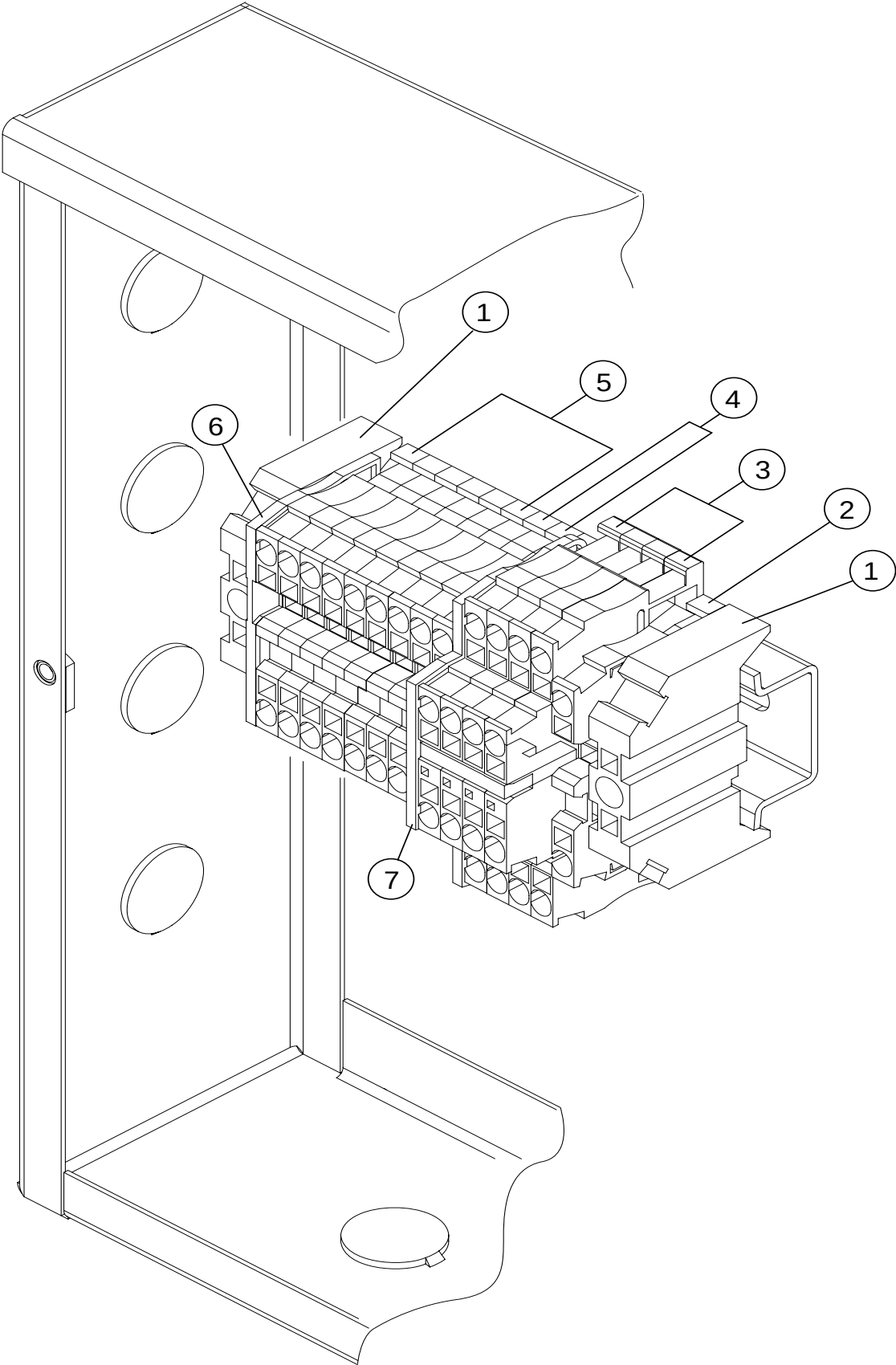
| Item No. | Part No. | Description                            | Qty. |
|----------|----------|----------------------------------------|------|
| 1        | 100212   | SCREW, 10-32 X 3/4" TRUSS HD.          | 4    |
| 2        | 108250   | ROD, CURTAIN                           | 1    |
| 3        | 109723   | CURTAIN, 24" X 6-1/4" (RINSE CURTAIN)  | 1    |
| 4        | 113691   | GUIDE, U-CHANNEL, DOOR                 | 2    |
| 5        | 113720   | CURTAIN, 24" X 20-1/4"                 | 2    |
| 6        | 113828   | ROD, CURTAIN, 5/16" DIA. X 21-1/2" LG. | 2    |
| 7        | 308133   | BRACKET, DOOR HOOK                     | 1    |
| 8        | 317345   | BRACKET, DOOR CATCH                    | 2    |
| 9        | 328023   | BAFFLE, CURTAIN WELDMENT               | 1    |
| 10       | 331981   | BRACKET, DOOR STOP                     | 1    |
| 11       | 333163   | PANEL, FRONT 44" MT-RACK               | 1    |
| 12       | 333539   | PANEL, END, FULL MT-RACK               | 1    |
| 13       | 333267   | DOOR, 26" (MD44)                       | 1    |
| 14       | 329985   | BRACKET, WATER DIVERTER                | 1    |
| 15       | 113937   | MAGNET, DOOR                           | 1    |
| 16       | 113486   | SCREW, 8-32 X 5/8" FLAT HD., SST       | 12   |
| 17       | 100141   | NUT, GRIP 1/4-20 SST                   | 4    |
| 18       | 100734   | BOLT, 1/4-20 X 1/2" SST                | 6    |
| 19       | 106482   | WASHER, LOCK 1/4" SPLIT                | 5    |
| 20       | 106026   | WASHER, 1/4" FLAT                      | 5    |
| 21       | 106382   | SCREW, 6-32 X 3/8" TRUSS HD., SST      | 6    |
| 22       | 108954   | GRIP NUT, 6-32 W/NYLON INSERT, SST     | 8    |
| 23       | 106014   | NUT, ACORN 1/4-20                      | 3    |
| 24       | 113719   | SWITCH, REED ALEPH                     | 1    |
| 25       | 114798   | HANDLE, DOOR (MD-44)                   | 1    |
| 26       | 108966   | HANDLE, DOOR (CH-44, I-44)             | 1    |
| 27       | 327857   | DOOR, 26" (CH-44, I-44)                | 1    |

# Wash Tank Heat, Float Switch, and Scrap Screens - All Models

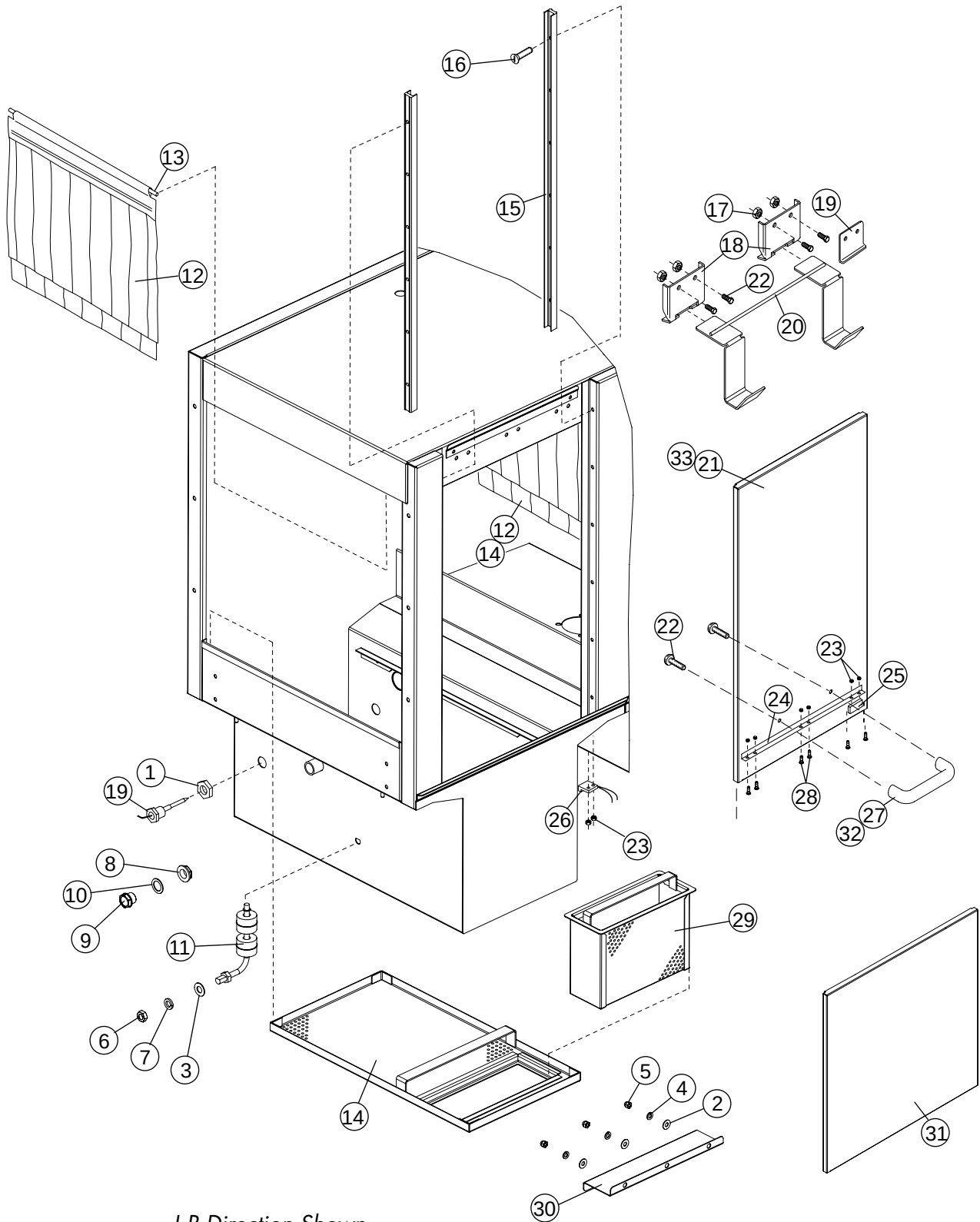




| Item No. | Part No. | Description                                                  | Qty. |
|----------|----------|--------------------------------------------------------------|------|
| 1        | 100154   | HEX PLAIN NUT, 5/16-18 SST                                   | 4    |
| 2        | 100547   | LOCKNUT, 1/2" NPT SST                                        | 3    |
| 3        | 100740   | BOLT, HEX HD., 5/16-18 X 1" SST                              | 4    |
| 4        | 102376   | WASHER, FLAT 5/16", SST                                      | 8    |
| 5        | 104882   | WASHER, FLAT SST                                             | 1    |
| 6        | 106013   | WASHER, LOCK 5/16" SPLIT                                     | 4    |
| 7        | 107089   | HEX PLAIN JAM NUT, 1/2-13 SST                                | 1    |
| 8        | 107589   | WASHER, LOCK 1/2" SPLIT SST                                  | 1    |
| 9        | 108345   | GASKET, HEATER                                               | 1    |
| 10       | 109069   | THERMOSTAT, CONTROL WITH CAPILLARY                           | 2    |
| 11       | 110561   | THERMOSTAT, HIGH LIMIT WITH CAPILLARY                        | 2    |
| 12       | 113291   | FLOAT SWITCH                                                 | 1    |
| —        | 111151   | C-CLIP                                                       | 4    |
| 13       | 113516   | HEATER, 15/18.1kW 200-220V/50-60/1 & 3PH<br>380V/50-60/3     | 1    |
| —        | 113517   | HEATER, 15/16.3kW 230-240V/50-60/1 & 3PH<br>400-415V/50-60/3 | 1    |
| —        | 113518   | HEATER, 15/16.3kW 460-480V/60/3                              | 1    |
| 14       | 201041   | WASHER, SPACER                                               | 3    |
| 15       | 323430   | BRACKET, THERMOSTAT                                          | 1    |
| 16       | 107440   | THERMOMETER, 8 FT. FLANGED                                   | 1    |
| 17       | 108418   | PLUG, 1/2" NPT PLASTIC                                       | 1    |
| 18       | 108417   | NUT, 1/2" NPT PLASTIC                                        | 1    |
| 19       | 109034   | GASKET, 1/2" PLUG                                            | 1    |
| 20       | 305163   | SCREEN SUPPORT WELDMENT                                      | 1    |
| 21       | 333078   | FILLER, SCREEN                                               | 2    |
| 22       | 305164   | SCREEN, SCRAP WELDMENT                                       | 4    |
| 23       | 100097   | SCREW, 10-32 X 1/2" TRUSS HD.                                | 2    |
| 24       | 107966   | NUT, GRIP 10-32 W/NYLON INSERT                               | 2    |
| 25       | 106014   | NUT, ACORN 1/4-20                                            | 6    |
| 26       | 106482   | WASHER, LOCK 1/4" SPLIT                                      | 6    |
| 27       | 106026   | WASHER, FLAT 1/4"                                            | 6    |
| 28       | 100141   | NUT, GRIP 1/4-20 SST                                         | 2    |
| 29       | 333088   | ENCLOSURE WELDMENT, 3.25" X 12" X 16"                        | 1    |

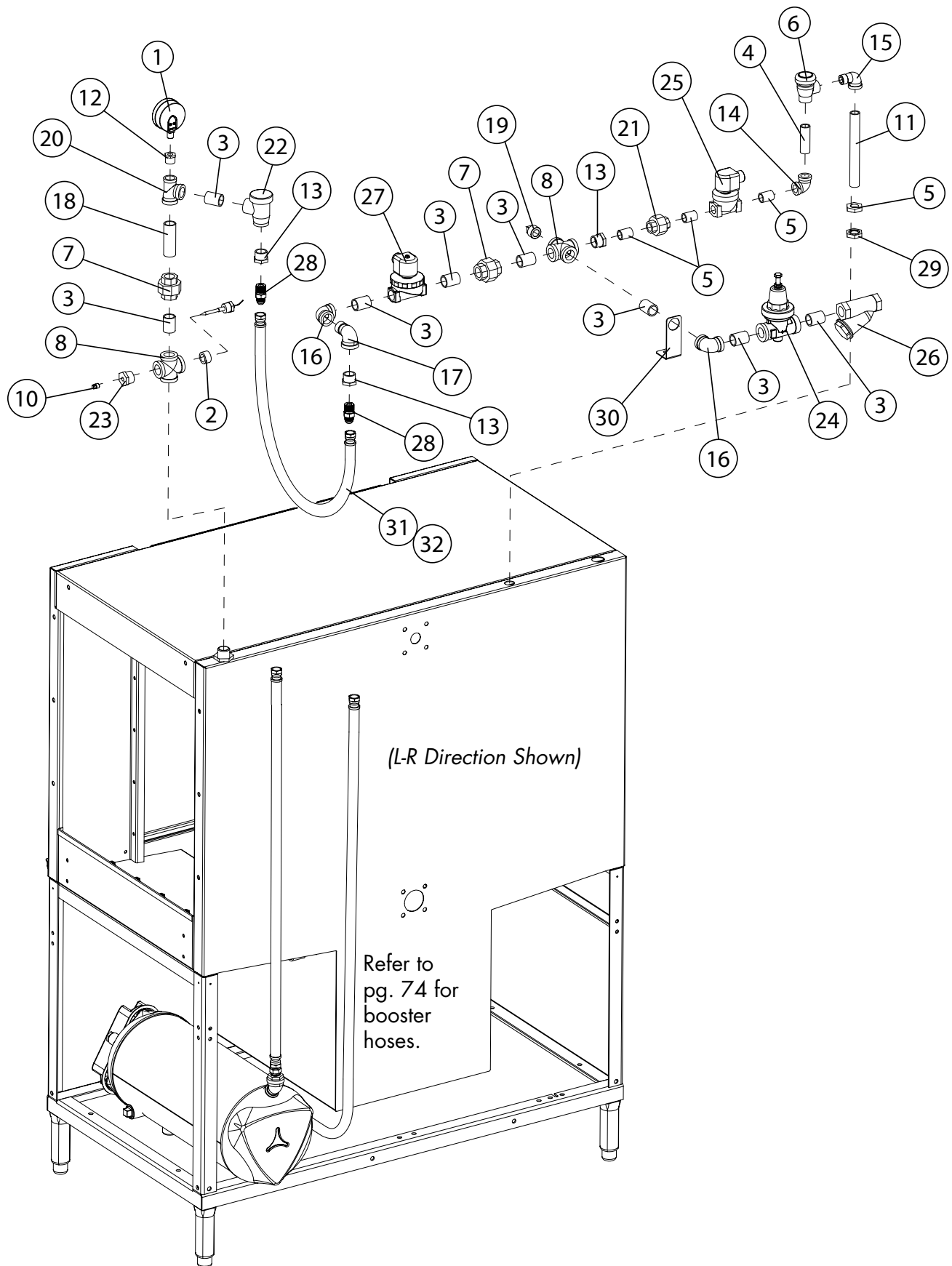


| Item No. | Part No. | Description                           | Qty. |
|----------|----------|---------------------------------------|------|
| 1        | 114519   | END BLOCK, E/NS 35N                   | 2    |
| 2        | 114514   | TERMINAL, SINGLE, ST 2.5 WH (WHITE)   | 1    |
| 3        | 114517   | TERMINAL, DOUBLE STTB 2.5             | 4    |
| 4        | 114512   | TERMINAL, SINGLE ST 2.5 (GRAY)        | 2    |
| 5        | 114513   | TERMINAL, SINGLE, ST 2.5 BU (BLUE)    | 6    |
| 6        | 114516   | END COVER, SINGLE TERMINAL D-ST 2.5   | 1    |
| 7        | 114518   | END COVER, DOUBLE TERMINAL D-STTB 2.5 | 1    |

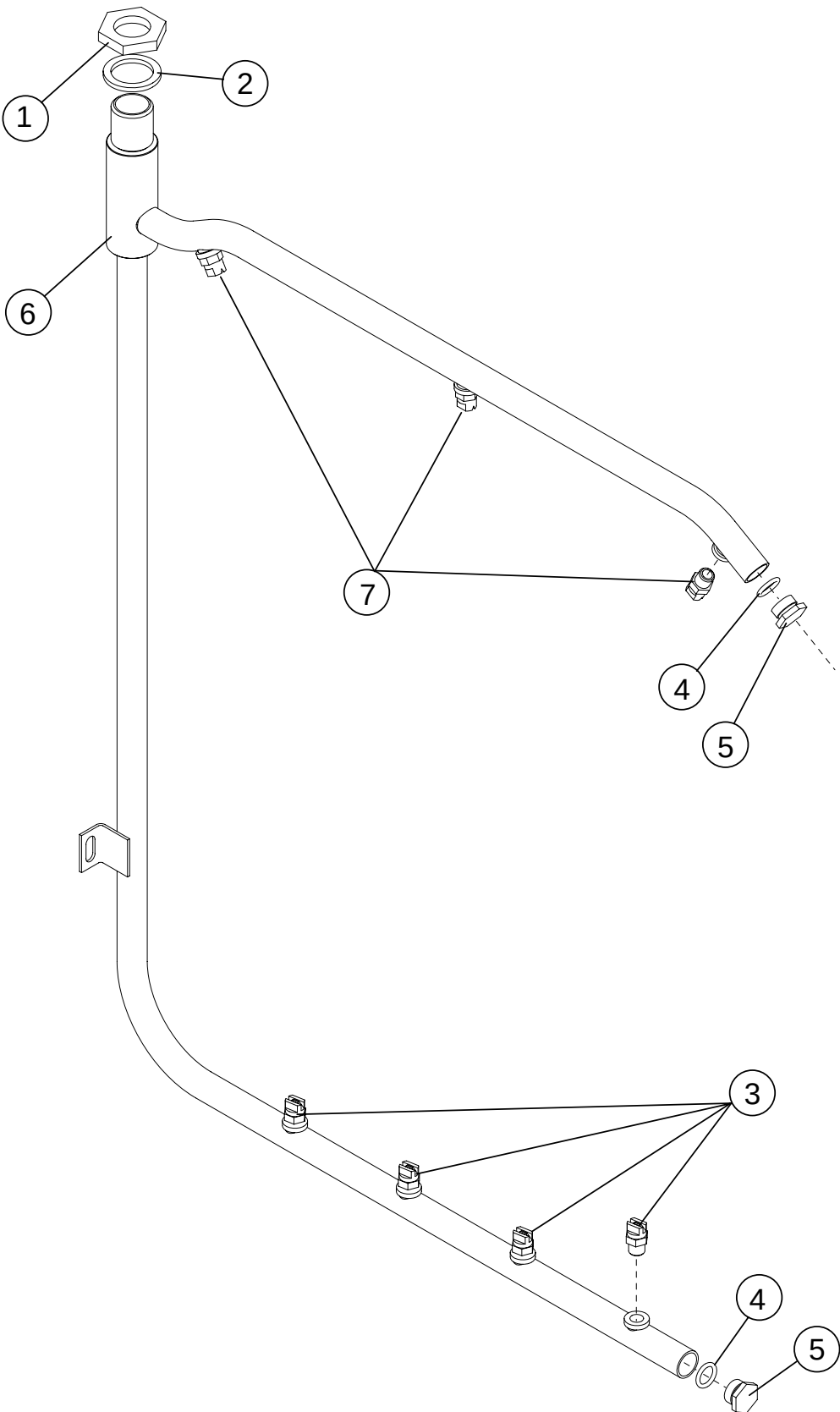


L-R Direction Shown

| Item No. | Part No. | Description                                | Qty. |
|----------|----------|--------------------------------------------|------|
| 1        | 100547   | LOCKNUT, 1/2" NPT SST                      | 1    |
| 2        | 102376   | WASHER, FLAT 5/16"                         | 3    |
| 12       | 108250   | ROD, CURTAIN, 5/16" DIA. x 24-5/8" LG.     | 1    |
| 3        | 104882   | WASHER, FLAT .531" X 1.06" X .062"         | 1    |
| 4        | 106013   | WASHER, LOCK 5/16" SPLIT SST               | 3    |
| 5        | 106014   | HEX ACORN PLAIN NUT, 1/4-20 SST            | 3    |
| 6        | 107089   | HEX PLAIN JAM NUT, 1/2-13 SST              | 1    |
| 7        | 107589   | WASHER, LOCK, 1/2" SPLIT SST               | 1    |
| 8        | 108417   | LOCKNUT, 1/2" NPT PLASTIC                  | 1    |
| 9        | 108418   | LOCKNUT, 1/4: PLASTIC                      | 1    |
| 10       | 109034   | GASKET (1/2" PLLUG)                        | 1    |
| 11       | 110854   | FLOAT SWITCH, RACK MACHINE                 | 1    |
| 12       | 113720   | CURTAIN, 24" X 20-1/4"                     | 2    |
| 13       | 113828   | ROD, CURTAIN, 5/16" DIA. X 21-1/2" LG.     | 1    |
| 14       | 333645   | SCREEN WELDMENT, PREWASH (CH66, i66, MD66) | 1    |
| 15       | 113691   | GUIDE, U-CHANNEL, DOOR                     | 2    |
| 16       | 113486   | SCREW, 8-32 X 5/8" FLAT HD., SST           | 12   |
| 17       | 100141   | NUT, GRIP 1/4-20 SST                       | 4    |
| 18       | 317345   | BRACKET, DOOR CATCH                        | 2    |
| 19       | 331981   | BRACKET, DOOR STOP                         | 1    |
| 20       | 308133   | BRACKET, DOOR HOOK                         | 1    |
| 21       | 333656   | DOOR, 15" PREWASH                          | 1    |
| 22       | 100734   | BOLT, 1/4-20 X 1/2" SST                    | 6    |
| 23       | 108954   | GRIP NUT, 6-32 W/NYLON INSERT, SST         | 8    |
| 24       | 329985   | BRACKET, WATER DIVERTER                    | 1    |
| 25       | 113937   | MAGNET, DOOR                               | 1    |
| 26       | 113719   | SWITCH, REED ALEPH                         | 1    |
| 27       | 114798   | HANDLE, DOOR (MD44)                        | 1    |
| 28       | 106382   | SCREW, 6-32 X 1/2" TRUSS HD., SST          | 6    |
| 29       | 333709   | REFUSE BASKET WELDMENT, MD66               | 1    |
| 30       | 333792   | SCREEN FILLER, FRT PW MD66                 | 1    |
| 31       | 333539   | PANEL, FULL END                            | 1    |
| 32       | 108966   | HANDLE, DOOR (CH66, i66)                   | 1    |
| 33       | 327952   | DOOR, 15" PREWASH (CH66, i66)              | 1    |

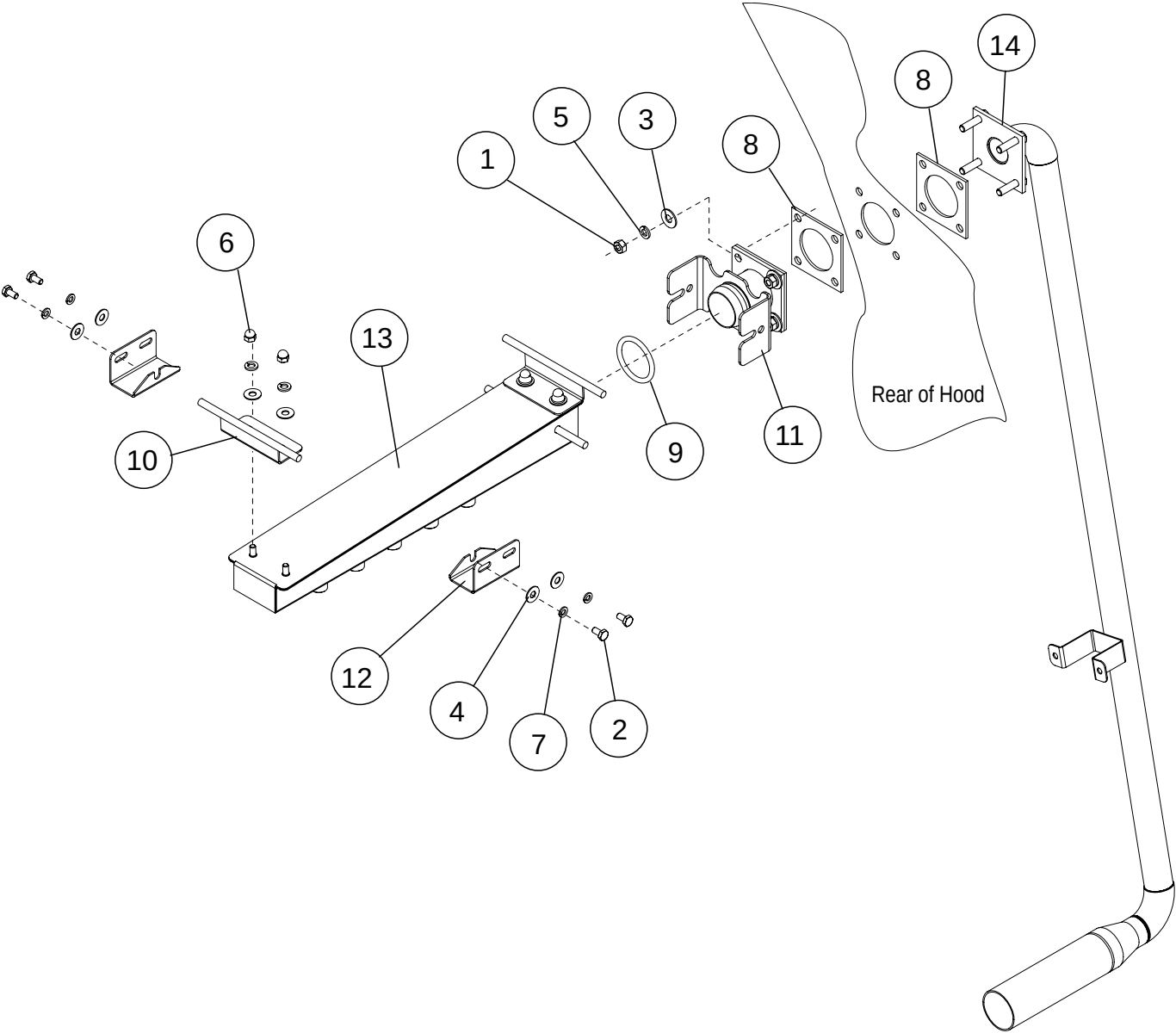


| Item No. | Part No. | Description                                                                     | Qty. |
|----------|----------|---------------------------------------------------------------------------------|------|
| 1        | 100135   | GAUGE, PRESSURE 0-60 PSI                                                        | 1    |
| 2        | 100171   | BUSHING, RED. FACE 3/4" X 1/2" BRASS                                            | 1    |
| 3        | 100184   | NIPPLE, 3/4" NPT X CLOSE BRASS                                                  | 8    |
| 4        | 100206   | NIPPLE, 1/2" NPT X 2-1/2" LG. BRASS                                             | 1    |
| 5        | 100209   | NIPPLE, 1/2" NPT X CLOSE BRASS                                                  | 3    |
| 6        | 100500   | VACUUM BREAKER, 1/2" NPT BRONZE                                                 | 1    |
| —        | 900836   | KIT, REPAIR 1/2" VACUUM BREAKER                                                 | A/R  |
| 7        | 100571   | UNION, 3/4" NPT BRASS                                                           | 2    |
| 8        | 100599   | CROSS, 3/4" NPT BRASS                                                           | 2    |
| 9        | 100709   | LOCKNUT, 1/2" NPT BRASS                                                         | 1    |
| 10       | 101259   | PLUG, 1/8" NPT SQ. HEAD BRASS                                                   | 1    |
| 11       | 101397   | NIPPLE, RTOE, 1/2" NPT X 7" LG. BRASS                                           | 1    |
| 12       | 102388   | BUSHING, REDUCING 1/2" NPT X 1/4" NPT BRASS                                     | 1    |
| 13       | 102392   | BUSHING, REDUCING, 3/4" NPT X 1/2" NPT BRASS                                    | 3    |
| 14       | 102435   | ELBOW, 1/2" NPT X 90° BRASS                                                     | 1    |
| 15       | 102438   | ELBOW, STREET, 1/2" NPT X 90° BRASS                                             | 1    |
| 16       | 102442   | ELBOW, 3/4" NPT X 90° BRASS                                                     | 2    |
| 17       | 102444   | ELBOW, STREET, 3/4" NPT X 90° BRASS                                             | 1    |
| 18       | 102470   | NIPPLE, 3/4" NPT 3" LG. BRASS                                                   | 1    |
| 19       | 102505   | PLUG, 3/4" NPT, SQ. HEAD BRASS                                                  | 1    |
| 20       | 102525   | TEE, RED, 3/4" X 1/2" X 3/4" NPT BRASS                                          | 1    |
| 21       | 102549   | UNION, 1/2" NPT BRASS                                                           | 1    |
| 22       | 104429   | VACUUM BREAKER, 3/4" BRASS                                                      | 1    |
| —        | 900837   | KIT, REPAIR VACUUM BREAKER 3/4"                                                 | A/R  |
| 23       | 105976   | BUSH, RED, 3/4" X 1/8" BRASS                                                    | 1    |
| 24       | 107550   | VALVE, PRESSURE REGULATING 3/4" NPT BRONZE                                      | 1    |
| 25       | 109886   | VALVE, SOLENOID, 1/2" NPT, HW PARKER                                            | 1    |
| —        | 109902   | KIT, REPAIR SOLENOID 1/2" HW                                                    | A/R  |
| —        | 108516   | COIL, SOLENOID, 1/2" HW, 120VAC                                                 | A/R  |
| 26       | 110768   | STRAINER, LINE 3/4" NPT BRONZE                                                  | 1    |
| 27       | 111437   | VALVE, SOLENOID 3/4" NPT HW                                                     | 1    |
| —        | 109903   | KIT, REPAIR 3/4" SOLENOID VALVE                                                 | A/R  |
| —        | 111472   | COIL, SOLENOID 3/4" 120VAC                                                      | A/R  |
| 28       | 115101   | CONNECTOR, MALE, 1/2" NPT X 1/2" 37° FLARE BRASS                                | 2    |
| 29       | 201029   | LOCKNUT, 1/2" NPT NP                                                            | 1    |
| 30       | 333654   | BRACKET, FILL PIPING SUPPORT                                                    | 1    |
| 31       | 419009   | FILL HOSE ASSY 1/2" ID X 22" LG FLARE JIC<br>(For L-R machines without booster) | 1    |
| 32       | 419005   | FILL HOSE ASSY 1/2" ID X 60" LG FLARE JIC<br>(For R-L machine without booster)  | 1    |

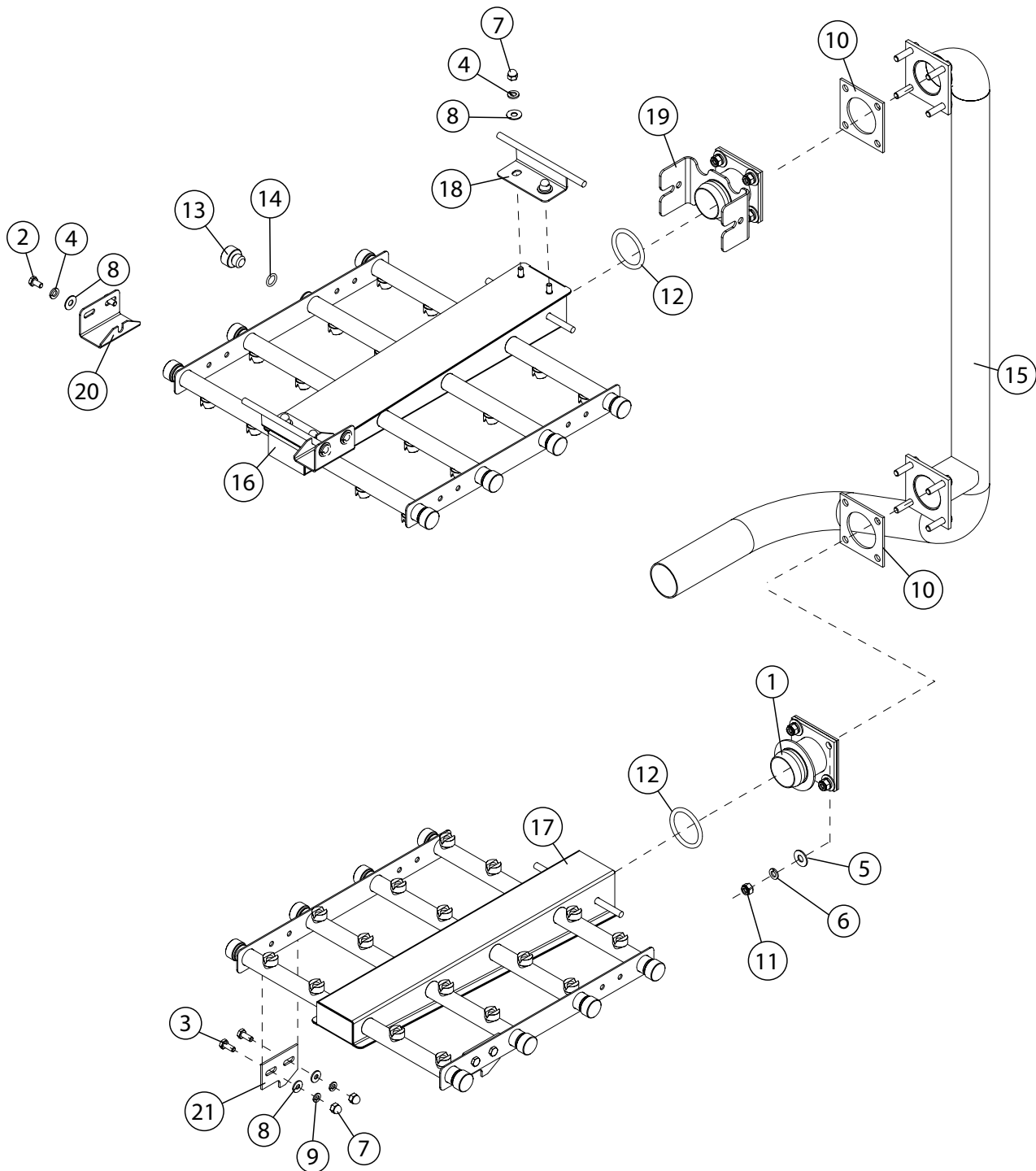




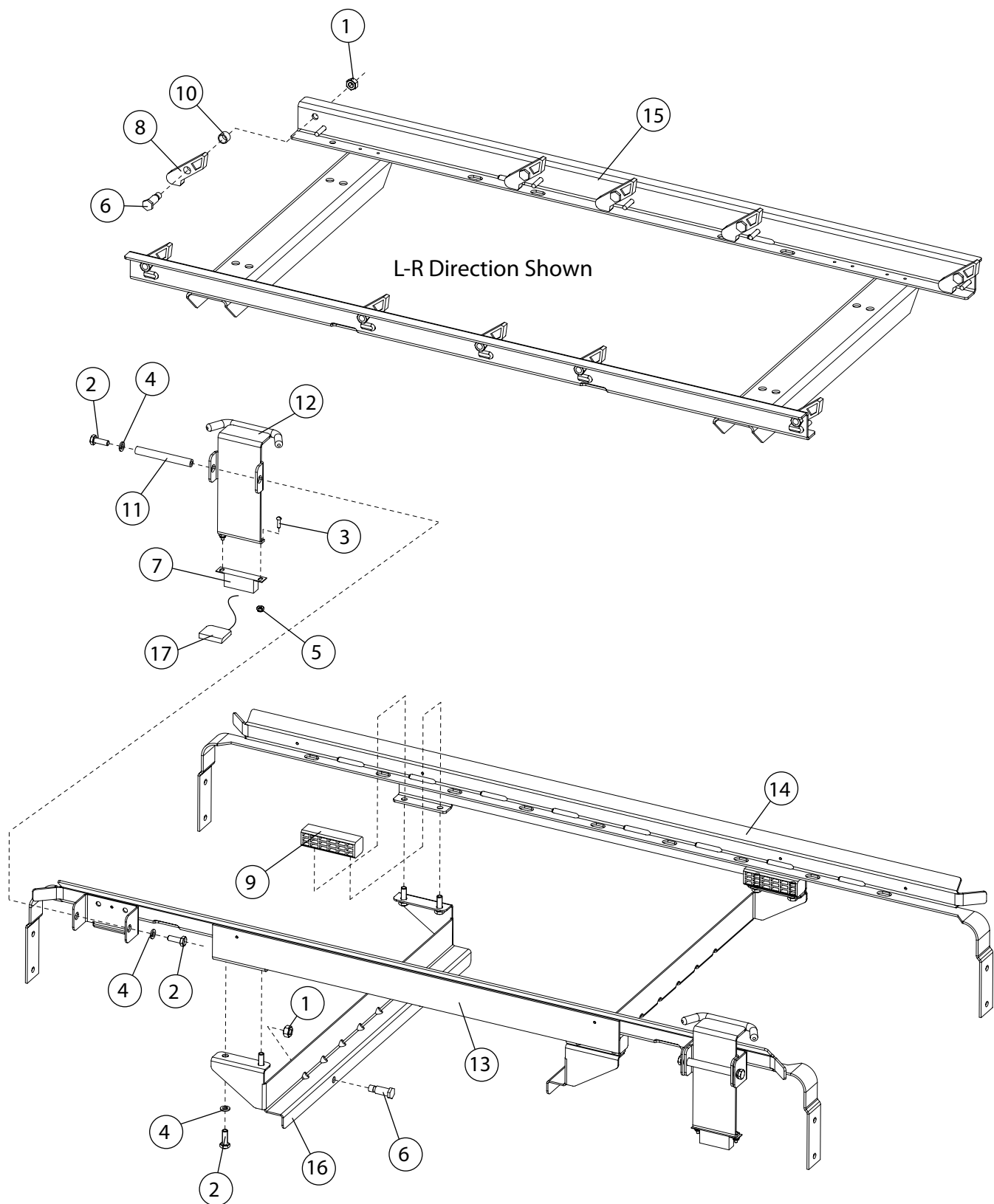
| Item No. | Part No. | Description                            | Qty. |
|----------|----------|----------------------------------------|------|
| 1        | 100156   | LOCKNUT, 3/4" BRASS                    | 1    |
| 2        | 108620   | GASKET, RINSE MANIFOLD, 3/4" PIPING    | 1    |
| 3        | 111203   | NOZZLE, VEE-JET #8003                  | 4    |
| 4        | 113716   | WASH ARM ORING                         | 2    |
| 5        | 113795   | PLUG, E FINAL RINSE MANIFOLD           | 2    |
| 6        | 114842   | MANIFOLD, WELDMENT FINAL RINSE MD-RACK | 1    |
| 7        | 0507451  | NOZZLE, RINSE SS H-1/8-W-SS 8006       | 3    |



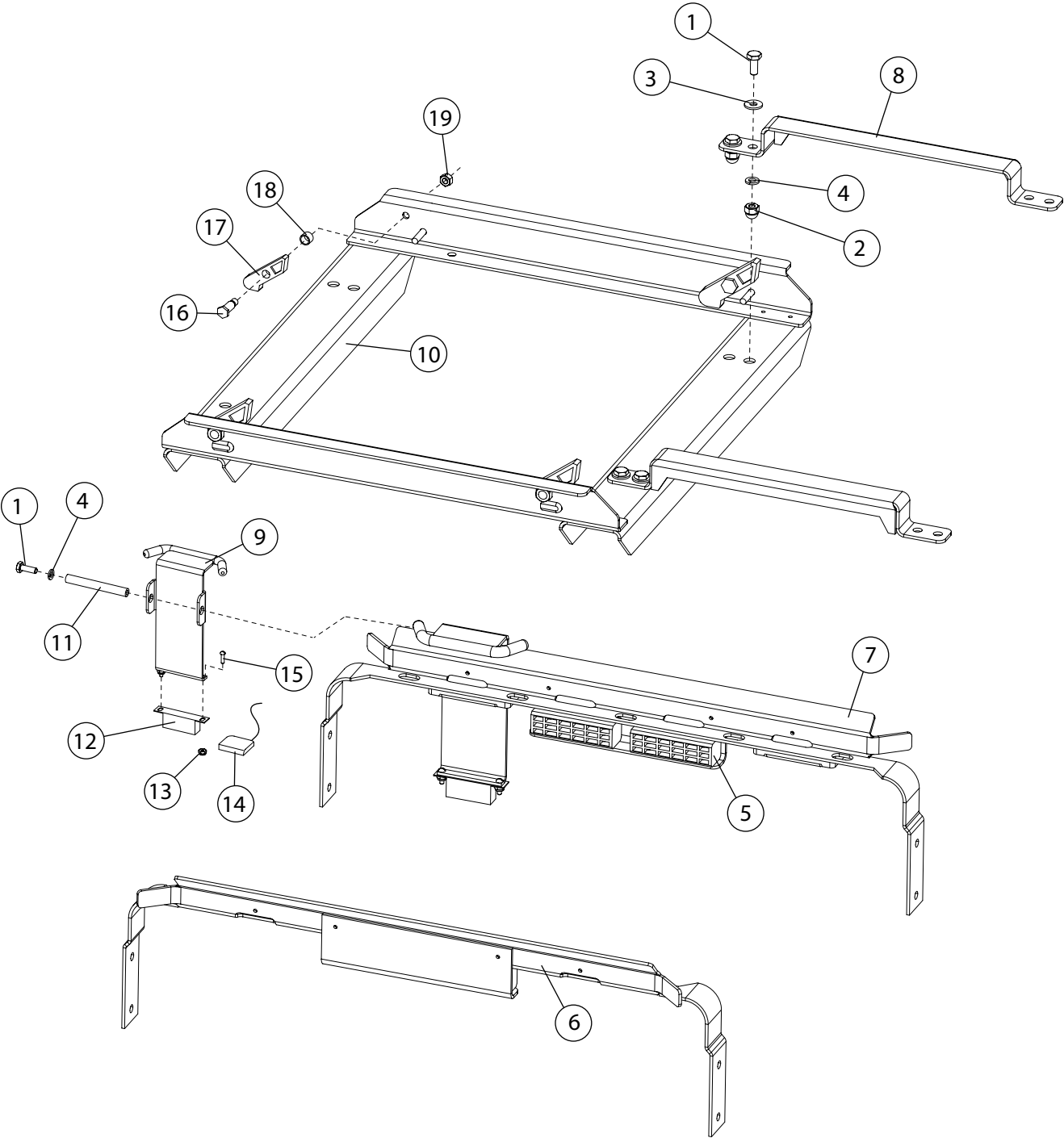
| Item No. | Part No. | Description                              | Qty. |
|----------|----------|------------------------------------------|------|
| 1        | 100154   | HEX PLAIN NUT, 5/16-18 SST               | 4    |
| 2        | 100734   | BOLT, HEX HEAD, 1/4-20 X 1/2" SST        | 4    |
| 3        | 102376   | WASHER, FLAT 5/16" SST                   | 4    |
| 4        | 106026   | WASHER, FLAT 1/4" SST                    | 4    |
| 5        | 106013   | WASHER, LOCK 5/16" SST                   | 4    |
| 6        | 106014   | HEX ACORN PLAIN NUT, 1/4-20 SST          | 4    |
| 7        | 106482   | WASHER, LOCK 1/4" SPLIT SST              | 4    |
| 8        | 108345   | GASKET, HEATER/STANDPIPE                 | 2    |
| 9        | 111532   | O-RING                                   | 1    |
| 10       | 333105   | UPPER WASH SUPPORT ROD WELDMENT, MT-RACK | 2    |
| 11       | 333460   | FLANGE WELDMENT, WASH SYS. CONN. MT-RACK | 1    |
| 12       | 333566   | BRACKET, UPPER WASH ADJUSTMENT MT-RACK   | 2    |
| 13       | 418238-S | MANIFOLD TAPER ASSY. UPPER WASH          | 1    |
| 14       | 333712   | STANDPIPE WELDMENT, 1-1/4" OD MD66       | 1    |



| Item No. | Part No. | Description                              | Qty. |
|----------|----------|------------------------------------------|------|
| 1        | 334979   | FLANGE WELDMENT, LOWER WASH              | 1    |
| 2        | 100734   | BOLT, HEX HEAD, 1/4-20 X 1/2" SST        | 4    |
| 3        | 100735   | BOLT, HEX HEAD, 1/4-20 X 5/8" SST        | 4    |
| 4        | 102376   | WASHER, FLAT 5/16" SST                   | 8    |
| 5        | 104618   | WASHER, FLAT 3/8" SST                    | 8    |
| 6        | 106013   | WASHER, LOCK 5/16" SPLIT SST             | 8    |
| 7        | 106014   | HEX ACORN PLAIN NUT, 1/4-20 SST          | 16   |
| 8        | 106026   | WASHER, FLAT 1/4" SST                    | 8    |
| 9        | 106482   | WASHER, LOCK 1/4" SPLIT SST              | 4    |
| 10       | 108345   | GASKET, HEATER/STANDPIPE                 | 2    |
| 11       | 109009   | HEX GRIP NUT, 5/16-18 SST W/NYLON        | 8    |
| 12       | 111532   | O-RING                                   | 2    |
| 13       | 113555   | THREADED PLUG, WASH ARM                  | 16   |
| 14       | 113716   | WASH-ARMS O-RING                         | 16   |
| 15       | 333096   | STANDPIPE WELDMENT 44" MT-RACK           | 1    |
| 16       | 417963-S | MANIFOLD TAPER ASSEMBLY, UPPER           | 1    |
| 17       | 333102   | MANIFOLD TAPER ASSEMBLY, LOWER           | 1    |
| 18       | 333105   | UPPER WASH SUPPORT ROD WELDMENT, MT-RACK | 2    |
| 19       | 333460   | FLANGE WELDMENT, WASH SYS. CONN. MT-RACK | 2    |
| 20       | 333566   | BRACKET, UPPER WASH ADJUSTMENT MT-RACK   | 2    |
| 21       | 334977   | BRACKET, LOWER WASH, LATCH               | 2    |

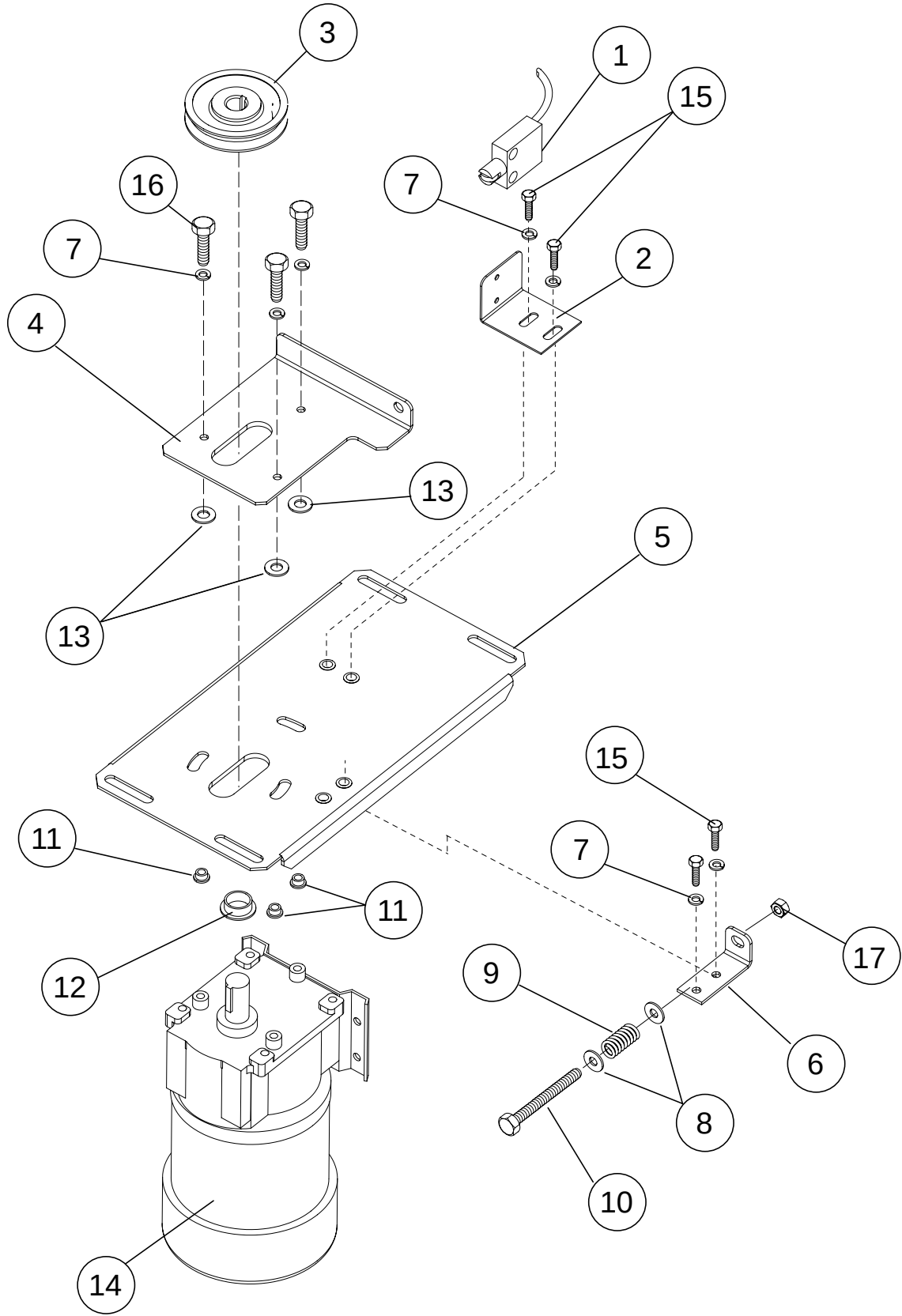


| Item No. | Part No. | Description                                           | Qty. |
|----------|----------|-------------------------------------------------------|------|
| 1        | 100142   | HEX GRIP NUT, 5/16-18 SST CTR LOCK                    | 12   |
| 2        | 100734   | BOLT, HEX HEAD 1/4-20 X 1/2" SST                      | 12   |
| 3        | 100764   | SCREW, ROUND HD., 6/32 X 1/2" SST                     | 4    |
| 4        | 106482   | WASHER, LOCK 1/4" SPLIT SST                           | 12   |
| 5        | 108954   | HEX, GRIP NUT, 6-32 SST W/NYLON INSERT                | 4    |
| 6        | 113692   | BOLT SHOULDER, 3/8" X 3/4" X 5/16-18 X 3/8"           | 12   |
| 7        | 113937   | MAGNET, REED SWITCH                                   | 2    |
| 8        | 204513   | PAWL, CRADLE                                          | 10   |
| 9        | 206343   | BEARING, E CRADLE SUPPORT                             | 4    |
| 10       | 206345   | SPACER, E CRADLE                                      | 10   |
| 11       | 327833   | PIN, IDLE PUMP SWITCH                                 | 2    |
| 12       | 333094   | SWITCH, WELDMENT                                      | 2    |
| 13       | 334919   | TRACK WELDMENT, FRONT 44"                             | 1    |
| 14       | 334920   | TRACK WELMENT, REAR 44"                               | 1    |
| 15       | 334922   | CRADLE, WELDMENT 44"                                  | 1    |
| 16       | 334923   | TRACK, LOWER WASH SUPPORT                             | 2    |
| 17       | 113719   | SWITCH, REED                                          | 2    |
| —        | 419065   | CRADLE ASSY COMPLETE (INCLUDES ITEMS 1, 6, 8, 10, 15) | 1    |





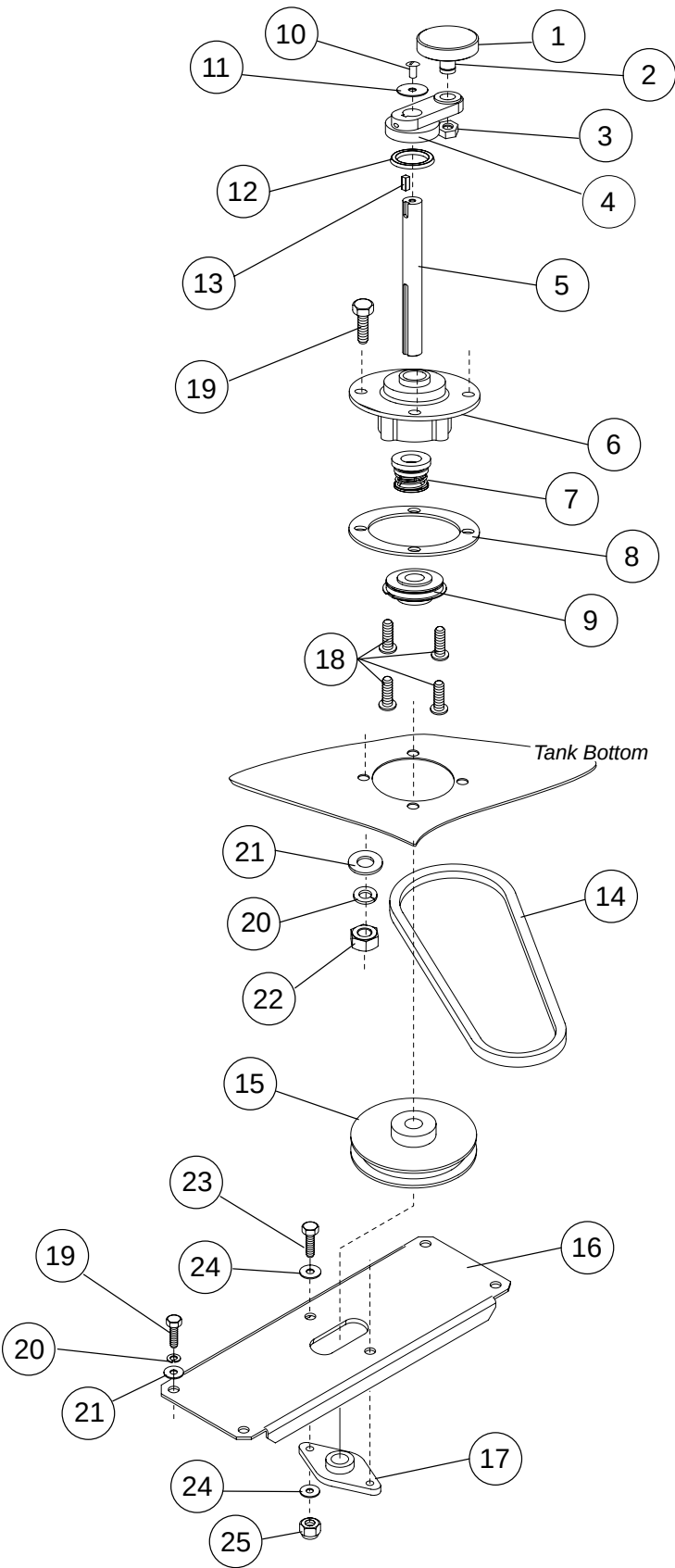
| Item No. | Part No. | Description                                               | Qty. |
|----------|----------|-----------------------------------------------------------|------|
| 1        | 100735   | BOLT, HEX HEAD 1/4-20 X 5/8" SST                          | 4    |
| 2        | 106014   | HEX ACORN PLAIN NUT 1/4-20 SST                            | 4    |
| 3        | 106026   | WASHER, FLAT 1/4" SST                                     | 4    |
| 4        | 106482   | WASHER, LOCK 1/4" SPLIT SST                               | 4    |
| 4        | 100764   | SCREW, ROUND HD., 6/32 X 1/2" SST                         | 4    |
| 5        | 206343   | BEARING, E CRADLE SUPPORT                                 | 4    |
| 6        | 334971   | TRACK WELDMENT, FRONT PW 66"                              | 1    |
| 7        | 334972   | TRACK WELDMENT, REAR PW 66"                               | 1    |
| 8        | 334976   | BRACKET, CRADLE CONNECTOR                                 | 2    |
| 9        | 333094   | SWITCH, WELDMENT                                          | 1    |
| 10       | 334973   | CRADLE, WELDMENT PW 66"                                   | 1    |
| 11       | 327833   | PIN, IDLE PUMP SWITCH                                     | 1    |
| 12       | 113937   | MAGNET, REED SWITCH                                       | 1    |
| 13       | 108954   | HEX, GRIP NUT, 6-32 SST W/NYLON INSERT                    | 2    |
| 14       | 113719   | SWITCH, REED                                              | 1    |
| 15       | 100764   | SCREW, ROUND HD., 6/32 X 1/2" SST                         | 2    |
| 16       | 113692   | BOLT SHOULDER, 3/8" X 3/4" X 5/16-18 X 3/8"               | 4    |
| 17       | 204513   | PAWL, CRADLE                                              | 4    |
| 18       | 206345   | SPACER, E CRADLE                                          | 4    |
| 19       | 100142   | HEX GRIP NUT, 5/16-18 SST CTR LOCK                        | 4    |
| —        | 419098   | CRADLE ASSY, COMPLETE (INCLUDES ITEMS 10, 16, 17, 18, 19) | 1    |



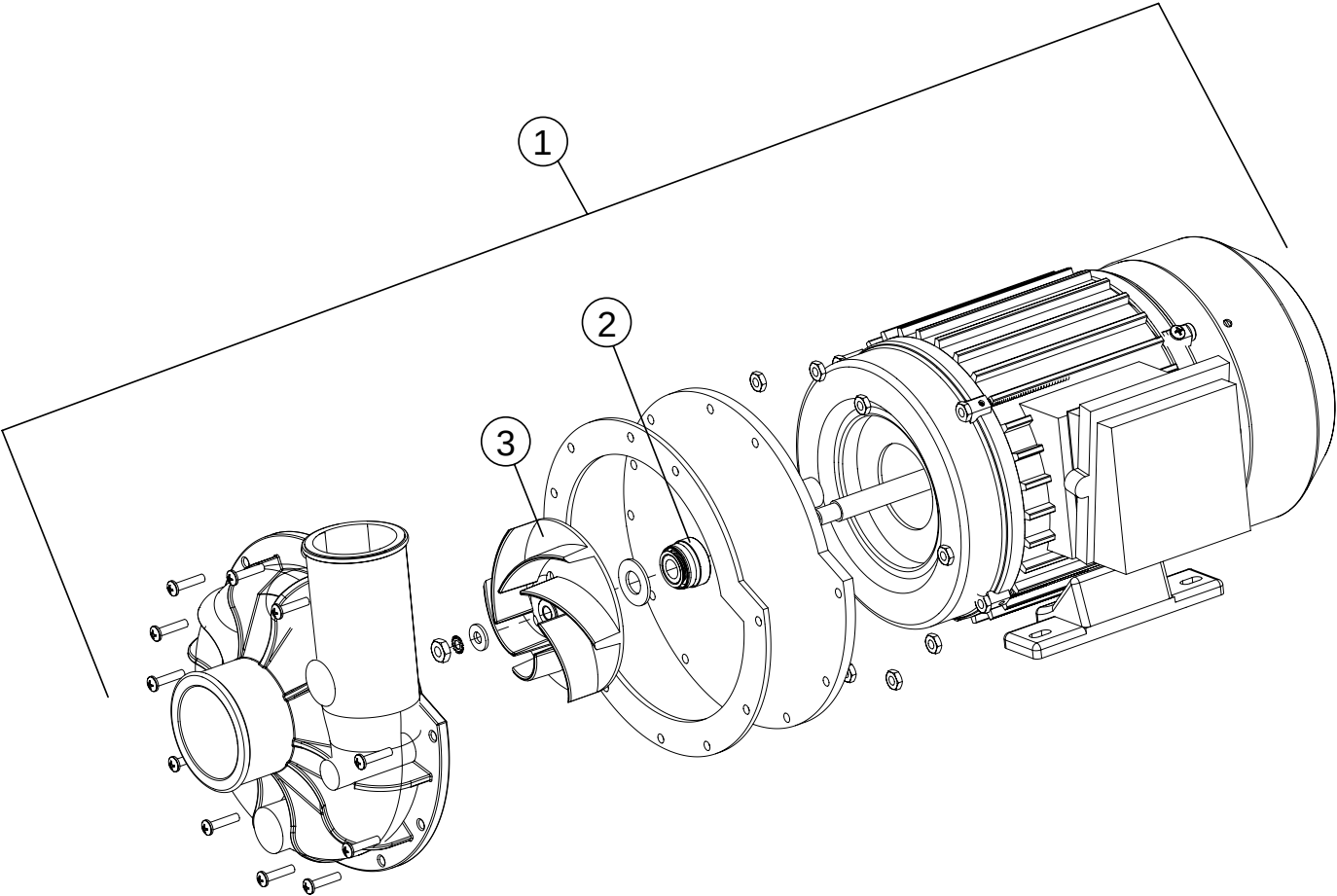
| Item No. | Part No. | Description                                                | Qty. |
|----------|----------|------------------------------------------------------------|------|
| 1        | 0509199  | DRIVE MOTOR SWITCH                                         | 1    |
| 2        | 327918   | BRACKET, MOTOR SWITCH MOUNTING                             | 1    |
| 3        | 113893   | SHEAVE, MOTOR AK51, 5/8" BORE                              | 1    |
| 4        | 327916   | PLATE, DRIVE MOTOR TORQUE                                  | 1    |
| 5        | 327920   | BRACKET, 44 MOTOR ASSY. MOUNTING                           | 1    |
| 6        | 327919   | BRACKET, MOTOR SPRING MOUNTING                             | 1    |
| 7        | 106482   | WASHER, LOCK 1/4" SPLIT SST                                | 4    |
| 8        | 102376   | WASHER, FLAT 5/16" SST                                     | 2    |
| 9        | 113702   | SPRING, 0.06" OD X 0.095" WIRE X 1.5" LG.                  | 1    |
| 10       | 113704   | BOLT, HEX HEAD, 5/16-18 X 3-1/2"                           | 1    |
| 11       | 113701   | BUSHING, FLANGE 1/4" X 3/8" X 1/4" LG. BRONZE OIL          | 1    |
| 12       | 113703   | BUSHING, FLANGE 5/8" X 7/8" X 3/8" X 1-1/8" LG. BRONZE OIL | 1    |
| 13       | 113700   | BUSHING, THRUST 3/8" X 3.4" X 1/16" BRONZE OIL             | 3    |
| 14       | 113732   | MOTOR, GEAR 1/6 HP, 3PH, MV                                | 1    |
| —        | 113679   | MOTOR, GEAR 1/6 HP 1PH, MV                                 | 1    |
| 15       | 100734   | BOLT, HEX HEAD 1/4-20 X 1/2" SST                           | 4    |
| 16       | 100736   | BOLT, HEX HEAD 1/4-20 X 3/4" SST                           | 3    |
| 17       | 109009   | NUT, GRIP 5/16-18 W/NYLON INSERT                           | 1    |

## International Models I-44, I-66 Drive Motors

| Item No. | Part No. | Description                                                                                         | Qty. |
|----------|----------|-----------------------------------------------------------------------------------------------------|------|
| 14       | 113679   | MOTOR, GEAR 1/6 HP, 1PH, MV<br>200-220/50-60/1PH, 230-240/50-60/1PH                                 | 1    |
| —        | 113732   | MOTOR, GEAR 1/6 HP, 1PH, MV<br>200-220/50-60/3PH, 230-240/50-60/3PH<br>380/50-60/3, 400-415/50-60/3 | 1    |



| Item No. | Part No. | Description                                       | Qty. |
|----------|----------|---------------------------------------------------|------|
| 1        | 202381   | ROLLER, CROSSHEAD                                 | 1    |
| 2        | 100868   | STUD, CROSSHEAD ROLLER                            | 1    |
| 3        | 107089   | NUT, JAM 1/2-13                                   | 1    |
| 4        | 206300   | CRANK, RACK ARM                                   | 1    |
| 5        | 206301   | SHAFT, 5/8" DIA. X 9" LG.                         | 1    |
| 6        | 206302   | HOUSING, BEARING/SEAL                             | 1    |
| 7        | 102244   | SEAL, 5/8"                                        | 1    |
| 8        | 206303   | GASKET, BEARING, HOUSING                          | 1    |
| 9        | 100382   | BEARING, JAF #ER104-10                            | 1    |
| 10       | 100007   | SCREW, 10-32 X 3/8" TRUSS HEAD                    | 1    |
| 11       | 104925   | WASHER, 1/4" X 1" X 16 GAUGE                      | 1    |
| 12       | 103180   | WIPER RING                                        | 1    |
| 13       | 104916   | KEY, 3/16" X 3/16" X 3/4" SST                     | 1    |
| 14       | 100796   | V-BELT, 4L330                                     | 1    |
| 15       | 113892   | SHEAVE, DRIVE AK59, 5/8" BORE                     | 1    |
| 16       | 333880   | BRACKET, 44 DRIVE BEARING MOUNTING                | 1    |
| 17       | 113860   | BEARING, SEALED 5/8" BORE                         | 1    |
| 18       | 104923   | SCREW, 1/4-20 X 3/8" ROUND HEAD SST               | 4    |
| 19       | 100739   | BOLT, HEX HEAD, 5/16-18 X 3/4" SST                | 8    |
| 20       | 106013   | WASHER, LOCK 5/16" SPLIT SST                      | 8    |
| 21       | 102376   | WASHER, FLAT 5/16" SST                            | 8    |
| 22       | 100154   | NUT, PLAIN HEX HEAD 5/16-18 SST                   | 8    |
| 23       | 100746   | BOLT, HEX HEAD, 3/8-16 X 1-1/4" SST               | 2    |
| 24       | 104618   | WASHER, FLAT 3/8" SST                             | 2    |
| 25       | 109010   | NUT GRIP, 3/8-16 W/NYLON INSERT                   | 2    |
| —        | 414316   | DRIVE INSTALL ASSY. (INCLUDES ITEMS 1-13, 18, 19) | 1    |

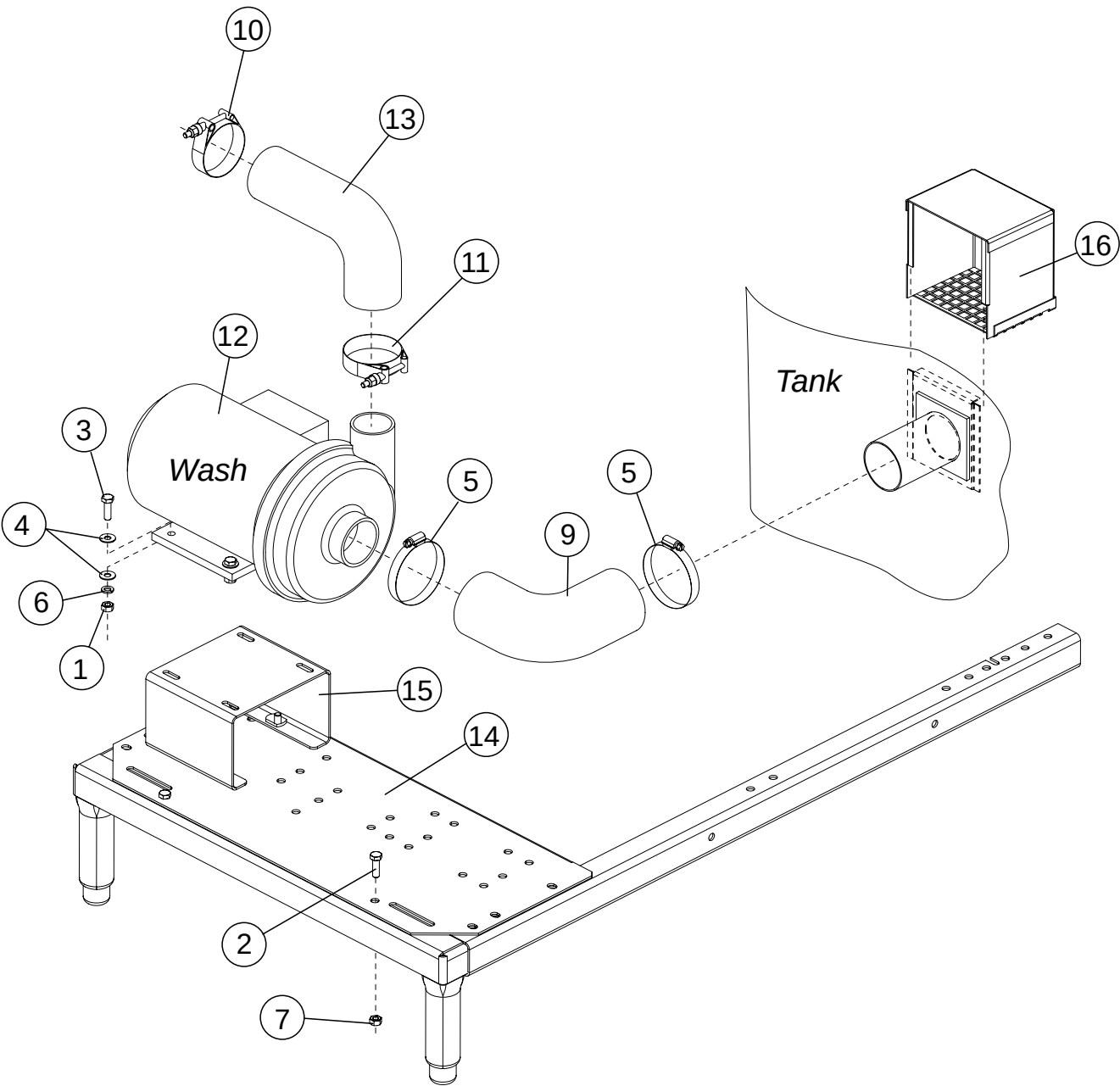


For Models: CH-44, CH-66, MD-44, MD-66

| Item No. | Part No. | Description                                                      | Qty. |
|----------|----------|------------------------------------------------------------------|------|
| 1        | 114347   | PUMP/MOTOR ASSEMBLY (INCLUDES ITEMS 2-3)<br>208-240/460/60/3 2HP | 1    |
| 2        | 108002   | PUMP SEAL ASSEMBLY                                               | 1    |
| 3        | 114793   | IMPELLER, 3.78" OD, COMPOSITE MTL                                | 1    |

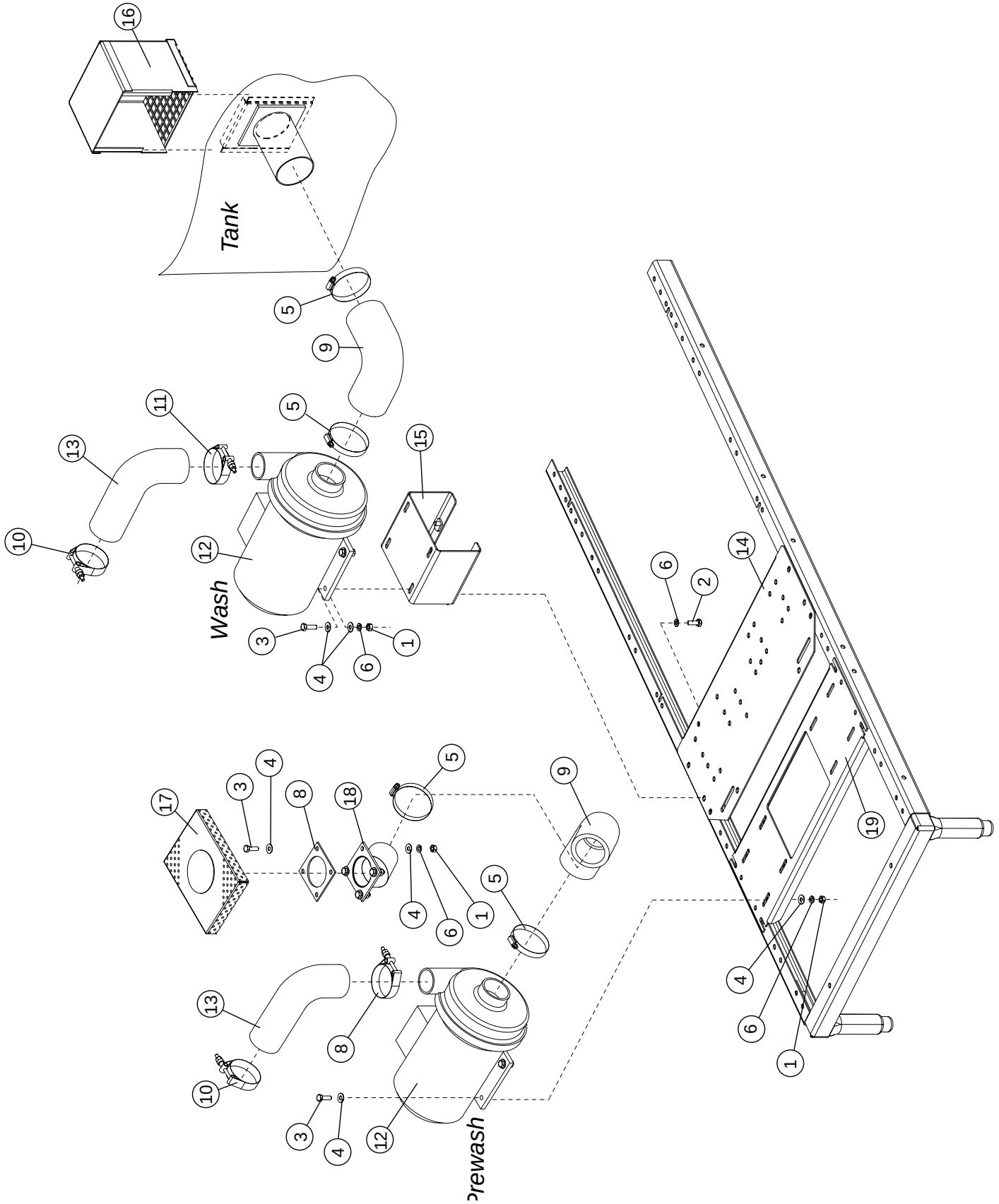
For International Models: I-44, I-66

| Item No. | Part No. | Description                                                                                                      | Qty. |
|----------|----------|------------------------------------------------------------------------------------------------------------------|------|
| 1        | 115338   | PUMP/MOTOR ASSEMBLY (INCLUDES ITEMS 2-3)<br>200-220/50/1PH & 3PH, 230-240/50/1PH & 3PH<br>380/50/3, 400-415/50/3 | 1    |
| —        | 115389   | PUMP/MOTOR ASSEMBLY (INCLUDES ITEMS 2-3)<br>200-220/60/1PH & 3PH, 230-240/60/1PH & 3PH<br>380/60/3, 400-415/60/3 | 1    |
| 2        | 108002   | PUMP SEAL ASSEMBLY                                                                                               | 1    |
| 3        | 114793   | IMPELLER, 3.78" OD, COMPOSITE MTL                                                                                | 1    |

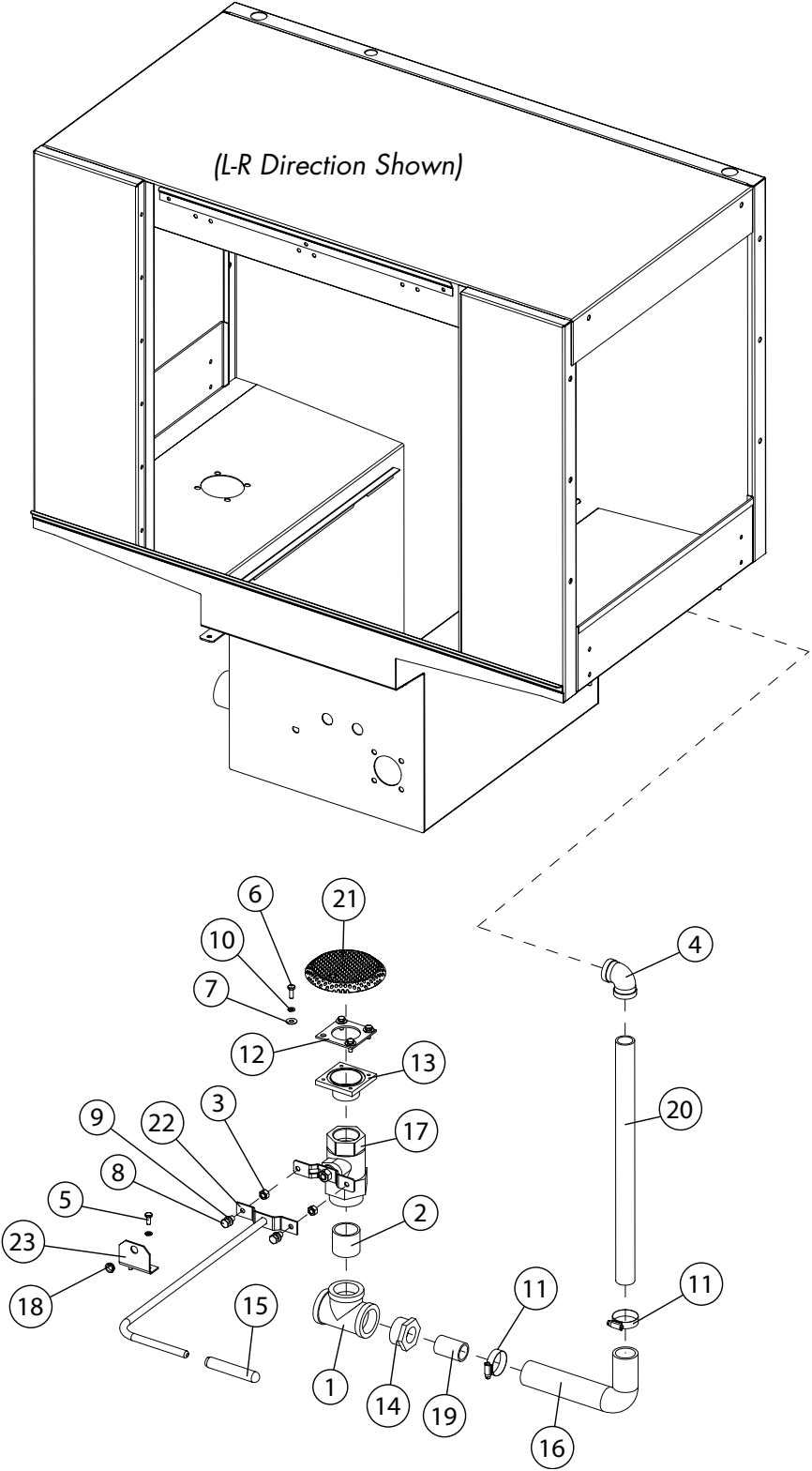




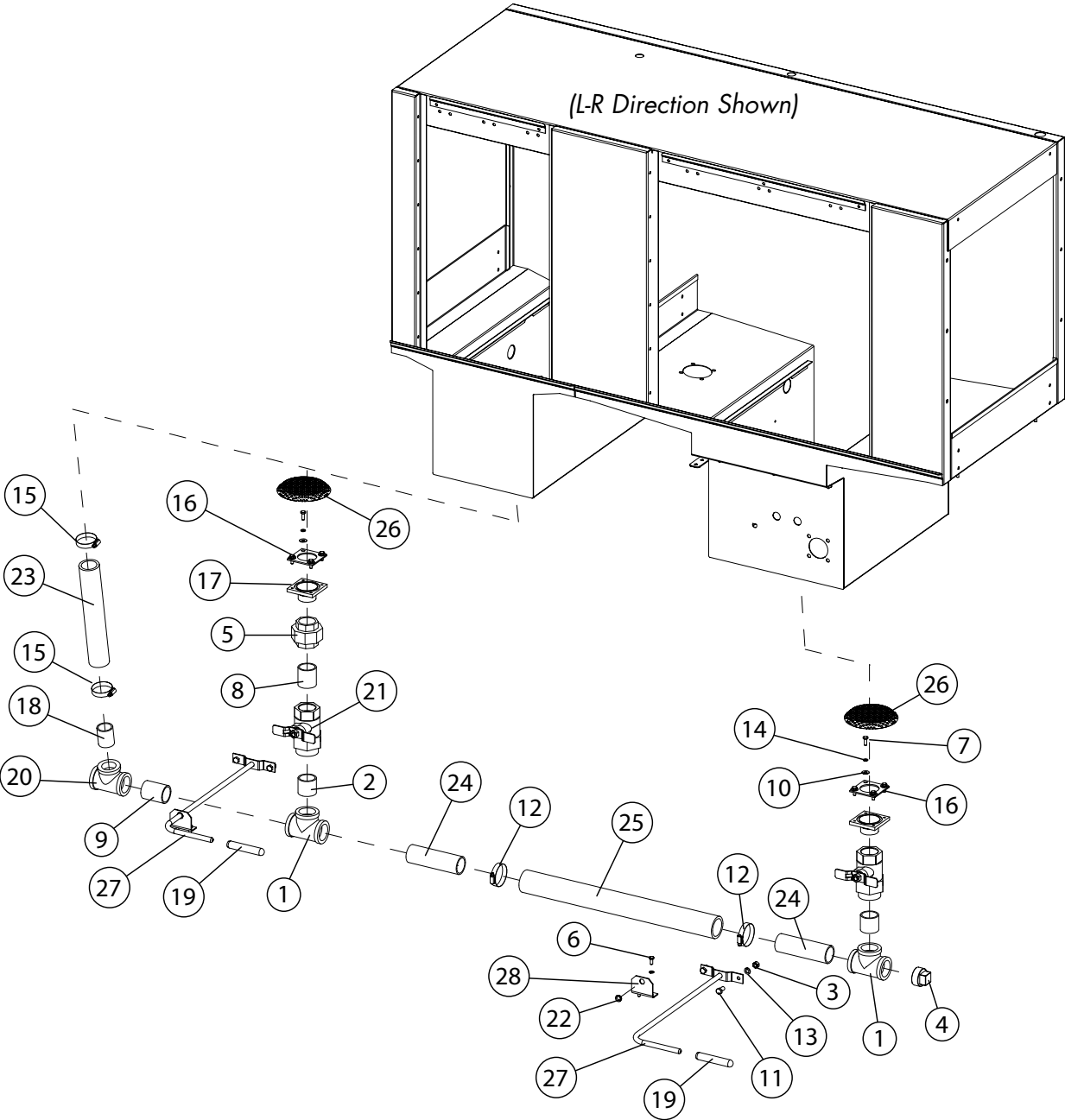
| Item No. | Part No. | Description                              | Qty. |
|----------|----------|------------------------------------------|------|
| 1        | 100154   | HEX PLAIN NUT, 5/16-18 SST               | 4    |
| 2        | 100739   | BOLT, HEX HD., 5/16-18 X 3/4" SST        | 4    |
| 3        | 100740   | BOLT, HEX HD., 5/16-18 X 1" SST          | 4    |
| 4        | 102376   | WASHER, FLAT 5/16" SST                   | 8    |
| 5        | 104203   | CLAMP, HOSE M52 SST GEAR TYPE            | 2    |
| 6        | 106013   | WASHER, LOCK 5/16" SPLIT SST             | 8    |
| 7        | 109009   | HEX GRIP NUT, 5/16-18 SST W/NYLON INSERT | 2    |
| 8        | 109568   | GASKET, PUMP SUCTION                     | 1    |
| 9        | 110372   | HOSE, PUMP SUCTION                       | 1    |
| 10       | 110858   | HOSE CLAMP, HIGH PRESSURE                | 1    |
| 11       | 111964   | HOSE CLAMP, HIGH PRESSURE                | 1    |
| 12       | 114347   | PUMP/MOTOR ASSY. 208-240/460/60/3 2HP    | 1    |
| 13       | 207022   | HOSE, PUMP DISCHARGE                     | 1    |
| 14       | 327869   | PLATE, E PUMP MOTOR SUPPORT              | 1    |
| 15       | 333076   | BRACKET WELDMENT, PUMP MOTOR SUPPORT     | 1    |
| 16       | 319742   | PUMP SUCTION SCREEN, MT-RACK             | 1    |



| Item No. | Part No. | Description                              | Qty. |
|----------|----------|------------------------------------------|------|
| 1        | 100154   | HEX PLAIN NUT, 5/16-18 SST               | 16   |
| 2        | 100739   | BOLT, HEX HD., 5/16-18 X 3/4" SST        | 4    |
| 3        | 100740   | BOLT, HEX HD., 5/16-18 X 1" SST          | 16   |
| 4        | 102376   | WASHER, FLAT 5/16" SST                   | 32   |
| 5        | 104203   | CLAMP, HOSE M52 SST GEAR TYPE            | 4    |
| 6        | 106013   | WASHER, LOCK 5/16" SPLIT SST             | 18   |
| 7        | 109009   | HEX GRIP NUT, 5/16-18 SST W/NYLON INSERT | 2    |
| 8        | 109568   | GASKET, PUMP SUCTION                     | 1    |
| 9        | 110372   | HOSE, PUMP SUCTION                       | 2    |
| 10       | 110858   | HOSE CLAMP, HIGH PRESSURE                | 1    |
| 11       | 111964   | HOSE CLAMP, HIGH PRESSURE                | 1    |
| 12       | 114347   | PUMP/MOTOR ASSY. 208-240/460/60/3 2HP    | 3    |
| 13       | 207022   | HOSE, PUMP DISCHARGE                     | 2    |
| 14       | 327869   | PLATE, E PUMP MOTOR SUPPORT (WASH)       | 2    |
| 15       | 333076   | BRACKET WELDMENT, PUMP MOTOR SUPPORT     | 1    |
| 16       | 319742   | PUMP SUCTION SCREEN, MT-RACK (WASH)      | 1    |
| 17       | 324580   | SUCTION SCREEN (PREWASH)                 | 1    |
| 18       | 333576   | FLANGE WELDMENT, PREWASH PUMP SUCTION    | 1    |
| 19       | 333075   | PLATE, PUMP MOTOR SUPPORT (PREWASH)      | 1    |



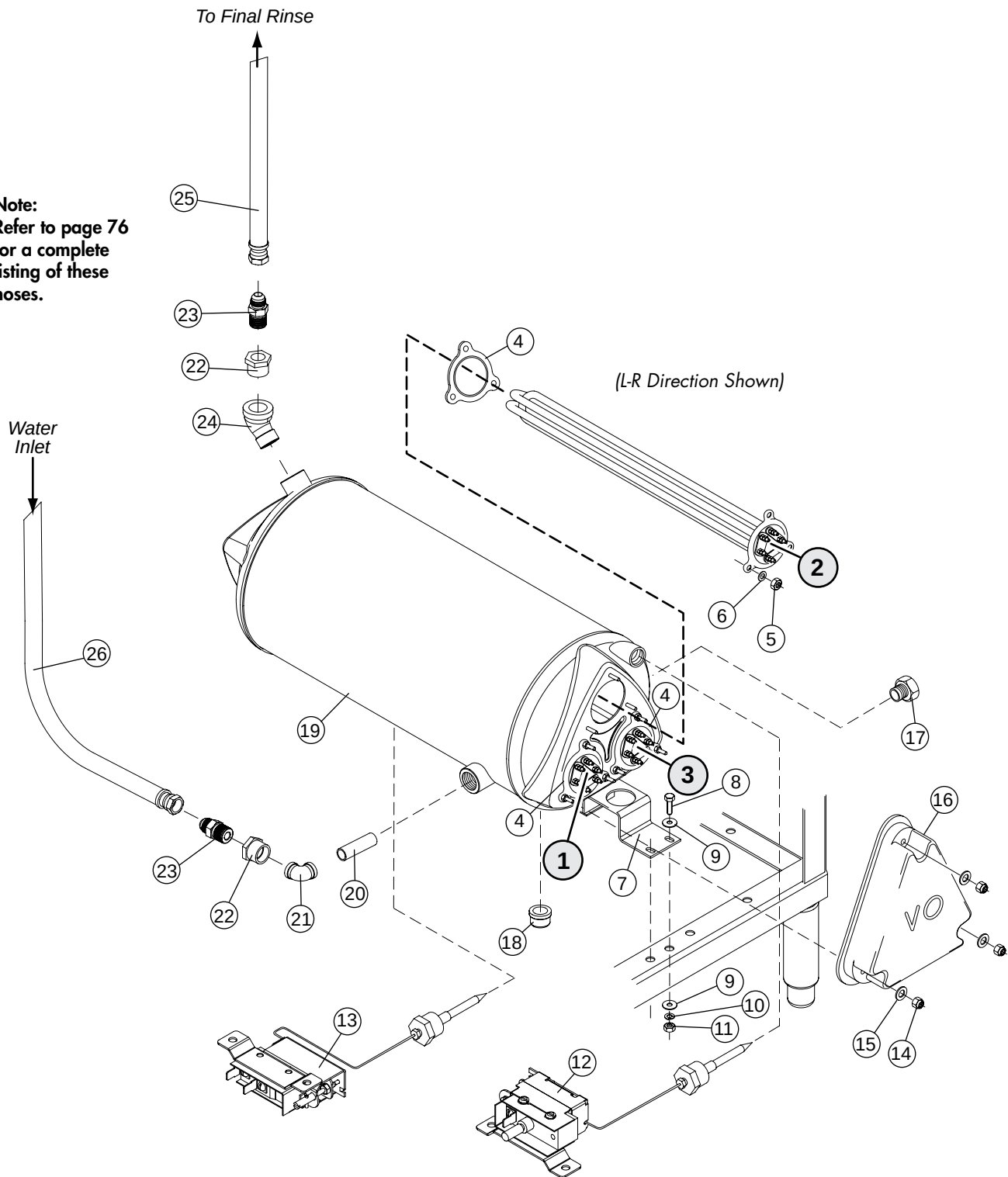
| Item No. | Part No. | Description                                                   | Qty. |
|----------|----------|---------------------------------------------------------------|------|
| 1        | 100036   | TEE, 1-1/2" NPT GALVANIZED                                    | 1    |
| 2        | 100043   | NIPPLE, 1-1/2" NPT X CLOSE, GALVANIZED                        | 1    |
| 3        | 100140   | HEX PLAIN NUT, 3/8-16 SST                                     | 2    |
| 4        | 100512   | ELBOW, 1" NPT X 90° SST                                       | 1    |
| 5        | 100735   | BOLT, HEX HD., 1/4-20 X 5/8" SST                              | 2    |
| 6        | 100736   | BOLT, HEX HD., 1/4-20 X 3/4" SST                              | 4    |
| 7        | 102376   | WASHER, FLAT 5/16" SST                                        | 4    |
| 8        | 102563   | BOLT, HEX HD., 3/8-16 X 3/4" SST                              | 2    |
| 9        | 106407   | WASHER, LOCK 3/8" SPLIT SST                                   | 2    |
| 10       | 106482   | WASHER, LOCK 1/4" SPLIT SST                                   | 6    |
| 11       | 107340   | CLAMP, HOSE M28 SST GEAR-TYPE                                 | 2    |
| 12       | 108345   | GASKET                                                        | 1    |
| 13       | 111034   | DRAIN BODY, MACHINED                                          | 1    |
| 14       | 111097   | BUSHING, REDUCING, 1-1/2" NPT X 1" NPT GALV.                  | 1    |
| 15       | 114185   | GRIP, DRAIN HANDLE                                            | 1    |
| 16       | 114472   | HOSE, OVERFLOW                                                | 1    |
| 17       | 114718   | VALVE, BALL 1-1/2" UNION END WITH TEE HANDLE                  | 1    |
| 18       | 114817   | SLEEVE BEARING, 1/2" DIA. ROD NYLON                           | 1    |
| 19       | 204383   | NIPPLE, TOE, 1"NPT X 2" LG., GALV.                            | 1    |
| 20       | 207190   | NIPPLE, TOE, 1" NPT X 19" LG., GALV.                          | 1    |
| 21       | 304816   | STRAINER, DRAIN                                               | 1    |
| 22       | 333268   | DRAIN HANDLE WELDMENT                                         | 1    |
| 23       | 333511   | BRACKET, DRAIN LEVER SUPPORT                                  | 1    |
| NS       | 114825   | NAMEPLATE, DRAIN                                              | 1    |
| —        | 602030   | PIPING ASSY, DRAIN<br>(INCLUDES ITEMS 1,2, 13-15, 17, 19, 22) | 1    |



| Item No. | Part No. | Description                                                               | Qty. |
|----------|----------|---------------------------------------------------------------------------|------|
| 1        | 100036   | TEE, 1-1/2" NPT GALVANIZED                                                | 2    |
| 2        | 100043   | NIPPLE, 1-1/2" NPT X CLOSE, GALVANIZED                                    | 2    |
| 3        | 100140   | HEX PLAIN NUT 3/8-16, SST                                                 | 4    |
| 4        | 100587   | PLUG, 1-1/2" NPT, SQ. HEAD, GALV.                                         | 1    |
| 5        | 100589   | UNION, 1-1/2" NPT GALVANIZED                                              | 1    |
| 6        | 100735   | BOLT, HEX HD., 1/4-20 X 5/8" SST                                          | 4    |
| 7        | 100736   | BOLT, HEX HD., 1/4-20 X 3/4" SST                                          | 8    |
| 8        | 101830   | NIPPLE, 1-1/2" NPT X 2" LG. GALV.                                         | 1    |
| 9        | 101833   | NIPPLE, 1-1/2" NPT X 2-1/2" LG. GALV.                                     | 1    |
| 10       | 102376   | WASHER, FLAT 5/16" SST                                                    | 8    |
| 11       | 102563   | BOLT, HEX HEAD, 3/8-16 X 3/4" SST                                         | 4    |
| 12       | 104165   | CLAMP, HOSE M40 SST GEAR-TYPE                                             | 2    |
| 13       | 106407   | WASHER, LOCK 3/8" SPLIT SST                                               | 4    |
| 14       | 106482   | WASHER, LOCK 1/4" SPLIT SST                                               | 12   |
| 15       | 107340   | CLAMP, HOSE M28 SST GEAR-TYPE                                             | 2    |
| 16       | 108345   | GASKET                                                                    | 2    |
| 17       | 111034   | DRAIN BODY, MACHINED                                                      | 2    |
| 18       | 111113   | NIPPLE TOE, 1-1/4" NPT X 2" LG. GALV.                                     | 1    |
| 19       | 114185   | GRIP, DRAIN HANDLE                                                        | 2    |
| 20       | 114216   | TEE, 1-1/2" NPT X 1-1/2" NPT X 1-1/4" NPT, GALV.                          | 1    |
| 21       | 114718   | VALVE, BALL 1-1/2" UNION END WITH TEE HANDLE                              | 2    |
| 22       | 114817   | SLEEVE BEARING, 1/2" DIA. ROD NYLON                                       | 2    |
| 23       | 202438   | HOSE, 1.625" ID X 12" LG.                                                 | 1    |
| 24       | 206430   | NIPPLE TOE, 1-1/2" NPT X 6" LG. GALV.                                     | 2    |
| 25       | 202438   | HOSE, 1.875" ID X 23" LG.                                                 | 1    |
| 26       | 304816   | STRAINER, DRAIN                                                           | 2    |
| 27       | 333268   | DRAIN HANDLE WELDMENT                                                     | 2    |
| 28       | 333511   | BRACKET, DRAIN LEVER SUPPORT MT-RACK                                      | 2    |
| —        | 601950   | PIPING ASSY, DRAIN (PREWASH)<br>(INCLUDES ITEMS 1, 2, 5, 8, 9, 17-21, 27) | 1    |
| —        | 602031   | PIPING ASSY, DRAIN (WASH)<br>(INCLUDES ITEMS 1, 2, 4, 17, 19, 21, 27)     | 1    |
| NS       | 114825   | NAMEPLATE, DRAIN, MD-RACK                                                 | 2    |

# Electric Booster Assembly - 40°F and 70°F Rise - All Models

**Note:**  
Refer to page 76  
for a complete  
listing of these  
hoses.





| Item No. | Part No. | Description | Qty. |
|----------|----------|-------------|------|
|----------|----------|-------------|------|

**16KW-40°F Rise** 208-220/60/3, 200-220/50-60/1 & 3PH, 380/50-60/3

|   |        |             |   |
|---|--------|-------------|---|
| 1 | 111235 | HEATER, 5kW | 1 |
| 2 | 107909 | HEATER, 6kW | 1 |
| 3 | 111235 | HEATER, 5kW | 1 |

**27kW-70°F Rise** - 208-220/60/3, 200-220/50-60/3, 380/50-60/3

|   |        |              |   |
|---|--------|--------------|---|
| 1 | 111232 | HEATER, 10kW | 1 |
| 2 | 112059 | HEATER, 12kW | 1 |
| 3 | 111235 | HEATER, 5kW  | 1 |

**16kW-40°F Rise** - 230-240/50-60/1 & 3PH, 400-415/50-60/3

|   |        |             |   |
|---|--------|-------------|---|
| 1 | 111236 | HEATER, 5kW | 1 |
| 2 | 114030 | HEATER, 6kW | 1 |
| 3 | 111236 | HEATER, 5kW | 1 |

**27kW-70°F Rise** - 230-240/50-60/3, 400-415/50-60/3

|   |        |              |   |
|---|--------|--------------|---|
| 1 | 111233 | HEATER, 10kW | 1 |
| 2 | 111334 | HEATER, 12kW | 1 |
| 3 | 111236 | HEATER, 5kW  | 1 |

**16KW-40°F Rise** - 460-480/60/3

|   |        |             |   |
|---|--------|-------------|---|
| 1 | 111237 | HEATER, 5kW | 1 |
| 2 | 114031 | HEATER, 6kW | 1 |
| 3 | 111237 | HEATER, 5kW | 1 |

**27kW-70°F Rise** - 460-480/60/3

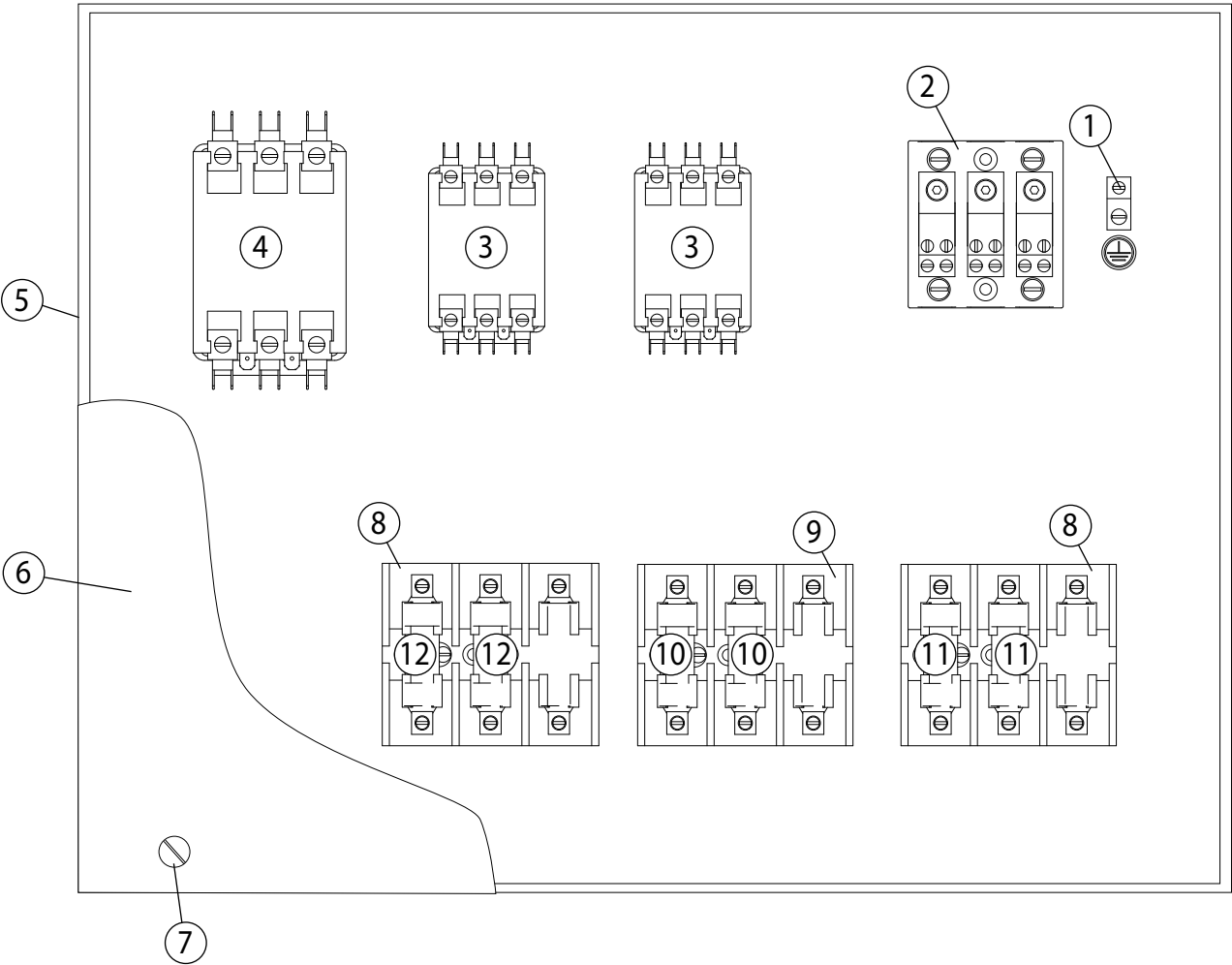
|   |        |              |   |
|---|--------|--------------|---|
| 1 | 111234 | HEATER, 10kW | 1 |
| 2 | 112060 | HEATER, 12kW | 1 |
| 3 | 111237 | HEATER, 5kW  | 1 |

Continued on next page

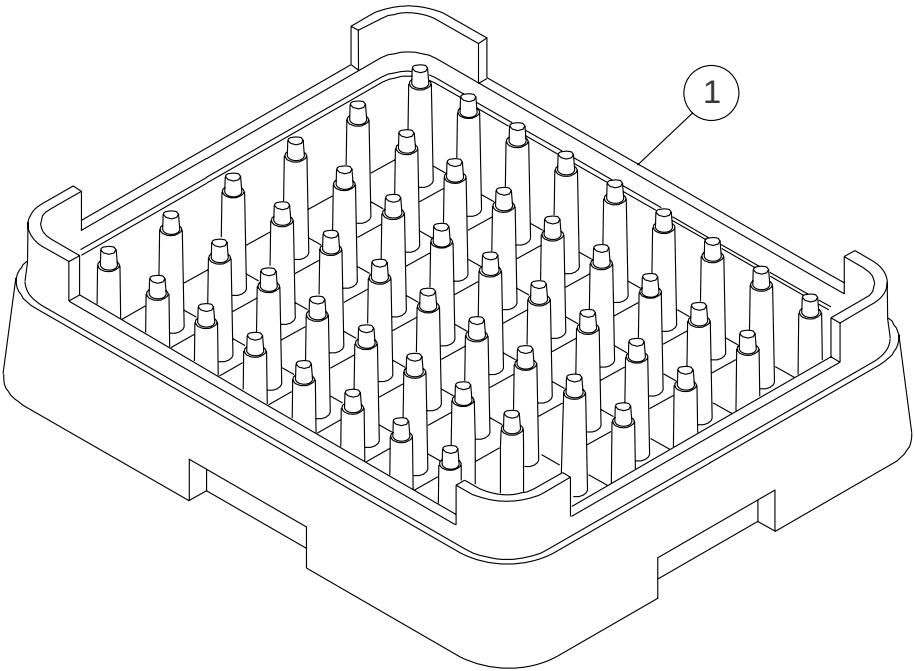
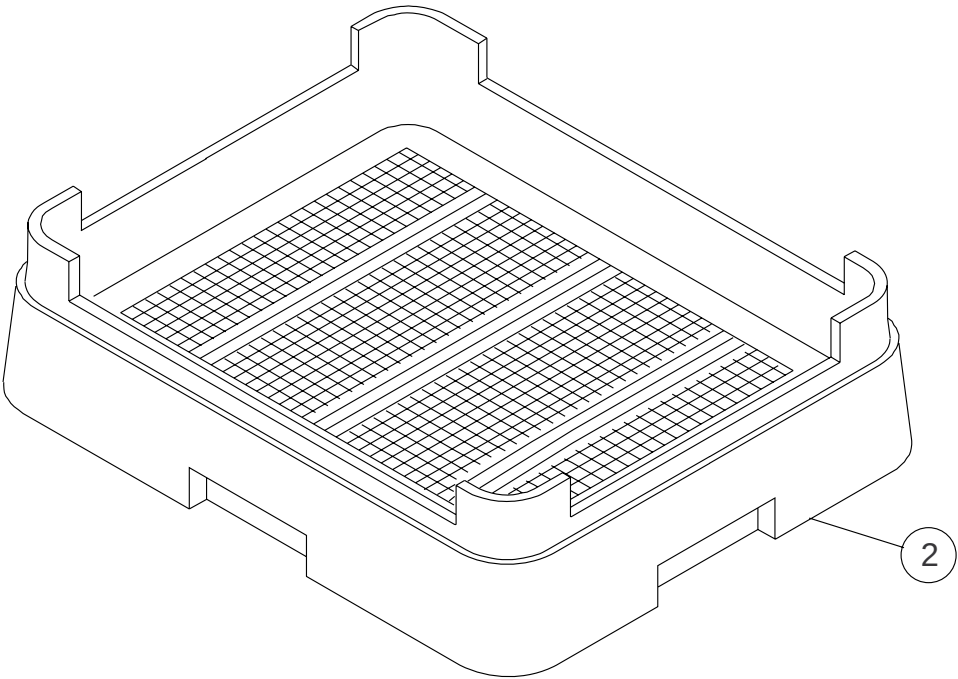
*Electric Booster Assembly - 40°F and 70°F Rise - All Models*

| Item No. | Part No. | Description                                            | Qty. |
|----------|----------|--------------------------------------------------------|------|
| 4        | 109985   | SEAL, ELECTRIC HEATER FLANGE                           | 3    |
| 5        | 100003   | NUT, PLAIN 1/4-20 SST                                  | 9    |
| 6        | 106482   | WASHER, LOCK 1/4" SPLIT                                | 9    |
| 7        | 328254   | BRACKET, FRONT BOOSTER                                 | 1    |
| 8        | 100739   | BOLT, 5/16-18 X 3/4" HEX HD. SST                       | 2    |
| 9        | 102376   | WASHER, FLAT 5/16"                                     | 4    |
| 10       | 106013   | WASHER, LOCK 5/16" SPLIT                               | 2    |
| 11       | 100154   | NUT, PLAIN 5/16-18 SST                                 | 2    |
| 12       | 110561   | THERMOSTAT, HIGH LIMIT WITH CAPILLARY                  | 1    |
| 13       | 109069   | THERMOSTAT, CONTROL WITH CAPILLARY                     | 1    |
| 14       | 107967   | NUT GRIP, 1/4-20 W/NYLON INSERT                        | 3    |
| 15       | 106026   | WASHER, FLAT 1/4"                                      | 3    |
| 16       | 108576   | COVER, BOOSTER, NO CUTOUT                              | 1    |
| 17       | 102505   | PLUG, 3/4" NPT BRASS                                   | 1    |
| 18       | 100113   | CAP, 3/4" NPT SST                                      | 1    |
| 19       | 414331   | TANK BOOSTER                                           | 1    |
| 20       | 102490   | NIPPLE, 3/4" NPT X 3-1/2" LG. BRASS                    | 1    |
| 21       | 102442   | ELBOW, 3/4" NPT X 90° BRASS                            | 1    |
| 22       | 102392   | BUSHING, REDUCING 3/4" NPT X 1/2" NPT BRASS            | 2    |
| 23       | 115101   | CONNECTOR, MALE 1/2" NPT X 1/2" 37° FLARE JIC BRASS    | 2    |
| 24       | 113706   | ELBOW, STREET 3/4" NPT X 45° BRASS                     | 1    |
| 25       | 419003   | HOSE ASSY 1/2" ID X 46" LG FLARE JIC (For L-R machine) | 1    |
| —        | 419012   | HOSE ASSY 1/2" ID X 60" LG FLARE JIC (For R-L machine) | 1    |
| 26       | 419012   | HOSE ASSY 1/2" ID X 60" LG FLARE JIC (For L-R machine) | 1    |
| —        | 419007   | HOSE ASSY 1/2" ID X 46" LG FLARE JIC (For R-L machine) |      |

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| Item No. | Part No. | Description                       | Qty. |
|----------|----------|-----------------------------------|------|
| 1        | 103310   | LUG, GROUND                       | 1    |
| 2        | 111833   | BLOCK, INPUT TERMINAL 175A 3-POLE | 1    |
| 3        | 111827   | CONTACTOR, 60 FLA                 | 2    |
| 4        | 111231   | CONTACTOR 75 FLA                  | 1    |
| 5        | 328421   | CABINET, BOOSTER CONTROL          | 1    |
| 6        | 328525   | COVER, CABINET BOOSTER CONTROL    | 1    |
| 7        | 100073   | SCREW, 1/4-20 X 1/2" TRUSS HEAD   | 2    |
| 8        | 108424   | BLOCK, FUSE 100A T-TYPE           | 2    |
| 9        | 180171   | BLOCK, FUSE 60A J-TYPE            | 1    |
| 10       | 180176   | FUSE, J-TYPE 60A                  | 2    |
| 11       | 111683   | FUSE, J-TYPE 30A                  | 2    |
| 12       | 180060   | FUSE, T-TYPE 70A                  | 2    |



| Item No. | Part No. | Description       | Qty. |
|----------|----------|-------------------|------|
| 1        | 101285   | RACK, PEG         | 1    |
| 2        | 101273   | RACK, COMBINATION | 1    |

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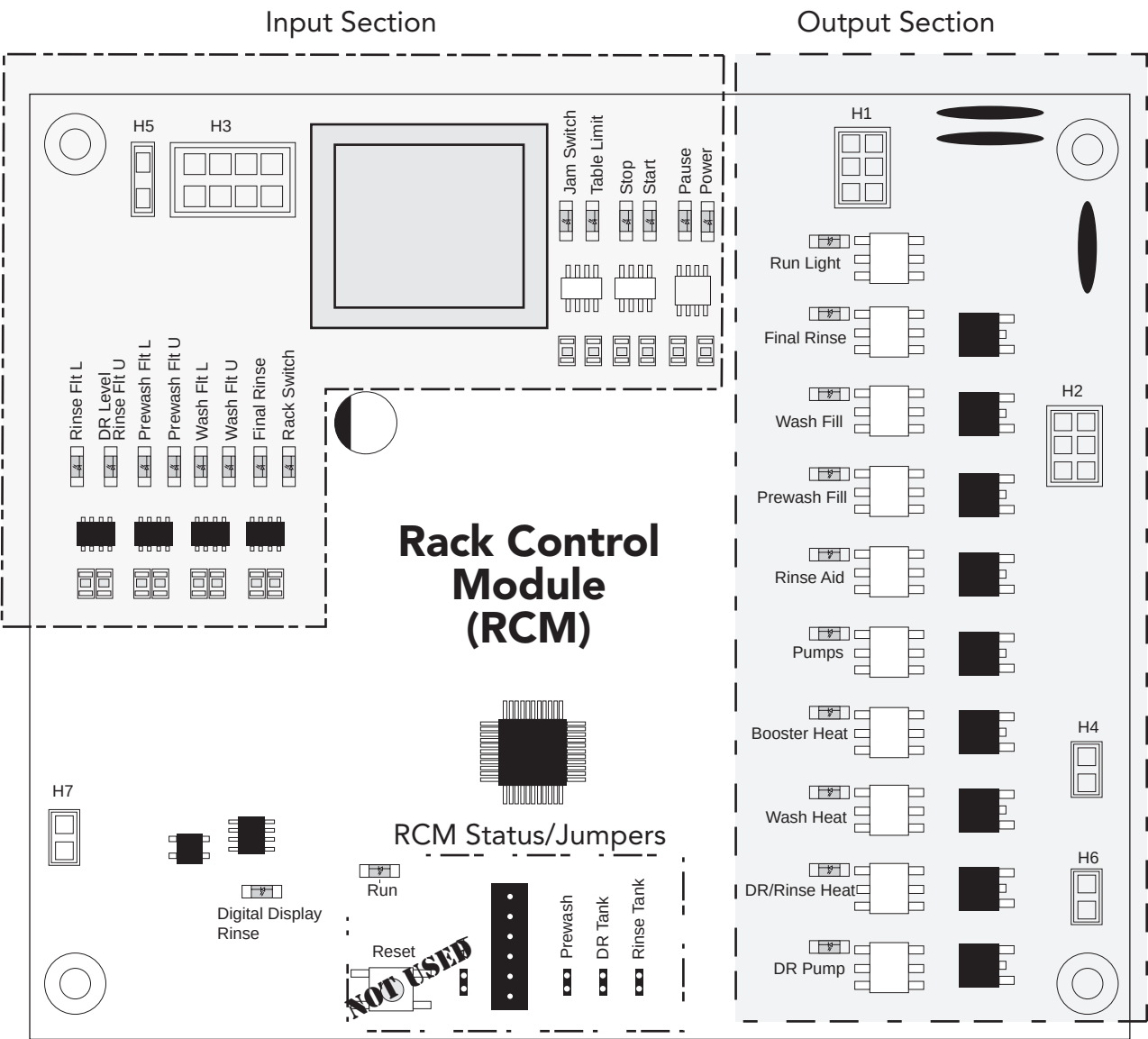


***Rack Control Module  
Operation***

***and***

***Electrical Schematics***

Rack Control Module (RCM)



RCM Sections

Refer to the illustration of the Rack Control Module (RCM) to the left and note the three outlined sections. The sections are labeled and the function of each section is explained below:

- Input Section —** Red lights show the state of the inputs to the RCM.
- Output Section —** Green lights show the outputs from the RCM to the machine.
- RCM Status/Jumpers —** The red RUN indicator shows that the RCM is energized. The jumpers set-up the inputs and outputs for various machine models.

RCM Harness Connectors

Note the illustration of the RCM at left and the details from the machine electrical schematic #702260 to the right. These illustrations show the location and connections from the RCM to the dishwasher. The function of the connectors are:

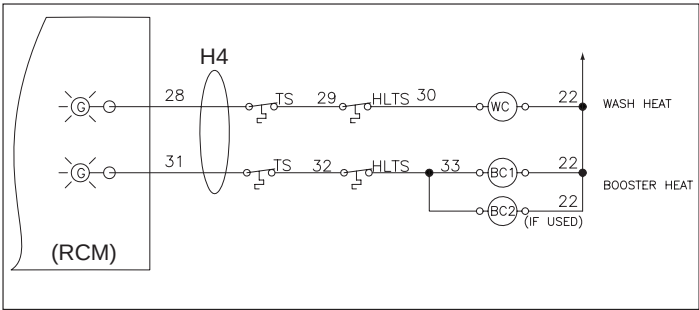
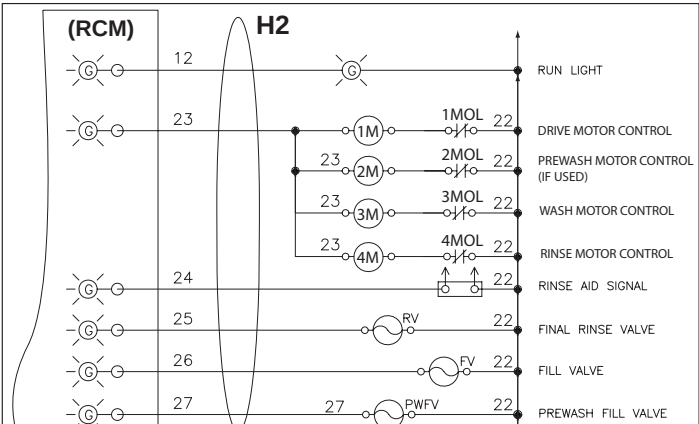
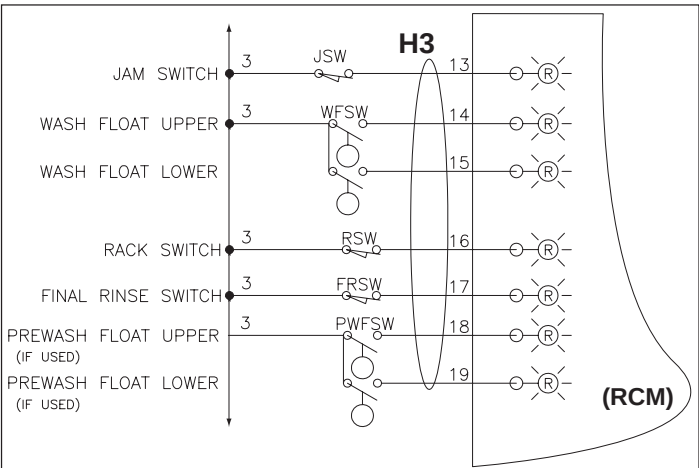
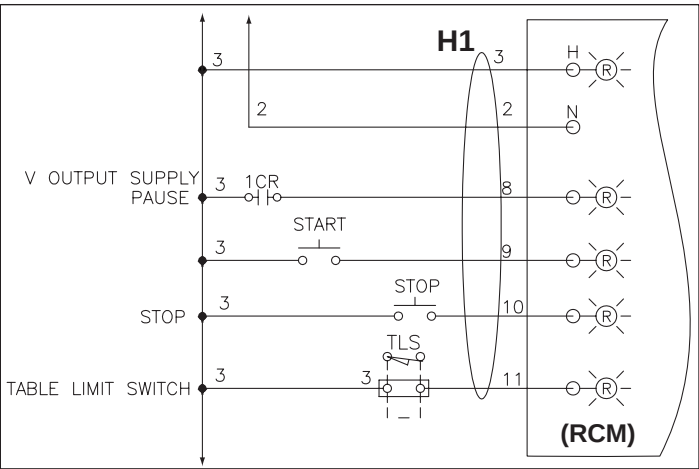
- H1 — Control Cabinet Inputs
- H2 — Control Cabinet Outputs
- H3 — Inputs from Junction Box
- H4 — Heat Outputs

RCM Operation

The insert following this page gives a detailed explanation of the inputs, outputs and the status of the indicator lights on the RCM during normal operation of the machine.

Refer to these instructions and the electrical schematic for the operation sequence of the RCM and to aid in the troubleshooting of the dishwasher.

Electrical Schematic 702260 — RCM Inputs



## Rack Control Module (RCM) Operation Input Section Red Indicator Lights

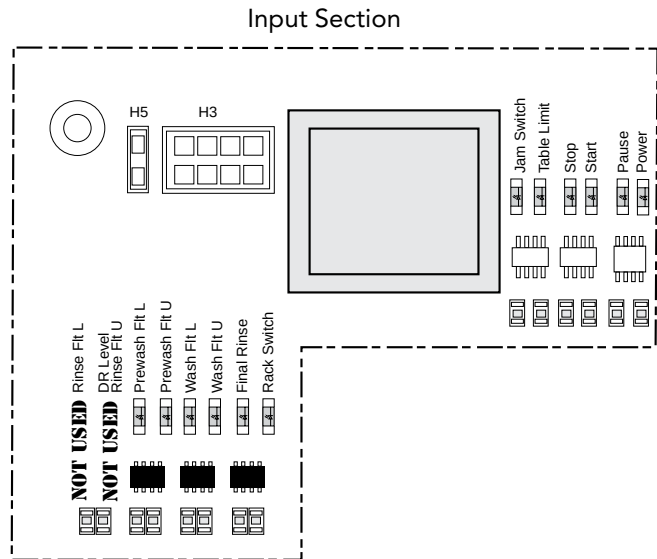
The indicator lights on the Rack Control Module (RCM) show the status of many dishwasher components and the corresponding operating condition of the dishwasher.

The explanation of the RCM is given below:

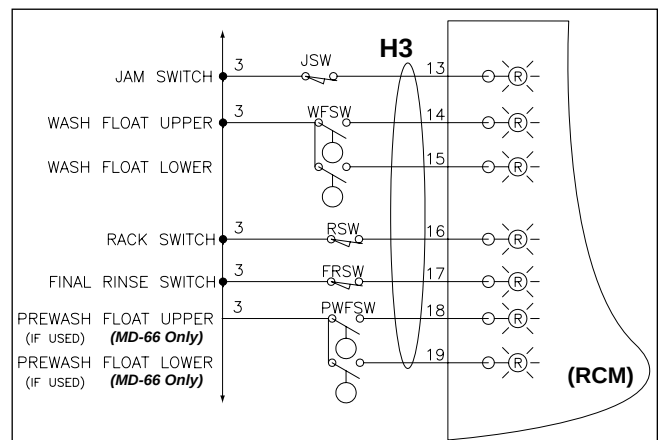
**NOTE:**

**Whenever a RED input light is ON its corresponding input circuit is CLOSED.**

- Power** - ON when the power switch on the dishwasher control panel is on.
- Pause** - ON when the door(s) are closed.  
OFF when a door(s) are open.
- Start** - ON when the front panel start push button is pressed, OFF when it is released.
- Stop** - ON when the front panel stop push button is pressed, OFF when it is released.
- Table Limit** - ON when the table limit switch is closed (The table limit switch is not activated).
- Jam Switch** - ON when the jam switch is closed (The jam switch is not activated).
- Rack Switch** - ON when the rack switch is closed.  
OFF when a dishrack is inserted into the dishwasher.
- Final Rinse** - ON when the final rinse switch is closed. OFF when the dishrack is in the Final Rinse Zone.
- Wash Flt U** - ON when the wash tank is full and the wash upper float is up.
- Wash Flt L** - ON when the wash tank is full and the wash lower float is up.
- Prewash Flt U** - ON when the prewash tank is full and the PW upper float is up.  
(Model MD-66 Only)
- Prewash Flt L** - ON when the prewash tank is full and the PW lower float is up.  
(Model MD-66 Only)



*Red lights show the status of inputs on the RCM.*



*Inputs to the RCM come from the dishwasher controls.*

**ATTENTION:**

The **DR Level/Rinse Flt U** and the **Rinse Flt L** are not used on the MD-44 and the MD-66 models.

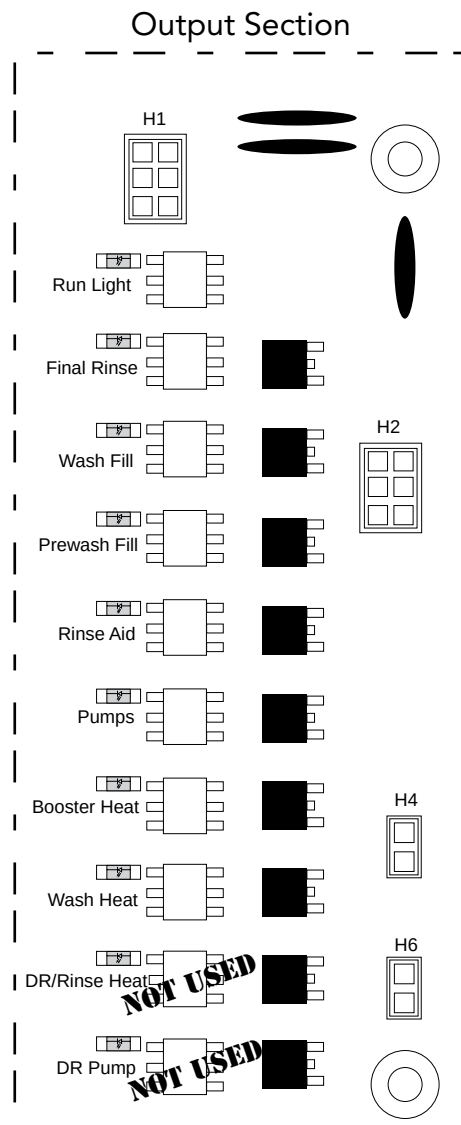
## Rack Control Module (RCM) Operation

### Output Section Green Indicator Lights

**NOTE:**

Whenever a Green output light is ON it means that the RCM is providing a signal to enable its corresponding circuit; however, the actual operation of the circuit may depend on other associated controls.

- Run Light** - ON when the start switch on the dishwasher front control panel is pressed. The run light on the dishwasher front panel is also comes on.
- SUSPENDED OPERATION** - In the event that the wash fill or prewash fill are on for more than 10 min., all inputs & outputs OFF. THE RUN LIGHT WILL FLASH ONCE PER SECOND. THE DISHWASHER POWER SWITCH MUST BE TURNED OFF AND ON TO CLEAR THIS CONDITION.
- Final Rinse** - ON whenever the final rinse switch is activated and the pumps are on or if the fill valve is ON when the machine is filling. Jam Switch, Table Limit and Pause output lights must be ON.
- Wash Fill** - If the Wash Flt L light is off, then the Wash Fill Green light is ON until the Wash Flt L and Wash Flt U red input lights come on. The Pause input light must be on. The Green Wash Fill light will stay ON for a maximum of 10 minutes.
- Prewash Fill** - If the Prewash Flt L red input light is off, then the Prewash Fill Green light is ON until the PW Flt L and PW Flt U red input lights come on. The Pause input light must be on. The Green Prewash Fill light will stay ON for a maximum of 10 minutes.  
MD-66 Only
- Rinse Aid** - ON when the final rinse switch is operated. OFF when the Jam Switch or Table Limit red output lights are OFF or if the Stop switch on the control cabinet is pressed. OFF when the RCM Timer has timed out during normal cycle.
- Pumps** - Output is controlled by the RCM Timer. ON when the rack switch is activated. OFF when the Jam Switch or Table Limit red output light are OFF or if the Stop switch on the control cabinet is pressed. OFF when the RCM Timer has timed out during normal cycle.
- Booster Heat** - Output to Booster Heat Contactor. ON if the Wash Flt L and Pause red input lights are on. OFF if Wash Flt L or Pause lights are off.
- Wash Heat** - Output to Wash Heat Contactor. ON if the Wash Flt L and Pause red input lights are on. OFF if Wash Flt L or Pause lights are off.
- DR/Rinse Heat** - NOT USED FOR MODELS MD-44 AND MD-66.
- DR Pump** - NOT USED FOR MODELS MD-44 AND MD-66.



## Rack Control Module (RCM) Operation

### RCM Timer Operation

The timer controls the drive motor, wash pump and the CH66, MD66, I66 prewash pump.

The timer is internal to the RCM. The PUMPS output indicator light shows the RCM Timer is active.

The timer output comes on when the rack switch opens, this turns on the PUMPS output. When the rack switch closes the timer counts for 90 seconds.

After 90 seconds, the PUMPS output is turned off.

Inserting another rack during the timing resets the timer to 90 seconds.

The red input lights below show how the RCM inputs affect the timer's operation,

START light ON - Enables automatic operation

STOP light ON - Disables automatic operation

RACK SWITCH light OFF - Starts the timer

TABLE LIMIT light OFF - Pauses the timer. Timing resumes when the light comes back on.

PAUSE light OFF - Pauses the timer. Timing resumes when the light comes back on.

JAM light OFF - Stops the machine and disables automatic operation. The front panel Start push button must be pressed to resume automatic operation and a dish rack must be inserted to start the timer.

### RCM Status and Jumper Section

The RCM Status/Jumper Section contains one red indicator light and four jumpers that are associated with the set-up of the RCM.

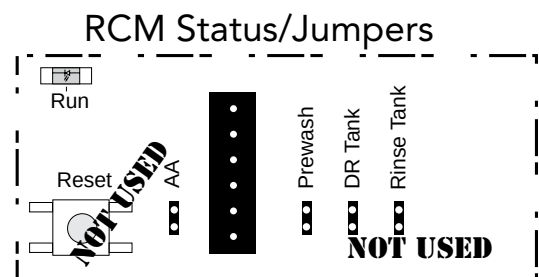
**RUN** - ON when power is turned on to the RCM.  
**THIS RUN LIGHT DOES NOT MEAN THAT THE DISHWASHER IS READY TO OPERATE.**

#### **JUMPERS:**

**Prewash** - Enables the prewash input and output functions.

**AA, DR Tank, Rinse Tank** - NOT USED

**Reset Button** - NOT USED



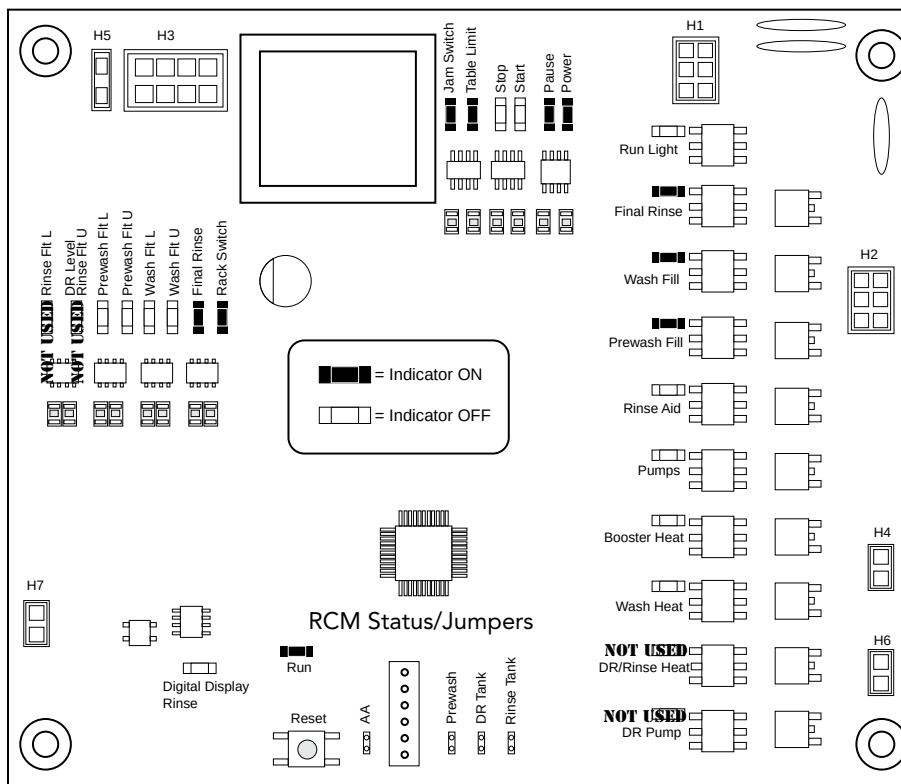
## Rack Control Module (RCM) Indicator Lights

### Normal Operation Sequence

The following illustration Plates show the condition of the indicator lights (On or Off) during the normal operation of the dishwasher. All lights are off when the dishwasher power is turned off.

#### Plate 1:

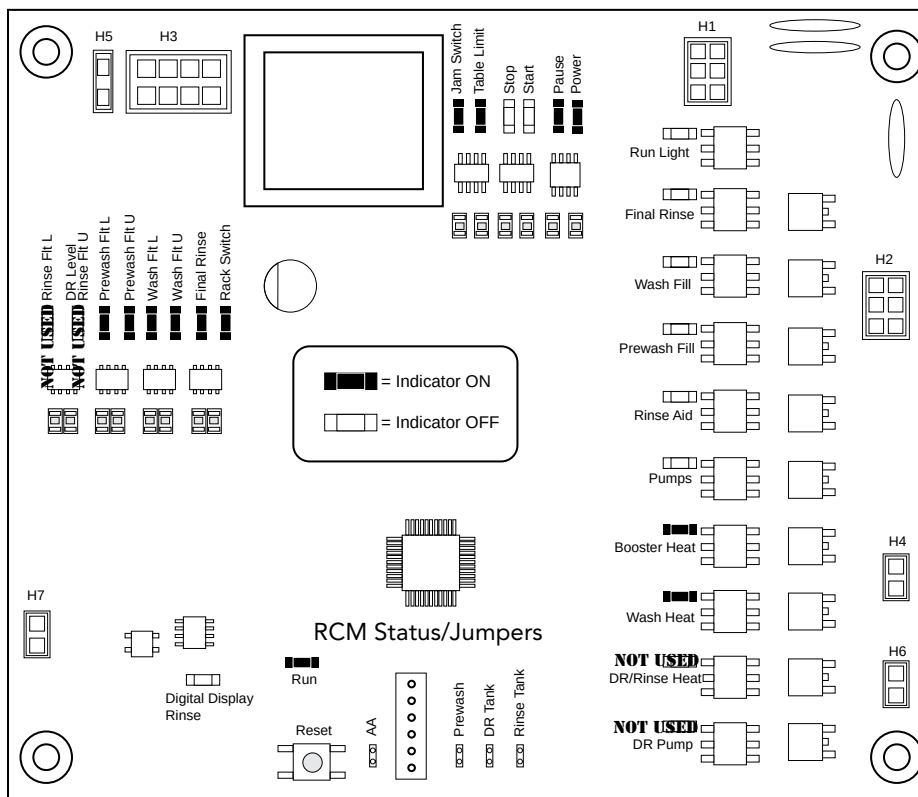
1. Doors closed.
2. Power switch ON.
3. POWER light ON.
4. Floats down.
5. Initial fill begins.



#### Plate 2:

1. Doors closed.
2. POWER light ON.
3. Floats up.
4. Tanks full.
5. Initial fill complete.
6. Tank heat enabled.\*
7. Booster heat enabled.\*

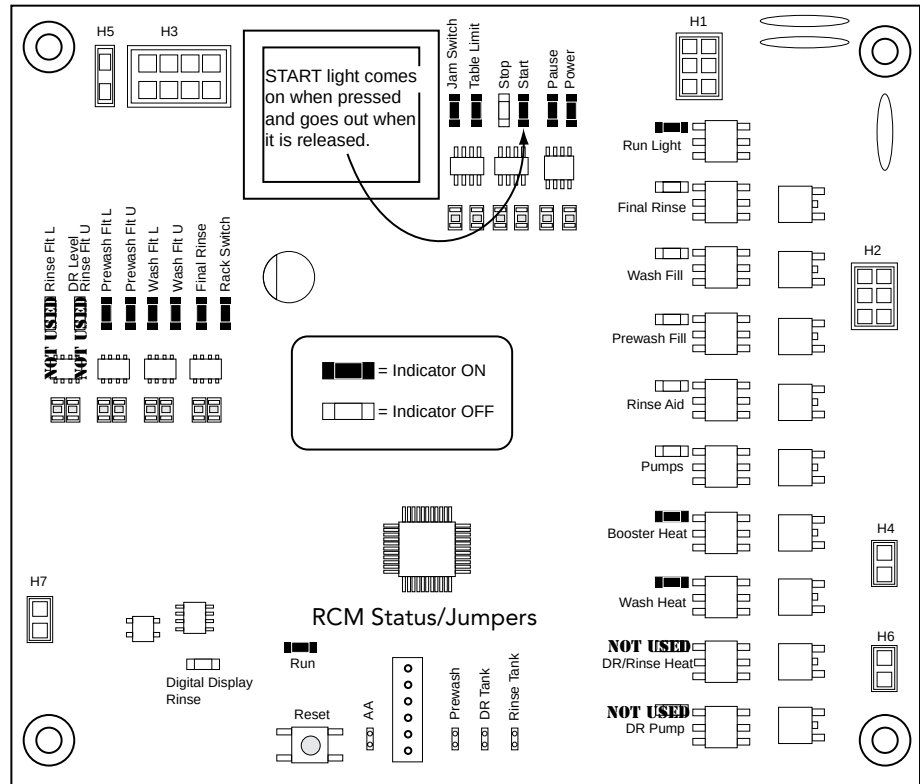
\* NOTE: Tank heat and booster tank heat were enabled as soon as the lower float in the wash tank was up.



## Rack Control Module Operation Normal Operation Sequence

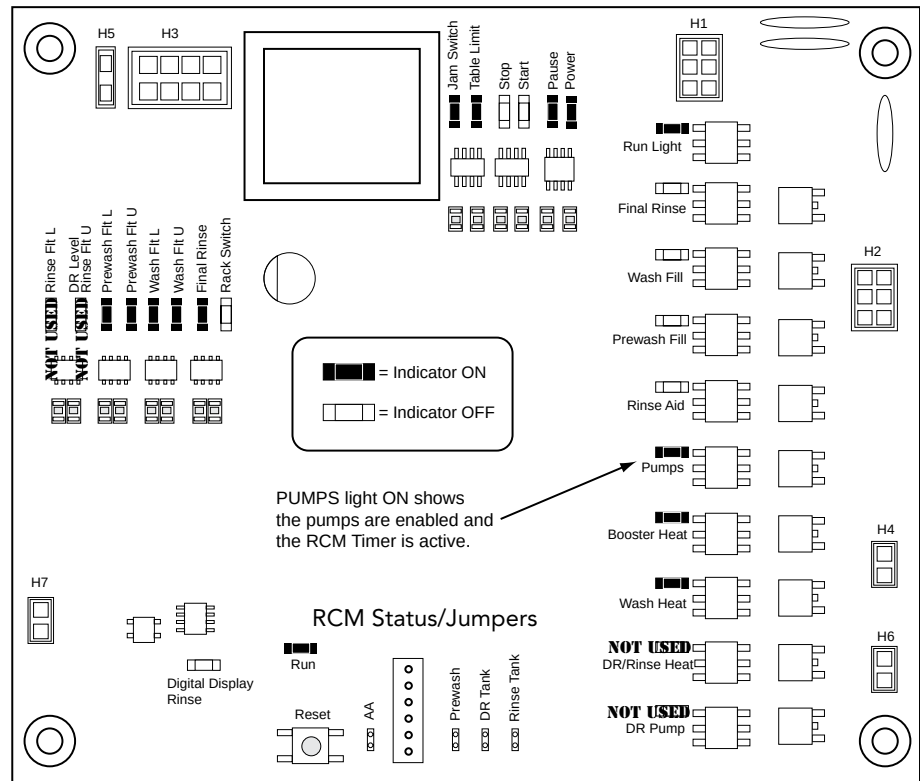
### Plate 3:

1. Doors closed.
2. POWER light ON.
3. Floats up.
4. Tanks full.
5. Tank heat enabled.
6. Booster heat enabled.
7. Initial fill complete.
8. Start push button pressed and released.
9. RUN LIGHT ON and Green light on front panel ON.

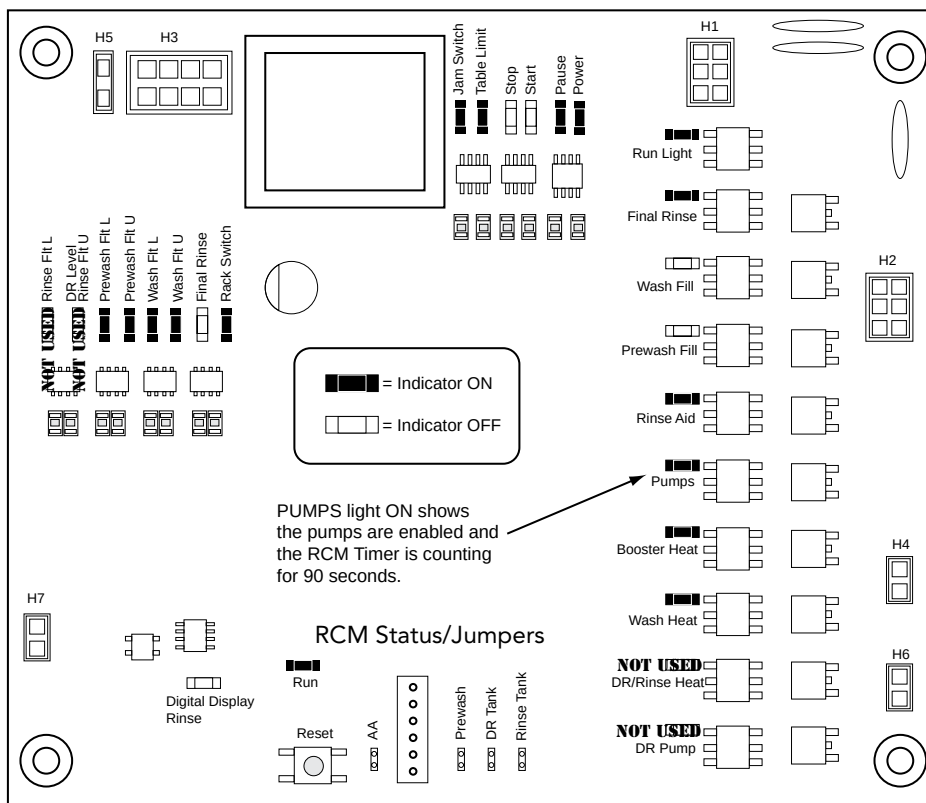


### Plate 4:

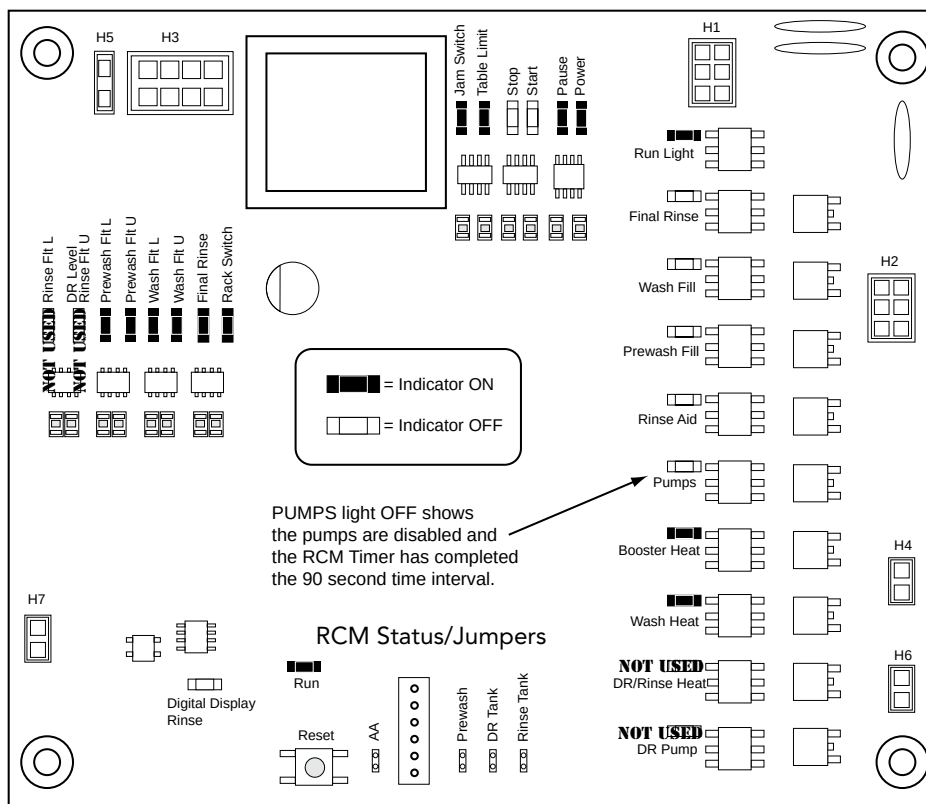
1. Doors closed.
2. POWER light ON.
3. Floats up.
4. Tanks full.
5. Tank heat enabled.
6. Booster heat enabled.
7. Rack inserted into machine/rack switch activated.
8. RCM timer active. Timer will begin counting down the 90 sec. interval when the rack switch is de-activated
9. Pumps and drive motor running.



1. Doors closed.
2. POWER light ON.
3. Floats up.
4. Tanks full.
5. Tank heat enabled.
6. Booster heat enabled.
7. Rack switch de-activated.
8. Rack has moved through the machine to the final rinse zone and activated the final rinse switch.
9. Final rinse valve enabled.
10. Rinse aid enabled.
11. RCM timer counting down the 90 second time interval.
12. Pumps and drive motor running.



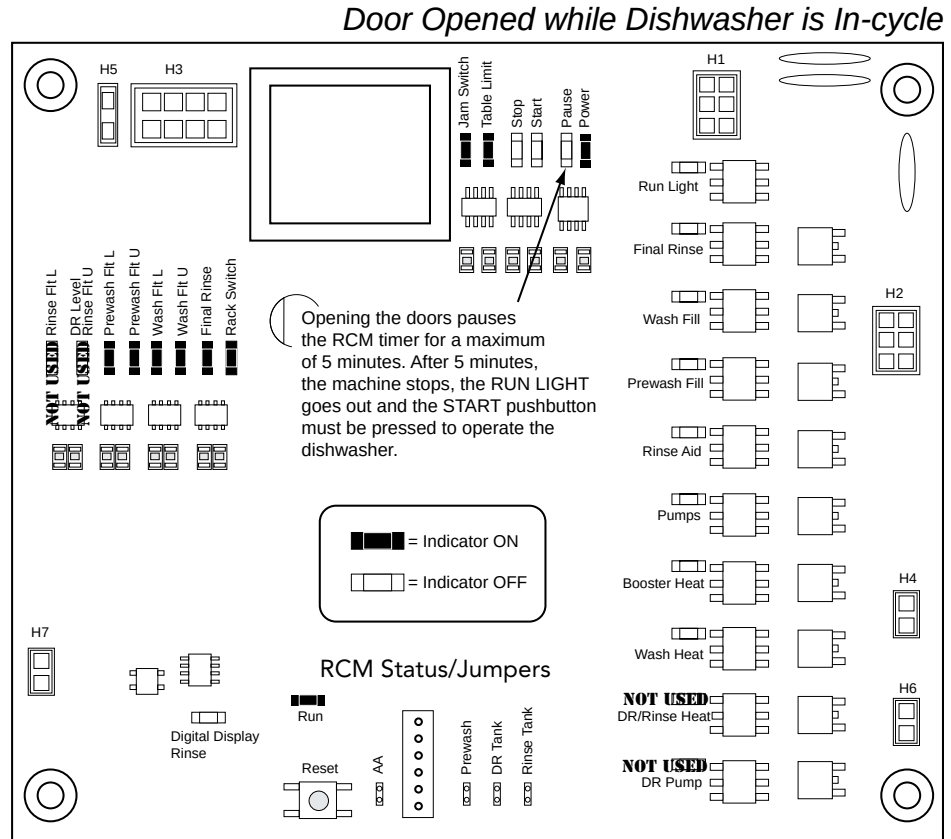
1. Doors closed.
2. Power light ON.
3. Floats up.
4. Tanks full.
5. Tank heat enabled.
6. Booster heat enabled.
7. Rack has exited the machine/final rinse switch de-activated.
8. Final rinse valve disabled.
9. Rinse aid disabled.
10. RCM timer 90 second interval complete.
11. Pumps and drive motor off.





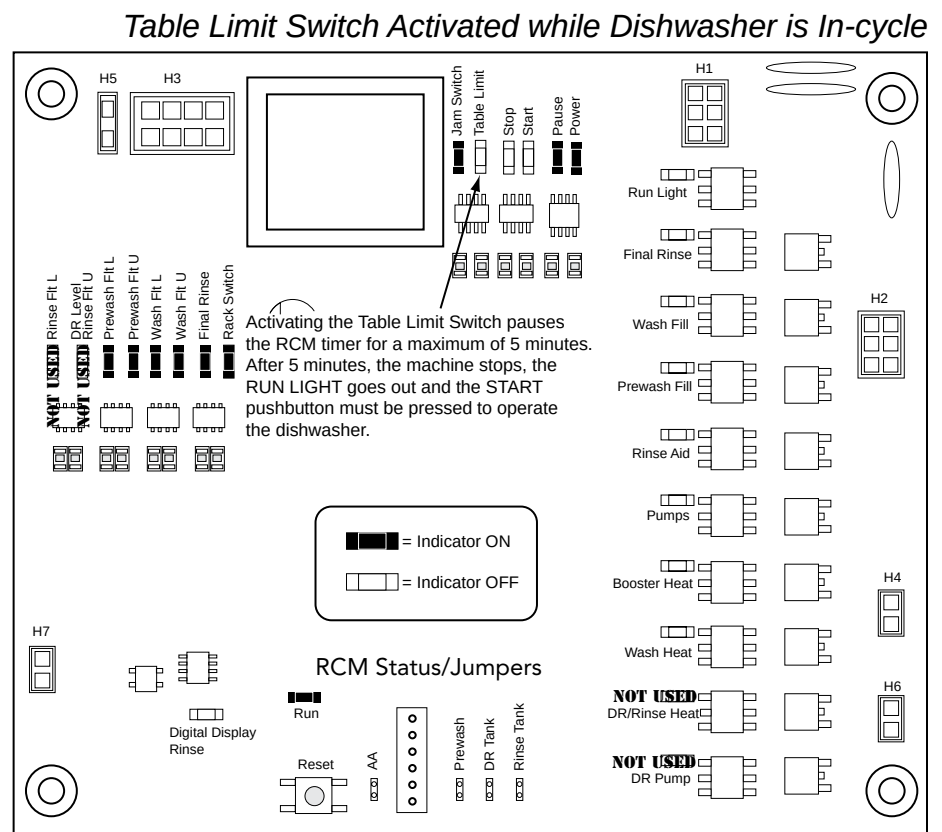
### Plate 7:

1. Door(s) Opened.
2. PAUSE light OFF.
3. POWER light ON.
4. RUN LIGHT OFF.
5. Tanks full.
6. Tank heat Disabled.
7. Booster heat disabled.
8. Pumps and drive disabled.
9. RCM timer paused for a maximum of 5 minutes. After 5 minutes, the machine stops and the START push button must be pressed to operate the dishwasher.
10. Closing the door(s) before 5 mins. has elapsed will resume operation where it left off. Run Light ON. Pause light ON.



### Plate 8:

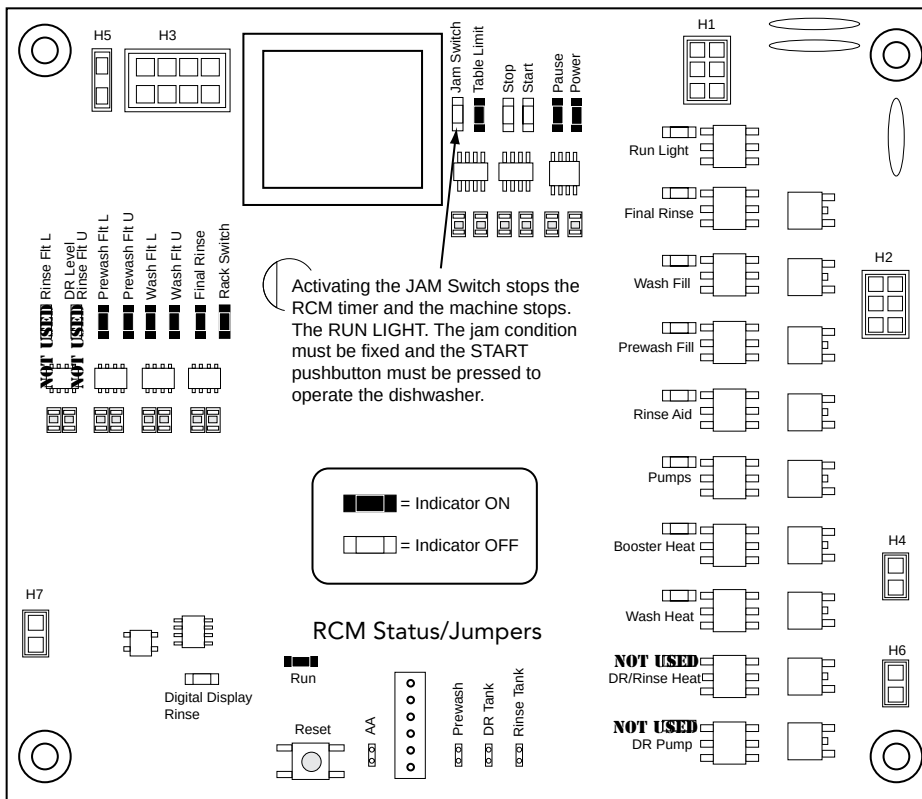
1. Door Closed.
2. POWER light ON.
3. Table limit switch activated.
4. TABLE LIMIT OFF.
5. RUN LIGHT OFF.
6. Tank heat ON.
7. Booster heat ON.
8. Pumps and drive disabled.
9. RCM timer paused for a maximum of 5 minutes. After 5 minutes, the machine stops and the START push button must be pressed to operate the dishwasher.
10. De-activating the table limit switch before 5 mins. has elapsed will resume operation where it left off. RUN LIGHT ON. TABLE LIMIT ON.



*Jam Switch Activated while Dishwasher is In-cycle*

**Plate 9:**

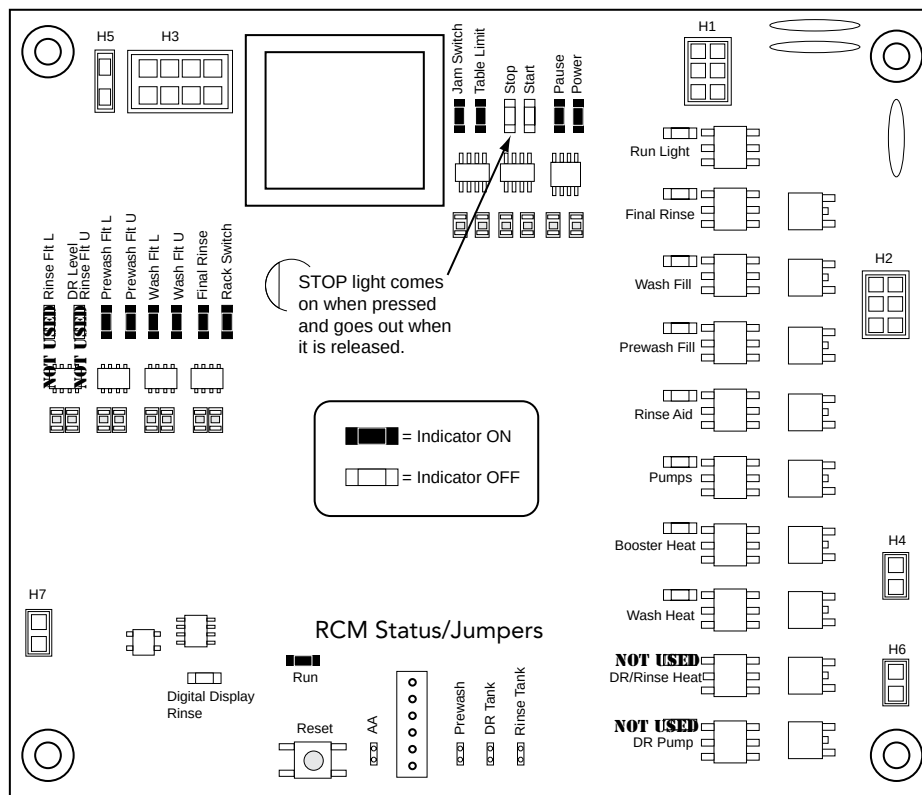
1. Doors closed.
2. POWER light ON.
3. RUN LIGHT OFF.
4. JAM SWITCH OFF.
5. Floats up.
6. Tanks full.
7. Tank heat Disabled.
8. Booster heat disabled.
9. Pumps and drive disabled.
10. RCM timer stopped.
11. Start push button on front panel must be pressed to operate the machine once the jam is fixed.  
RUN LIGHT ON.  
JAM SWITCH ON.



*Stop Switch Pressed while Dishwasher is In-cycle*

**Plate 10:**

1. Door closed.
2. POWER light ON.
3. Floats up.
4. Tanks full.
5. Tank heat Disabled.
6. Booster heat disabled.
7. Pumps and drive disabled.
8. RUN LIGHT OFF.
9. RCM timer stopped.
10. Automatic operation stopped.
11. Start push button on front panel must be pressed to operate the machine.  
RUN LIGHT ON.



## Float Operation During Initial Fill

### Plate 11:

1. Door closed.
2. POWER light ON.
3. Tanks empty.

### ALL FLOATS DOWN

4. Final rinse enabled.
5. Wash fill enabled.
6. Prewash fill enabled  
MD-66 Only
7. Tank heat disabled.
8. Booster heat disabled.

### LOWER FLOATS UP

9. Tank heat enabled.
10. Booster heat enabled.
11. Final rinse, wash fill, prewash fill enabled.

### UPPER FLOATS UP

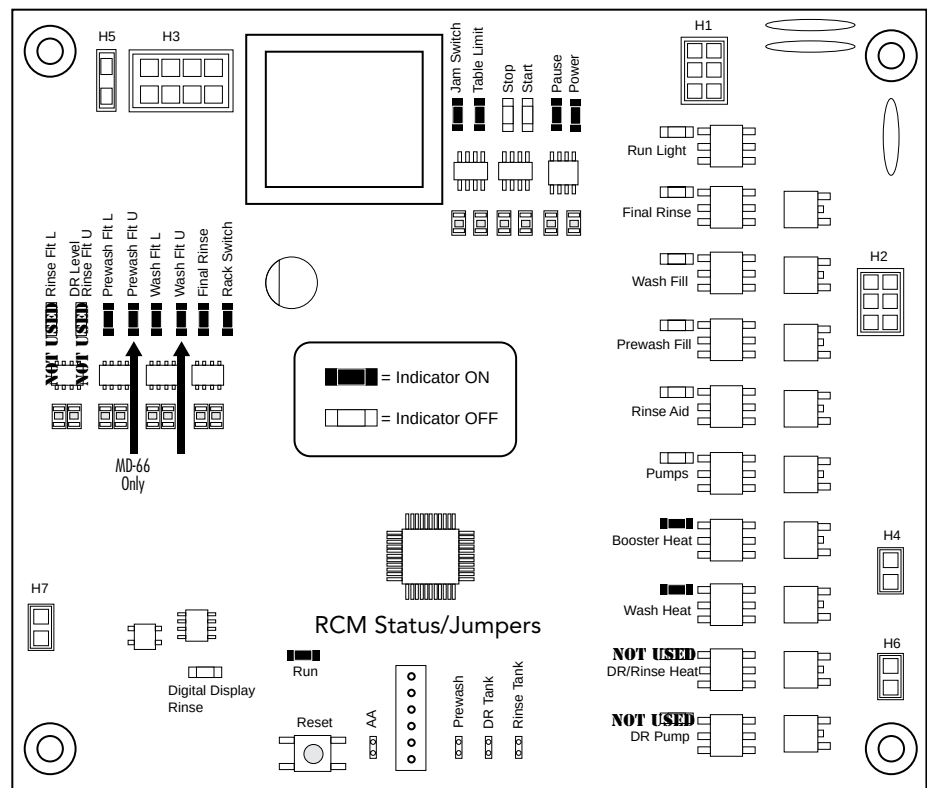
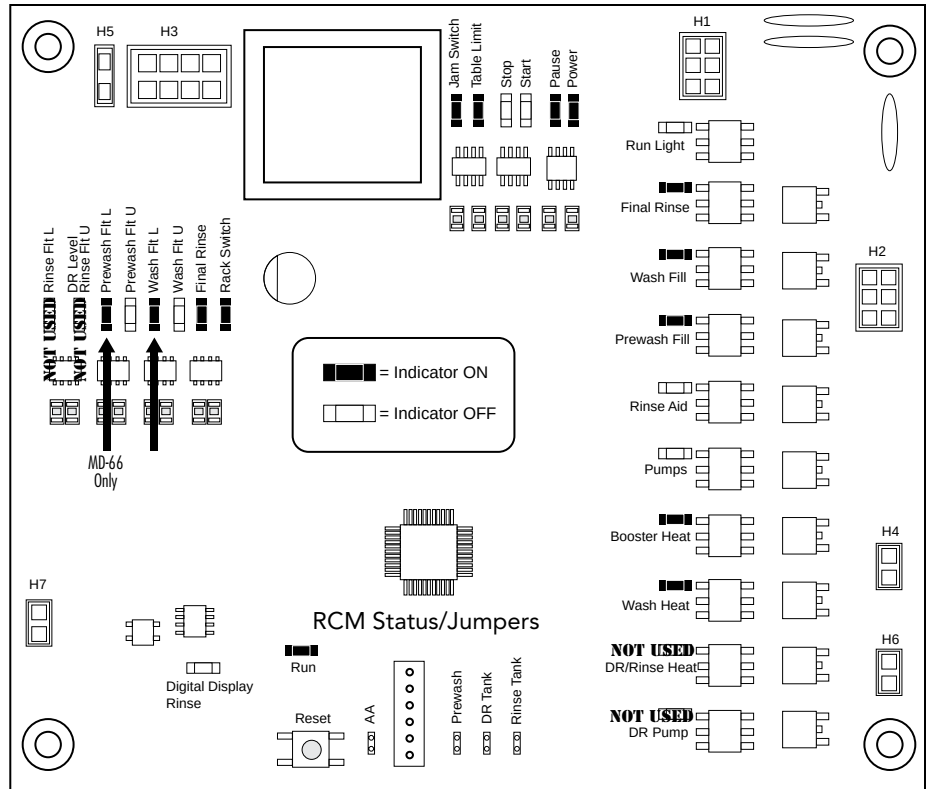
12. Final rinse disabled.
13. Wash fill disabled.
14. Prewash fill disabled.  
MD-66 Only

### !! IMPORTANT !!

If upper floats do not close within 10 minutes of initial fill, then the machine enters a state of SUSPENDED OPERATION.

The final rinse, wash fill and prewash fill are disabled and the run light on the RCM and the front panel will flash on once per second.

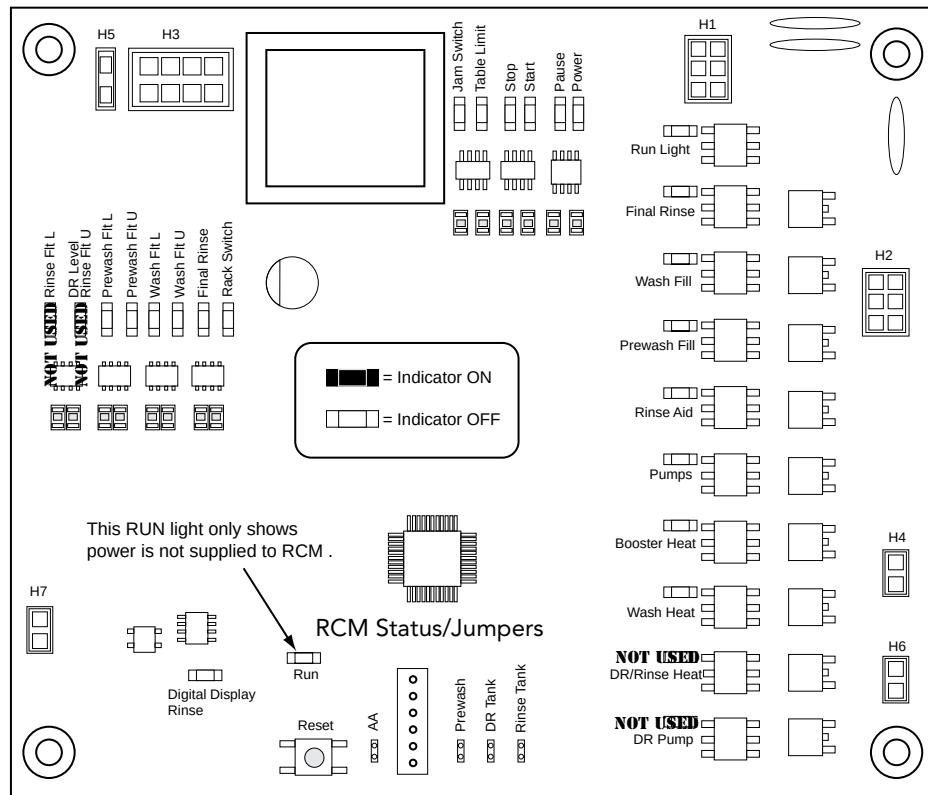
The machine power switch must be turned off and then back on to clear this condition.



Front Panel Power Switch Turned Off while Dishwasher is In-cycle

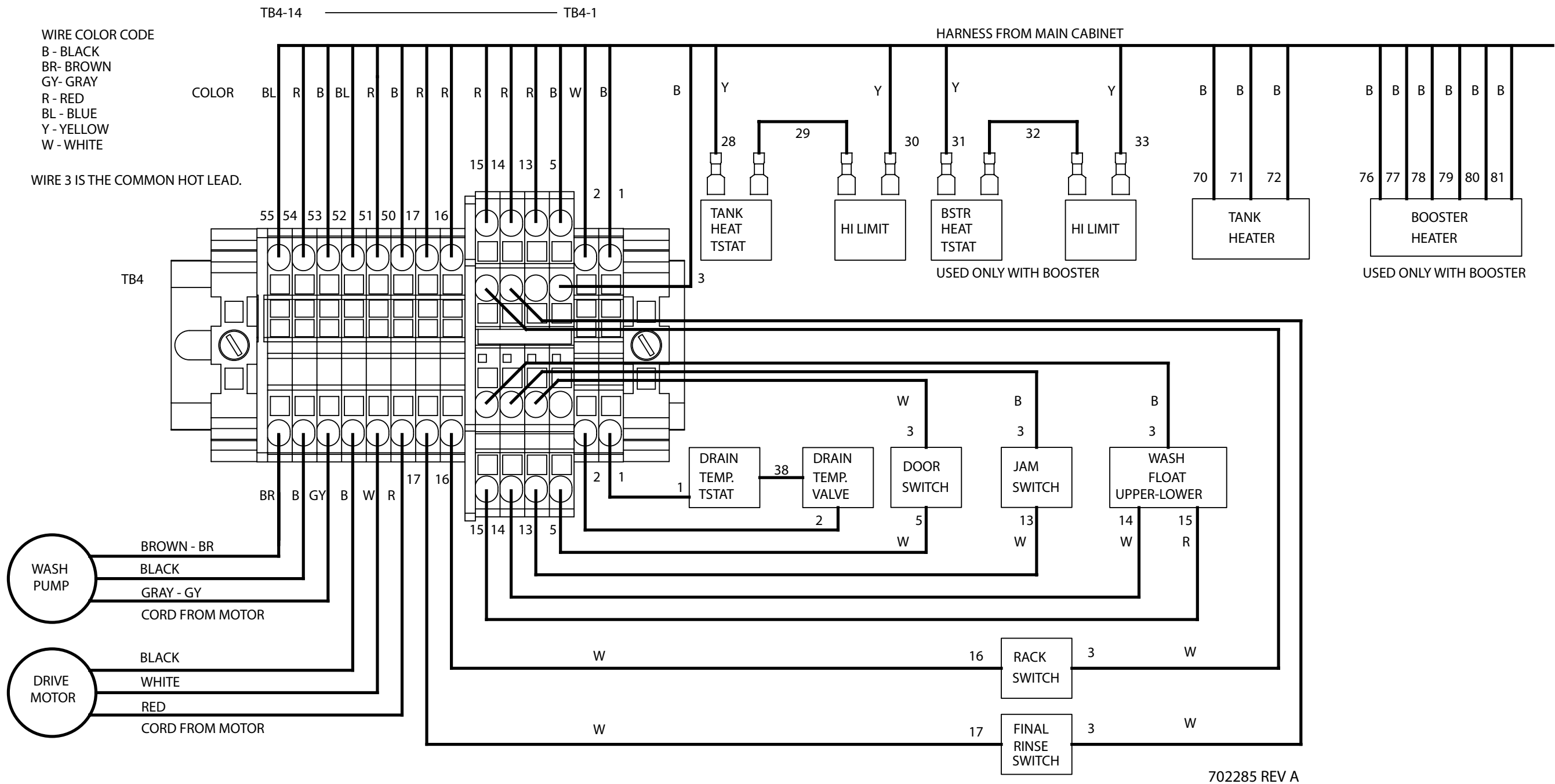
**Plate 12:**

1. Door closed.
2. POWER Switch OFF.
3. Floats up.
4. Tanks full.
5. Tank heat Disabled.
6. Booster heat disabled.
7. Pumps and drive disabled.
8. RUN LIGHT OFF.
9. RCM Status light OFF.
10. All Input and Output lights OFF.





CH-44/MD-44 JUNCTION BOX WIRING DIAGRAM



|                                                                                                                                                                                          |               |       |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|--------|
| CUSTOMER TO SUPPLY RATED VOLTAGE/PHASE/Hz,<br>AS SPECIFIED PER ORDER, TO DISCONNECT SWITCH.<br>ALL POWER SUPPLIED TO EACH CONNECTION POINT<br>MUST COMPLY WITH ALL LOCAL ELECTRIC CODES. |               |       |        |
| DR. BY                                                                                                                                                                                   | J. MCALLISTER | SCALE | NONE   |
| DATE                                                                                                                                                                                     | 26 MAY 10     | SHEET | 1 OF 1 |

| REV. | DESCRIPTION                         | DATE     | BY  |
|------|-------------------------------------|----------|-----|
| A    | MOVED DRAIN TEMPERING VALVE CIRCUIT | 4 MAY 10 | JAM |
|      |                                     |          |     |
|      |                                     |          |     |

| REV. | DESCRIPTION | DATE | BY |
|------|-------------|------|----|
|      |             |      |    |
|      |             |      |    |
|      |             |      |    |

**Champion**  
The Dishwashing Machine Specialists

CH-44//MD44  
JUNCTION BOX WIRING DIAGRAM  
B 702285 REV. A

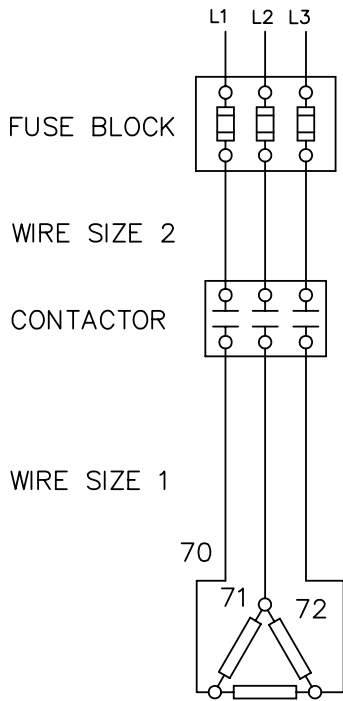
CH44, MD44, CH66, MD66 Rack Machine Tank and Booster Heater Wiring

15 KW TANK HEATER

|                  |      |
|------------------|------|
| VOLTAGE 200–220V | WASH |
| KW/CONTACTOR     | 15KW |
| CONTACTOR RATING | 60A  |
| FUSE RATING      | 50A  |
| WIRE SIZE        | #8   |

|                  |      |
|------------------|------|
| VOLTAGE 230–240V | WASH |
| KW/CONTACTOR     | 15KW |
| CONTACTOR RATING | 60A  |
| FUSE RATING      | 45A  |
| WIRE SIZE        | #8   |

|                  |      |
|------------------|------|
| VOLTAGE 460–480V | WASH |
| KW/CONTACTOR     | 15KW |
| CONTACTOR RATING | 60A  |
| FUSE RATING      | 25A  |
| WIRE SIZE        | #12  |



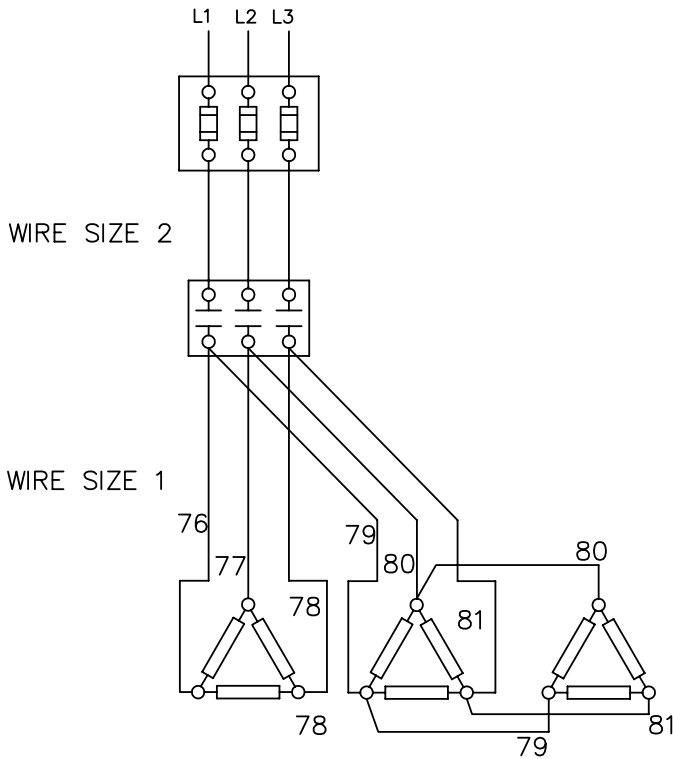
TOTAL  
15KW      15KW

16 KW BOOSTER HEATER

|                  |       |
|------------------|-------|
| VOLTAGE 208V     | DELTA |
| TOTAL KW         | 16KW  |
| KW/CONTACTOR     | 16KW  |
| CONTACTOR #      | 1     |
| CONTACTOR RATING | 60A   |
| WIRE SIZE 1      | 10    |
| WIRE SIZE 2      | 8     |
| FUSE RATING      | 60A   |

|                  |       |
|------------------|-------|
| VOLTAGE 240V     | DELTA |
| TOTAL KW         | 16KW  |
| KW/CONTACTOR     | 16KW  |
| CONTACTOR #      | 1     |
| CONTACTOR RATING | 60A   |
| WIRE SIZE 1      | 10    |
| WIRE SIZE 2      | 8     |
| FUSE RATING      | 50A   |

|                  |       |
|------------------|-------|
| VOLTAGE 480V     | DELTA |
| TOTAL KW         | 16KW  |
| KW/CONTACTOR     | 16KW  |
| CONTACTOR #      | 1     |
| CONTACTOR RATING | 60A   |
| WIRE SIZE 1      | 10    |
| WIRE SIZE 2      | 8     |
| FUSE RATING      | 25A   |



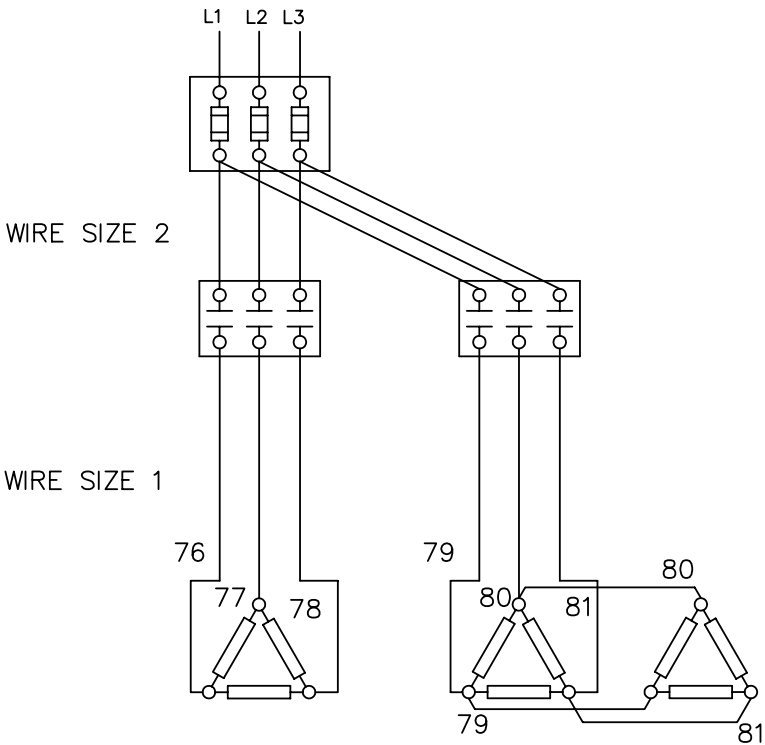
TOTAL  
16KW      6KW      5KW      5KW

27 KW BOOSTER HEATER

|                  |              |
|------------------|--------------|
| VOLTAGE 208V     | DELTA        |
| TOTAL KW         | 27KW         |
| KW/CONTACTOR     | 12KW    15KW |
| CONTACTOR #      | 1        2   |
| CONTACTOR RATING | 60A    60A   |
| WIRE SIZE 1      | 8            |
| WIRE SIZE 2      | 8            |
| FUSE RATING      | 90A          |

|                  |              |
|------------------|--------------|
| VOLTAGE 240V     | DELTA        |
| TOTAL KW         | 27KW         |
| KW/CONTACTOR     | 12KW    15KW |
| CONTACTOR #      | 1        2   |
| CONTACTOR RATING | 60A    60A   |
| WIRE SIZE 1      | 8            |
| WIRE SIZE 2      | 8            |
| FUSE RATING      | 80A          |

|                  |             |
|------------------|-------------|
| VOLTAGE 480V     | DELTA       |
| TOTAL KW         | 27KW        |
| KW/CONTACTOR     | 12KW    15W |
| CONTACTOR #      | 1        2  |
| CONTACTOR RATING | 60A    60A  |
| WIRE SIZE 1      | 10          |
| WIRE SIZE 2      | 10          |
| FUSE RATING      | 40A         |



TOTAL  
27KW      12KW      10KW      5KW

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